

Chapter 6: Shunt FACTS Controllers: TCR, TSR, TSC, SVC and STATCOM

Learning Objectives

After working through this chapter, the reader should be able to:

- explain, from the Thevenin equivalent of a bus and from the two-machine power-angle relationship, why controllable shunt reactive current is the most direct means of supporting voltage and raising transfer capability;
- derive the instantaneous current, fundamental susceptance, total RMS current, individual harmonics and THD of a thyristor-controlled reactor, and translate correctly between the zero-crossing and peak-referenced firing-angle conventions found in the literature;
- analyse six-pulse, twelve-pulse and segmented TCR arrangements, including the sequence behaviour of their harmonics and the cancellation produced by phase-shifting transformers;
- design thyristor-switched reactor and capacitor stages, including the series detuning reactor, the magnification factor, transient-free and minimum-transient firing instants, valve voltage stress after turn-off, and binary-weighted banks;
- construct the V-I and susceptance characteristics of FC-TCR, TSC-TCR and transformer-coupled SVCs, locate their operating points on a network characteristic, and design the voltage regulator against the effective short-circuit ratio;
- derive the voltage-source model of the STATCOM, its V-I and P-Q capability, its converter waveforms for multi-pulse, three-level and PWM topologies, its dq current control and its DC capacitor requirement;
- compare the five devices quantitatively under voltage depression, harmonic, loss, response and cost criteria, and relate them to current research questions.

The five controllers treated in this chapter are classified in Table 6.1 by power stage, control action and the nature of their output.

Table 6.1. Classification of shunt FACTS controllers.

Device	Power stage	Control action	Output range	Nature of output
TCR	Reactor + antiparallel thyristors	Phase (firing-angle) control	Continuous, inductive only	Variable susceptance, harmonics generated
TSR	Reactor + antiparallel thyristors	On/off at voltage peak	Stepped, inductive	Fixed susceptance per stage
TSC	Capacitor + small reactor + thyristors	On/off at voltage match or appropriate crest	Stepped, capacitive	Fixed susceptance per stage
SVC	Combinations of TCR, TSR, TSC, FC, filters	Coordinated stage switching + TCR trim	Continuous, capacitive and inductive	Variable susceptance; $Q \propto V^2$ at limits
STATCOM	VSC or MMC with DC capacitor	Converter voltage magnitude and phase	Continuous, symmetric about zero	Controlled voltage/current within limits; $Q \propto V$ at current limit

6.1 Reactive Power, Voltage and the Case for Shunt Compensation

6.1.1 The voltage-reactive current coupling at a bus

Seen from any bus, a transmission network can be reduced to a Thevenin source E_{th} behind an impedance $R_{th} + jX_{th}$. For transmission networks $X_{th} \gg R_{th}$, and if a shunt device draws a purely reactive current I (taken positive when inductive, that is, lagging the bus voltage by 90°), the bus voltage magnitude obeys

$$V = E_{th} - X_{th} I \quad (6.1)$$

where V is the RMS bus voltage, E_{th} the open-circuit Thevenin voltage, X_{th} the Thevenin reactance and I the signed RMS reactive-current component drawn by the device, positive for inductive absorption. All are in per unit on a common base. For an ideal lossless Thevenin circuit and purely reactive current, Equation (6.1) is the load line on the positive-voltage branch. Its gradient is $-X_{th}$: capacitive current ($I < 0$) raises voltage. A correction ΔV requires current magnitude $|\Delta V|/X_{th}$, so a weak bus has greater voltage sensitivity, and for the same reason its voltage-control loop is more prone to interaction. The device characteristic and the load line together locate the steady operating point.

The more general approximate expression for voltage drop across a short line, $\Delta V \approx (RP + XQ)/V$, shows the same thing from the power side: where X/R is large, the reactive term dominates, and local reactive injection is the most effective lever on voltage magnitude.

6.1.2 Ideal midpoint compensation of a two-machine system

Consider the lossless two-machine system of Figure 6.1, with equal end voltages $V_s = V_r = V$, a total series reactance X , and a shunt compensator at the electrical midpoint.

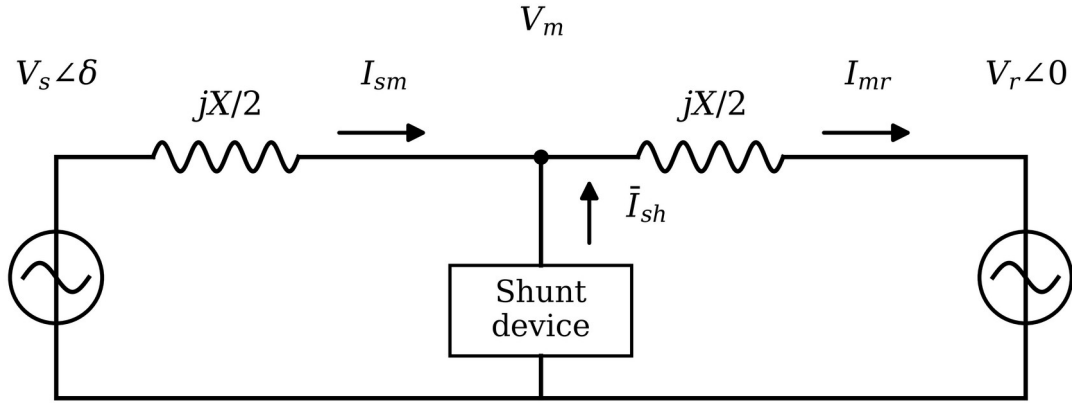


Figure 6.1. Two-machine system with midpoint shunt compensation. Line currents are directed from sending to receiving end; I_{sh} is injected upward into the midpoint bus.

Without compensation the transmitted power is

$$P_u = \frac{V^2}{X} \sin \delta \quad (6.2)$$

where P_u is the uncompensated active power, δ the angle between the end voltages and X the total line reactance. The uncompensated midpoint voltage follows from the phasor average of the two end voltages,

$$V_{m0} = V \cos \frac{\delta}{2} \quad (6.3)$$

where V_{m0} is the midpoint voltage magnitude with no shunt current. This sag of the midpoint as δ rises is the physical reason the uncompensated line saturates at $\delta = 90^\circ$.

Now let the compensator hold $V_m = V$. Because the compensator exchanges no active power, the same P flows through each half-line, and by symmetry each half carries angle $\delta/2$ across reactance $X/2$. Applying Equation (6.2) to one half,

$$P_c = \frac{V \cdot V}{X/2} \sin \frac{\delta}{2} = \frac{2V^2}{X} \sin \frac{\delta}{2} \quad (6.4)$$

where P_c is the transmitted power with ideal midpoint regulation. The maximum, $2V^2/X$, occurs at $\delta = 180^\circ$: in this idealized symmetric case the steady-state limit doubles. The phasor diagram in Figure 6.2 shows the geometry.

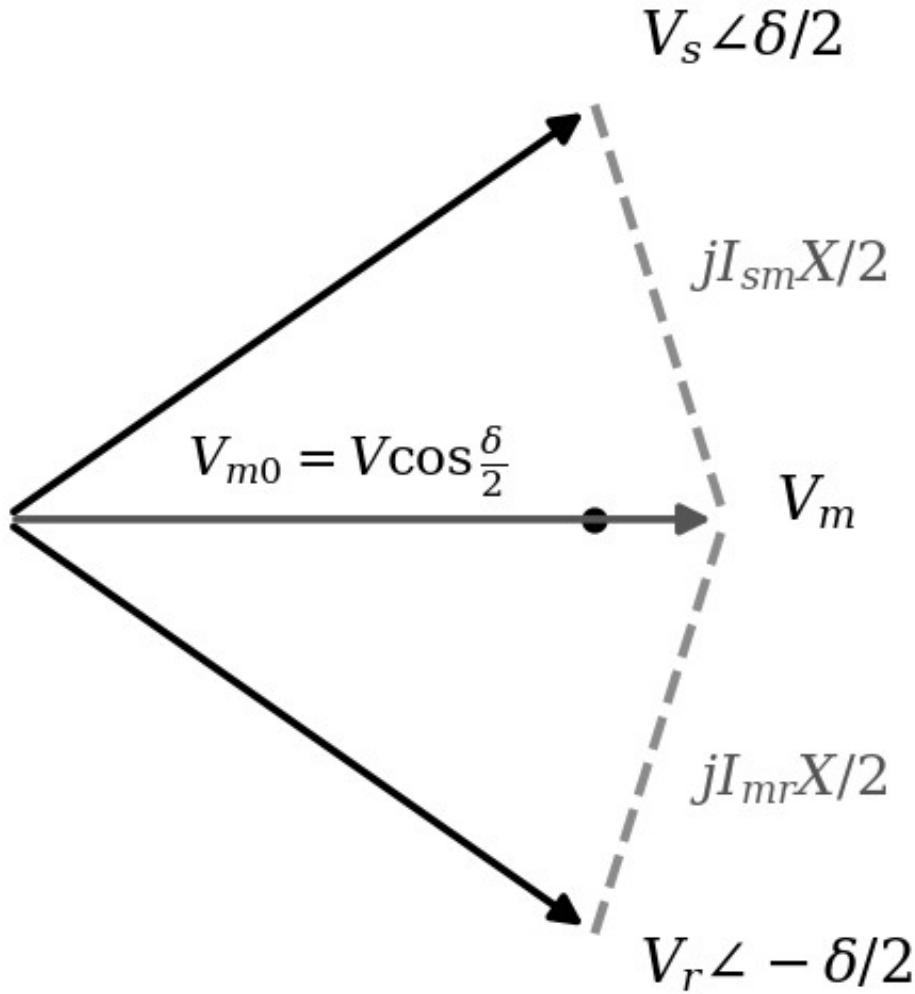


Figure 6.2. Midpoint-compensation phasors on a symmetric angle reference: $\underline{V}_s = V \angle (\delta/2)$, $\underline{V}_r = V \angle (-\delta/2)$ and $\underline{V}_m = V \angle 0$. This is the same circuit as Figure 6.1 after rotating all phasors by $-\delta/2$.

The current entering the midpoint from the sending side is $I_{sm} = (V e^{j\delta/2} - V)/(jX/2)$, and the current leaving towards the receiving end is $I_{mr} = (V - V e^{-j\delta/2})/(jX/2)$. Their difference must be supplied by the compensator. Carrying out the subtraction,

$$I_{sh} = I_{mr} - I_{sm} = -j \frac{4V}{X} \left(1 - \cos \frac{\delta}{2} \right) \quad (6.5)$$

The phasor I_{sh} is purely reactive and lags the midpoint voltage by 90° when directed from converter to bus. Its injected complex power is $S_{sh} = \underline{V}_m \underline{I}_{sh}^*$, so its reactive part is positive: the device supplies capacitive vars. The bus-to-device current has the opposite direction and leads voltage. In Equation (6.6), I_{sh} without an overbar denotes the positive RMS magnitude, not the complex phasor.

$$I_{sh} = \frac{4V}{X} \left(1 - \cos \frac{\delta}{2} \right) = \frac{8V}{X} \sin^2 \frac{\delta}{4}, Q_{sh} = \frac{4V^2}{X} \left(1 - \cos \frac{\delta}{2} \right) \quad (6.6)$$

where I_{sh} is the RMS shunt current and Q_{sh} the capacitive reactive power the compensator must inject. At $\delta = 180^\circ$ these reach $4V/X$ and $4V^2/X$, finite but large: four times the maximum power of the uncompensated line. Real compensators are therefore rated well below this ideal point, and the shape of their limiting characteristic (constant susceptance for an SVC, constant current for a STATCOM) decides how the power-angle curve bends once the limit is reached. Figure 6.3 anticipates that comparison, which is derived in Sections 6.5.8 and 6.6.9.

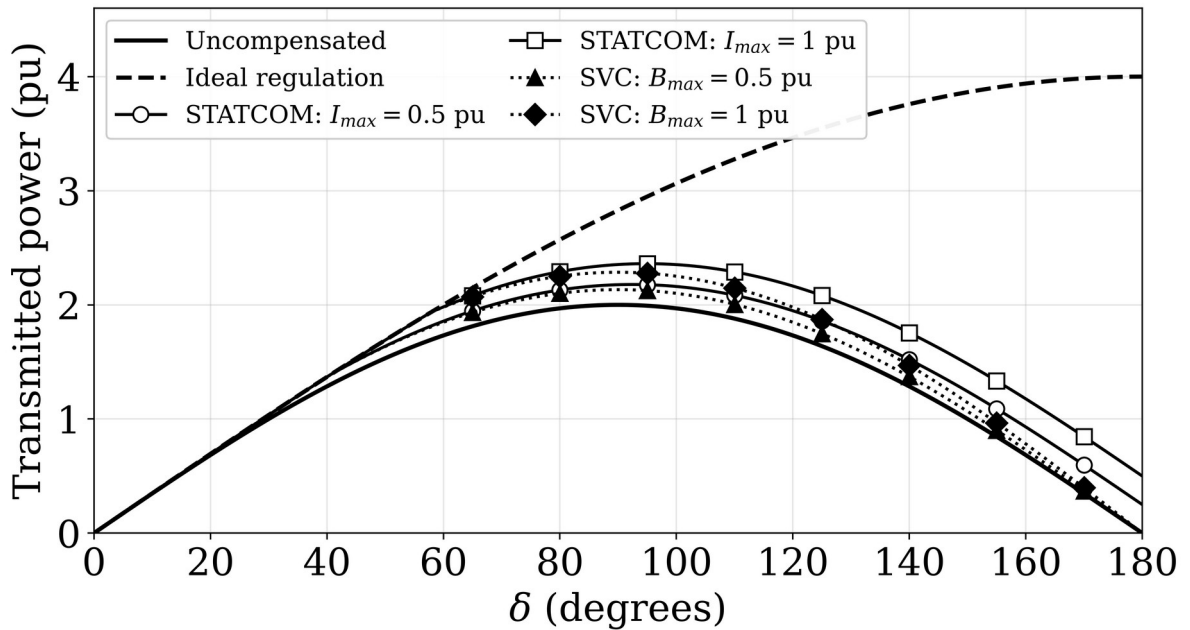


Figure 6.3. Power-angle curves for $X = 0.5$ pu: uncompensated line, ideal midpoint regulation, and midpoint SVC and STATCOM with finite ratings.

6.1.3 Voltage stability, transient stability and damping

For a radial corridor, a P-V nose marks the limit of the stated power-flow model. Local reactive compensation can increase the loadability margin while it remains within its limits. At its capacitive susceptance limit an SVC behaves as a fixed capacitor with $Q \propto V^2$; it can therefore fail to arrest a severe voltage decline. Whether a controller changes the collapse boundary depends on network, load and protection dynamics. A current-limited STATCOM has $Q \propto V$ within its feasible operating envelope. This distinction underpins Section 6.6.11.

For transient stability, midpoint capacitive support raises the post-fault power-angle curve and enlarges the decelerating area of the equal-area construction; how much it gains depends on the clearing time, the device limits and the post-fault network. For small-signal oscillations, a supplementary controller modulates the reactive output through phase compensation chosen to produce damping torque. The required phase depends

on the measured signal and the operating point, and an unshaped speed signal with the wrong phase can reduce damping instead of adding it. Chapter 11 develops these controls.

6.1.4 Evolution of shunt compensation: a short literature survey

Mechanically switched capacitors and reactors, and rotating synchronous condensers, were the only options until the late 1960s. Saturated-reactor compensators, built for arc-furnace flicker reduction, were the first static devices, but their slow, fixed slope and high losses limited them. The line-commutated thyristor changed the picture in the early 1970s. Gyugyi's systematic treatment of thyristor var generators [1] and the collection edited by Miller [2] established the TCR-FC and TSC-TCR configurations and the analytical tools still used today, including the susceptance-versus-firing-angle law, the harmonic formula, and the transient-free switching conditions for capacitors. Mathur's edited volume [3] consolidated SVC application and control practice.

The second wave began with forced-commutated converters. A 20 MVA force-commutated var generator was field tested in Japan from 1980 [4], an 80 MVA GTO-based installation entered commercial service at Inuyama in 1991 [5], and the ± 100 MVA Sullivan STATCOM of the Tennessee Valley Authority, built under EPRI sponsorship and commissioned in 1995, became the reference case for transmission STATCOMs in North America [6]. Hingorani and Gyugyi [7] gave these devices the vocabulary used in this book (STATCOM, SSG, SVG) and introduced the V-I and loss comparisons reproduced in Section 6.6. Mathur and Varma [8], Song and Johns [9], Padiyar [10], Acha *et al.* [11], Zhang *et al.* [12] and Sen and Sen [13] extend the modelling, control and power-flow representation. Since about 2010, modular multilevel converters (Chapter 5) have displaced multi-pulse transformers in new STATCOMs, and research has moved towards energy-storage-equipped and grid-forming STATCOMs (Section 6.7).

6.1.5 Thyristor valves and firing-angle conventions

Thyristor-based shunt devices use bidirectional valves formed from antiparallel thyristor strings. Series devices provide blocking voltage, and parallel paths may be used for current capability. Snubbers and grading networks control dynamic and static voltage sharing. Triggering may be electrical or optical according to valve design; protection, insulation and cooling are selected for the station duty. Example 6.21 gives a simplified blocking-voltage and current calculation, not a complete valve design.

A source of genuine confusion in textbooks, lecture notes and examination papers is the reference for the firing angle. Two conventions coexist, illustrated in Figure 6.4. In this chapter α is always measured from the zero crossing of the voltage across the branch, so that full conduction of a reactor corresponds to $\alpha = 90^\circ$ and blocking to $\alpha = 180^\circ$. Several classic texts and many course notes instead measure a delay β from the voltage peak, with $0 \leq \beta \leq 90^\circ$. The two are related by

$$\beta = \alpha - 90^\circ \quad (6.7)$$

where α is the zero-crossing-referenced firing angle and β the peak-referenced delay angle. Every formula must be used with the reference for which it was derived. Table 6.2 lists the equivalent forms of the main TCR expressions, and Example 6.19 shows the numerical consequences of mixing them.

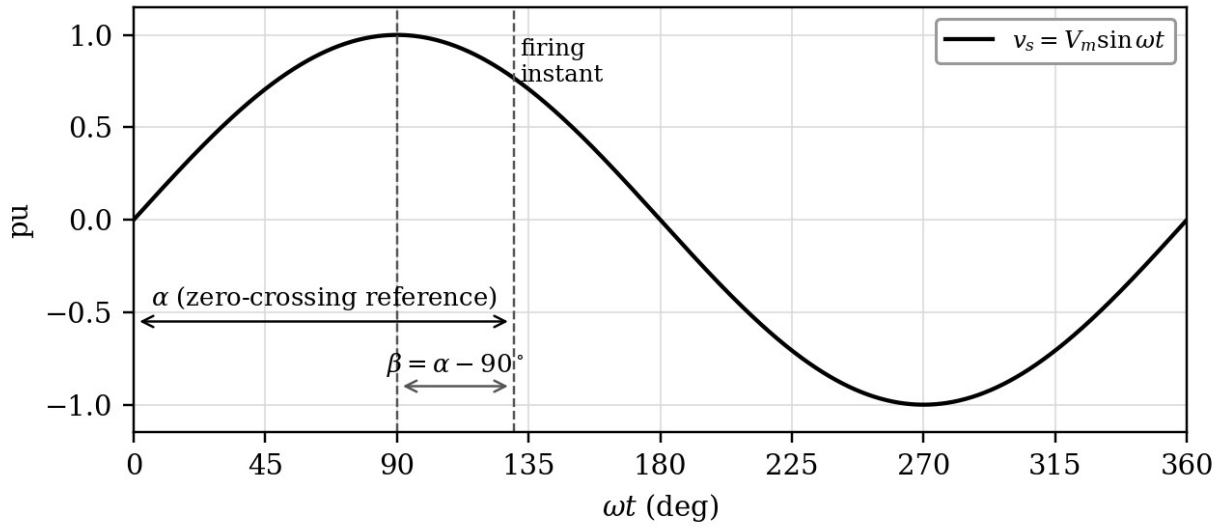
Figure 6.4. Zero-crossing (α) and peak-referenced (β) firing-angle conventions.

Table 6.2. Equivalent TCR expressions in the two firing-angle conventions.

Quantity	Zero-crossing form ($90^\circ \leq \alpha \leq 180^\circ$)	Peak-referenced form ($0 \leq \beta \leq 90^\circ$)
Conduction angle	$\sigma = 2(\pi - \alpha)$	$\sigma = \pi - 2\beta$
Branch current	$\frac{V_m}{\omega L} (\cos \alpha - \cos \omega t)$	$\frac{V_m}{\omega L} (\sin \omega t' - \sin \beta)$, $v = V_m \cos \omega t'$
B_{TCR}/B_{max}	$\frac{2(\pi - \alpha) + \sin 2\alpha}{\pi}$	$1 - \frac{2\beta}{\pi} - \frac{\sin 2\beta}{\pi}$
Full conduction / blocked	$90^\circ / 180^\circ$	$0^\circ / 90^\circ$
Third harmonic maximum	$\alpha = 120^\circ$	$\beta = 30^\circ$

6.2 The Thyristor-Controlled Reactor (TCR)

Reading conventions. Unless explicitly marked peak, V and I denote RMS fundamental quantities. Angles inside trigonometric derivatives and integrals are in radians. Firing-angle results may be reported in degrees. Harmonic percentages in Table 6.3 use the full-conduction fundamental as their fixed base; THD uses the actual fundamental at the selected firing angle. These percentages are different quantities.

6.2.1 Operating principle

A single-phase TCR (Figure 6.5) is a linear reactor of inductance L in series with an antiparallel thyristor valve. Each thyristor is gated in its appropriate half-cycle and turns off at natural current zero. The firing angle sets the start of a pulse; the circuit determines its end. Splitting the reactor into two windings on either side of the valve is common. This provides a symmetric arrangement and retains some current-limiting impedance if one winding is shorted, but does not replace fault protection or the valve's full blocking-voltage rating.

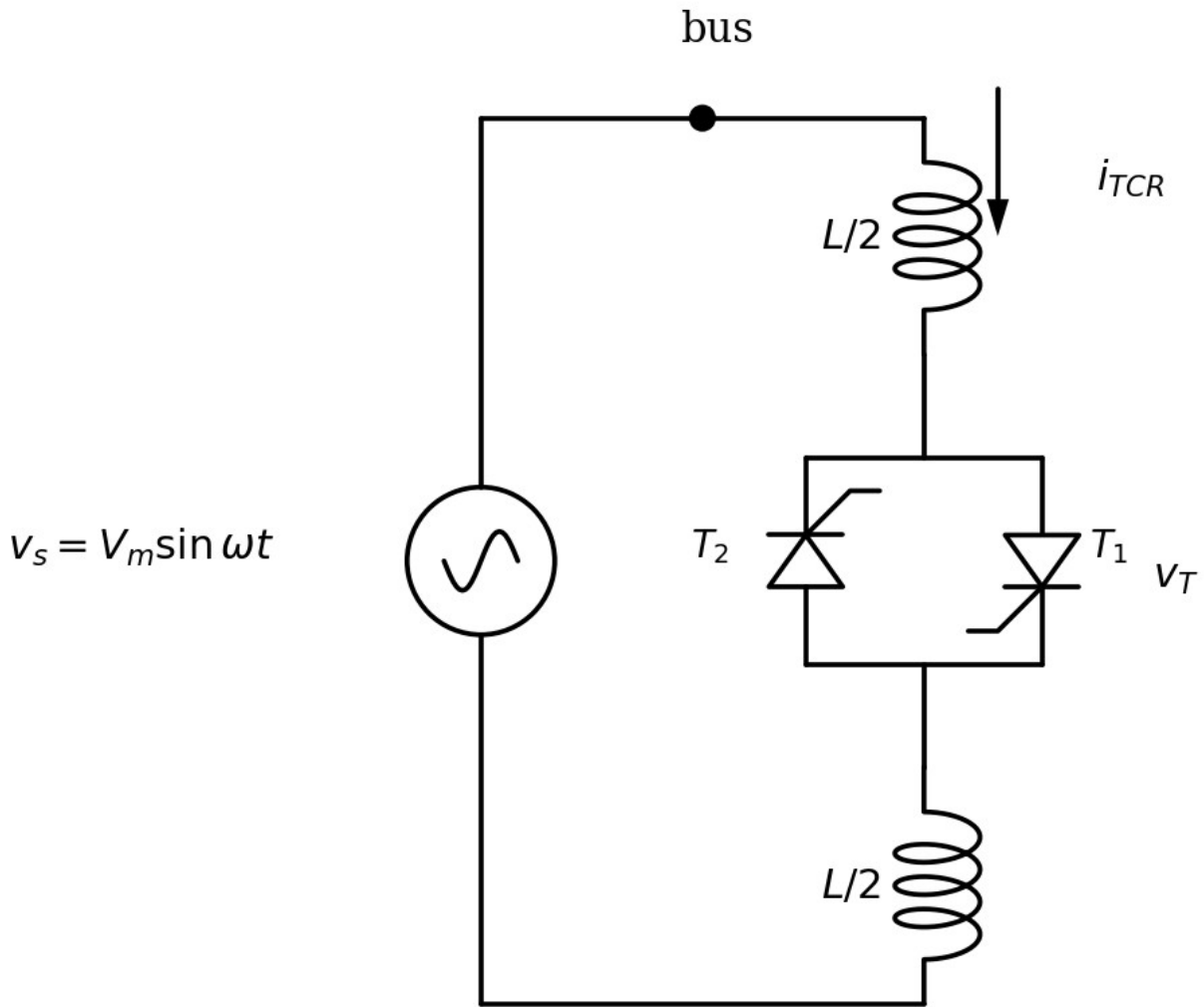


Figure 6.5. Single-phase TCR with split reactor and antiparallel thyristor valve.

Figure 6.6 shows the branch current and valve voltage at four firing angles. At $\alpha = 90^\circ$ the pulses from T_1 and T_2 join into a continuous sinusoid lagging the voltage by 90° , exactly the current of the bare reactor. As α increases, conduction starts later and, by symmetry of the inductive circuit, ends equally early, so each pulse is centred on the voltage zero crossing at $\omega t = \pi$ (and 2π for the negative pulse) and shrinks towards zero as $\alpha \rightarrow 180^\circ$. Firing earlier than 90° is not used: conduction would then extend past the next firing instant of the opposite thyristor, the currents would become asymmetric, and a DC component would appear.

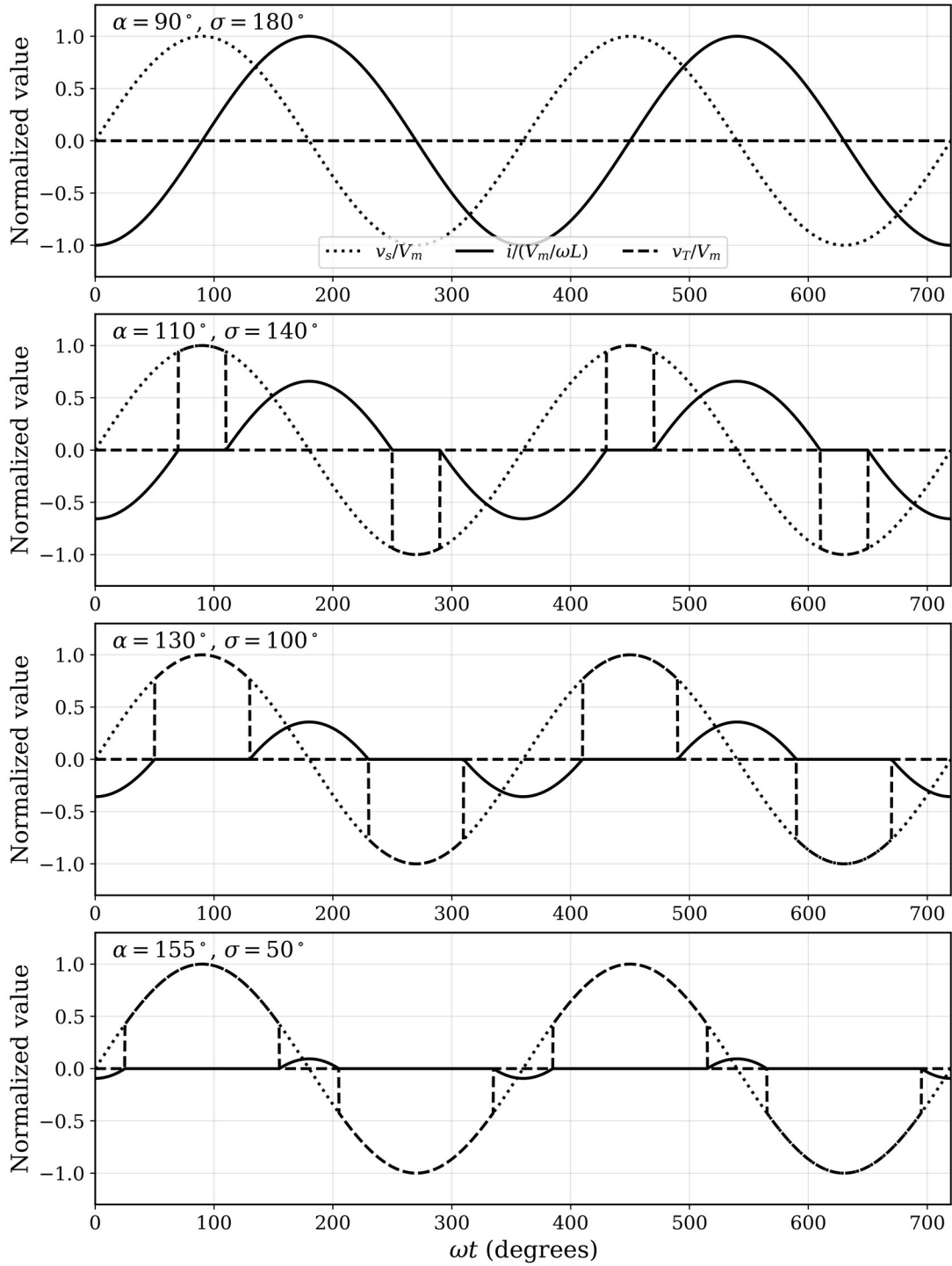


Figure 6.6. Ideal TCR source voltage, branch current and valve voltage at four firing angles. Current base is the full-conduction peak; voltage base is the supply peak.

The valve voltage v_T equals the source voltage whenever both thyristors are off and is zero while either conducts. At the instant of firing the valve therefore blocks $V_m \sin \alpha$, which is the full peak value when α approaches 90° ; the valve must be rated for the full peak branch voltage plus switching overvoltage margins.

6.2.2 Instantaneous current

Let the branch voltage be $v_s = V_m \sin \omega t$, where V_m is its peak value and $\omega = 2\pi f$ the angular frequency. While T_1 conducts, the circuit is a pure inductance:

$$L \frac{di}{dt} = V_m \sin \omega t \quad (6.8)$$

where L is the branch inductance and i the instantaneous branch current. Integrating, $i = -(V_m/\omega L) \cos \omega t + K$. In periodic steady state the pulse starts from zero at $\omega t = \alpha$, which fixes K :

$$i(t) = \frac{V_m}{\omega L} (\cos \alpha - \cos \omega t), \alpha \leq \omega t \leq 2\pi - \alpha \quad (6.9)$$

Setting Equation (6.9) to zero gives the second root $\omega t = 2\pi - \alpha$, the natural extinction angle. The conduction angle of each thyristor is therefore

$$\sigma = 2(\pi - \alpha) \quad (6.10)$$

where σ is the conduction angle in radians. The peak instantaneous current, at $\omega t = \pi$, is $(V_m/\omega L)(1 + \cos \alpha)$, which falls from $V_m/\omega L$ at full conduction to zero at blocking. The negative pulse from T_2 is the mirror image half a cycle later, so the waveform has half-wave odd symmetry, $i(\omega t + \pi) = -i(\omega t)$, and contains only odd harmonics.

6.2.3 Fundamental current and equivalent susceptance

Because the positive pulse is symmetric about $\omega t = \pi$ and the waveform has half-wave odd symmetry, the fundamental contains only a cosine term. Its coefficient is twice the integral over one pulse:

$$a_1 = \frac{2}{\pi} \int_{\alpha}^{2\pi-\alpha} \frac{V_m}{\omega L} (\cos \alpha - \cos \theta) \cos \theta d\theta \quad (6.11)$$

where $\theta = \omega t$ and a_1 is the peak cosine coefficient of the fundamental. The two integrals are $\int_{\alpha}^{2\pi-\alpha} \cos \theta d\theta = -2 \sin \alpha$ and $\int_{\alpha}^{2\pi-\alpha} \cos^2 \theta d\theta = (\pi - \alpha) - \frac{1}{2} \sin 2\alpha$. Substituting and collecting terms,

$$a_1 = -\frac{V_m}{\omega L} \cdot \frac{2(\pi - \alpha) + \sin 2\alpha}{\pi} \quad (6.12)$$

The negative cosine coefficient means the fundamental lags $v_s = V_m \sin \omega t$ by 90° : the TCR is purely inductive at fundamental frequency. Writing the peak fundamental as $I_{1m} = V_m B_{TCR}(\alpha)$ defines the equivalent susceptance

$$B_{TCR}(\alpha) = B_{max} \frac{2(\pi - \alpha) + \sin 2\alpha}{\pi} = B_{max} \frac{\sigma - \sin \sigma}{\pi}, B_{max} = \frac{1}{\omega L} \quad (6.13)$$

where B_{TCR} is the fundamental-frequency inductive susceptance magnitude and B_{max} its value at full conduction. The second form uses Equation (6.10) and $\sin 2\alpha = -\sin \sigma$. The RMS fundamental current and the reactive power absorbed by one branch of RMS voltage V are

$$I_1 = V B_{TCR}(\alpha), Q_{TCR} = V^2 B_{TCR}(\alpha) \quad (6.14)$$

where I_1 is the RMS fundamental current and Q_{TCR} the absorbed reactive power (a positive magnitude). For a delta-connected three-phase TCR, V is the line-to-line voltage and the three-phase absorption is $3V^2 B_{TCR}$; for a star connection V is the phase voltage.

Figure 6.7 plots B_{TCR}/B_{max} . The curve is steep near $\alpha = 90^\circ$ and flat near 180° . Differentiating Equation (6.13),

$$\frac{d}{d\alpha} \left(\frac{B_{TCR}}{B_{max}} \right) = -\frac{4 \sin^2 \alpha}{\pi} \quad (6.15)$$

The derivative is per radian; multiply by $\pi/180$ to obtain the gain per degree. Its magnitude changes by about 3.97 between 95° and 150° , and by about 14.8 between 95° and 165° . Practical regulators commonly command susceptance and use an inverse of Equation (6.13), implemented as a lookup table or numerical linearizer, instead of treating firing angle as a constant-gain actuator. This removes the main firing-law nonlinearity, but does not remove network and limit effects.

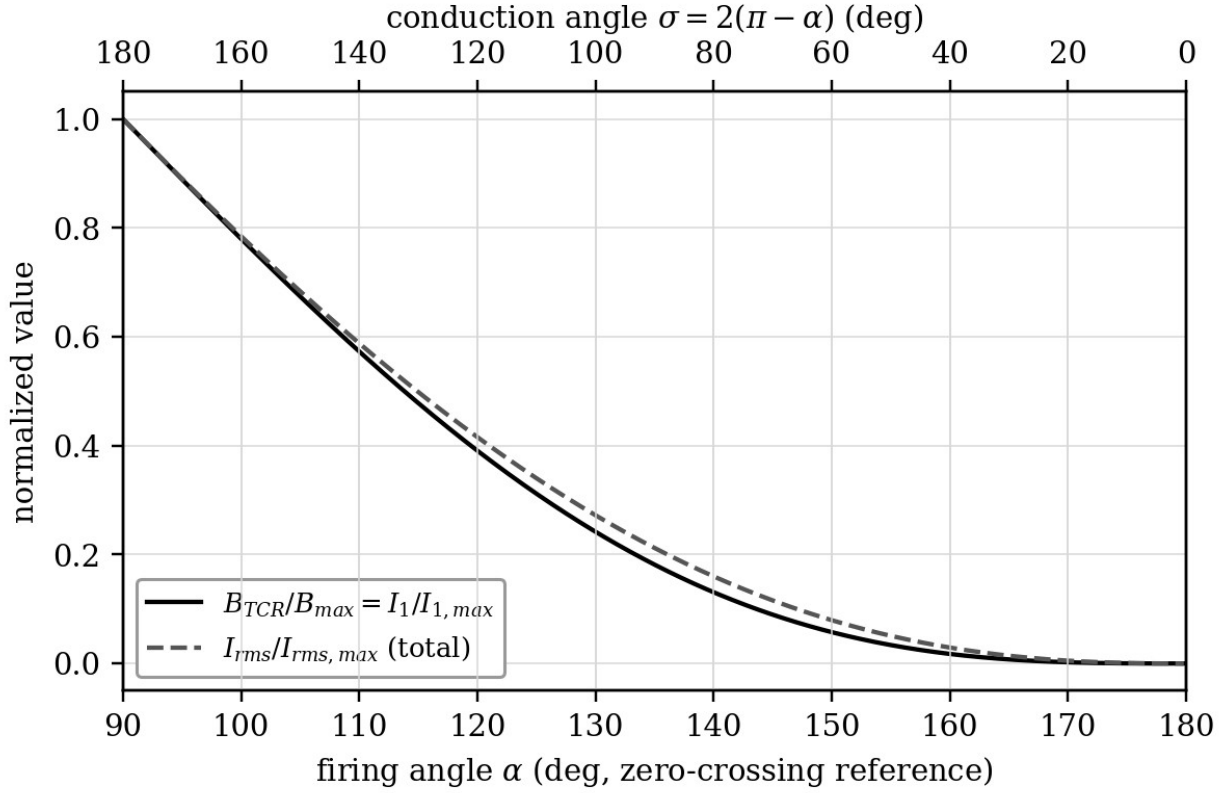


Figure 6.7. Normalized fundamental susceptance (equivalently fundamental current) and total RMS current of the TCR versus firing angle, with the conduction angle on the upper axis.

6.2.4 Total RMS current and total harmonic distortion

Thermal design of the reactor and the valve is governed by total RMS current, not by its fundamental. Squaring Equation (6.9), integrating over both pulses and dividing by 2π :

$$I_{rms}^2 = \frac{1}{\pi} \left(\frac{V_m}{\omega L} \right)^2 \int_{\alpha}^{2\pi-\alpha} (\cos \alpha - \cos \theta)^2 d\theta \quad (6.16)$$

To evaluate it, expand $(\cos \alpha - \cos \theta)^2 = \cos^2 \alpha - 2 \cos \alpha \cos \theta + (1 + \cos 2\theta)/2$; half-wave symmetry makes the two pulses contribute equally, which gives the factor $1/\pi$ and the limits α to $2\pi - \alpha$.

The integral evaluates to $2(\pi - \alpha) \cos^2 \alpha + 4 \sin \alpha \cos \alpha + (\pi - \alpha) - \frac{1}{2} \sin 2\alpha$, which simplifies to

$$I_{rms} = \frac{V_m}{\omega L} \sqrt{\frac{(\pi - \alpha)(1 + 2 \cos^2 \alpha) + \frac{3}{2} \sin 2\alpha}{\pi}} \quad (6.17)$$

where I_{rms} is the total RMS branch current. At $\alpha = 90^\circ$ this reduces to $V_m/(\sqrt{2} \omega L)$, as it must. In the peak-referenced convention the same expression reads $(V_m/\omega L) \sqrt{\left[(\pi - 2\beta) \left(\frac{1}{2} + \sin^2 \beta \right) - \frac{3}{2} \sin 2\beta \right] / \pi}$, which is the form found in several lecture notes. The current THD follows from

$$THD_I = \sqrt{\left(\frac{I_{rms}}{I_1}\right)^2 - 1} \quad (6.18)$$

where $I_1 = a_1/\sqrt{2}$ in magnitude. Figure 6.8 shows that branch-current THD rises *monotonically* with α : about 11% at 100° , 36.3% at 120° , 95% at 150° , and it grows without bound as the pulses narrow towards zero width. There is therefore no finite “maximum THD” of a TCR branch current. What does have a maximum is each individual harmonic, measured on a fixed base, which is the quantity that matters for filter design. The delta line current, from which triplen harmonics are absent, has a much lower THD over the useful range.

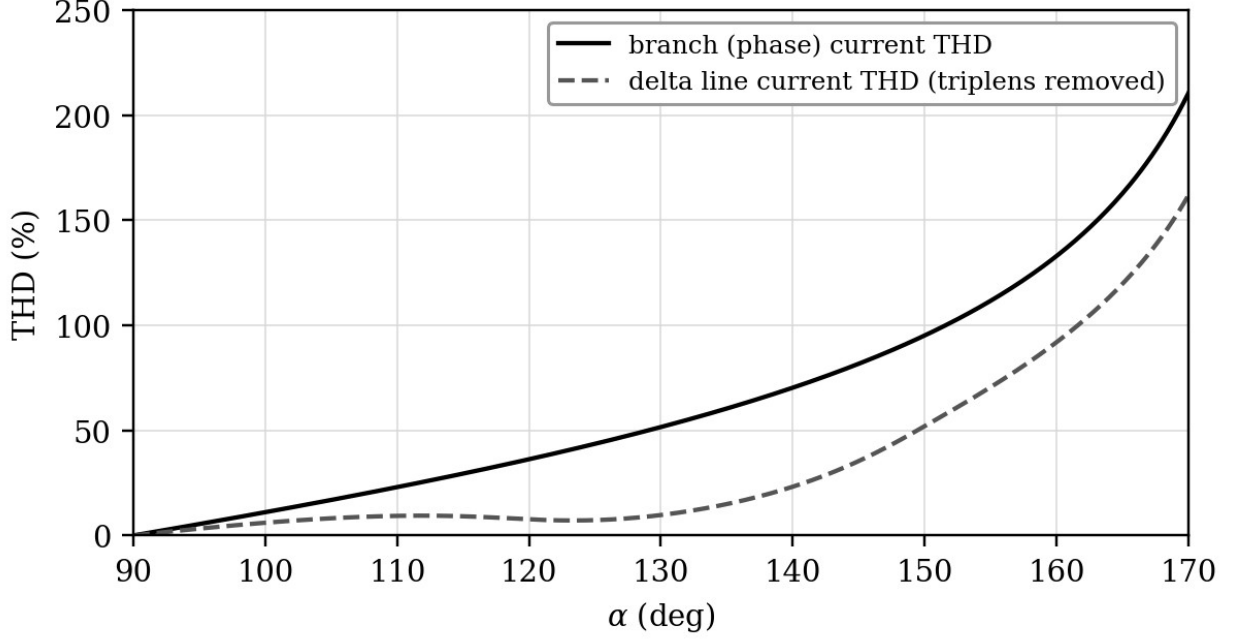


Figure 6.8. THD of TCR branch current and of delta line current versus firing angle.

6.2.5 Harmonic spectrum

For odd $n \geq 3$ the cosine coefficient is obtained exactly as for the fundamental, now with $\cos n\theta$ inside the integral.

Here a_n is $2V_m/(\pi\omega L)$ times the integral of $(\cos\alpha - \cos\theta)\cos n\theta$ from α to $2\pi - \alpha$. With the product-to-sum identity $\cos\theta\cos n\theta = [\cos(n-1)\theta + \cos(n+1)\theta]/2$, each term integrates directly over these limits, giving

$$a_n = \frac{4V_m}{\pi\omega L} \left[\frac{\sin(n+1)\alpha}{2(n+1)} + \frac{\sin(n-1)\alpha}{2(n-1)} - \frac{\cos\alpha\sin n\alpha}{n} \right] \quad (6.19)$$

where a_n is the signed peak cosine coefficient of the n th harmonic; a harmonic rating uses its magnitude. Combining the terms over the common denominator $n(n^2 - 1)$ and taking the magnitude gives

$$I_n = \frac{4V_m}{\pi\omega L} \left| \frac{\cos\alpha\sin n\alpha - n\sin\alpha\cos n\alpha}{n(n^2 - 1)} \right|, n=3, 5, 7, \dots \quad (6.20)$$

where $I_n = |a_n|$ is the peak harmonic magnitude. Divide by $\sqrt{2}$ for RMS current. In the peak-referenced convention, the magnitude is $\frac{4V_m}{\pi\omega L} \left| \frac{\sin\beta\cos(n\beta) - n\cos\beta\sin(n\beta)}{n(n^2 - 1)} \right|$ for odd $n \geq 3$; the sign of an individual Fourier coefficient also depends on the chosen time origin.

Where is each harmonic largest? Differentiating the numerator of Equation (6.20) with respect to α gives $(n^2 - 1) \sin \alpha \sin n\alpha$, so the extrema lie at $\sin n\alpha = 0$, that is $\alpha = k\pi/n$ within $90^\circ < \alpha < 180^\circ$. For the third harmonic this is $\alpha = 120^\circ$ ($\beta = 30^\circ$), where I_3 reaches 13.78% of the full-conduction fundamental; for the fifth it is 108° and 144° . Figure 6.9 plots the low-order harmonics, and Table 6.3 lists their maxima, which are the numbers used to rate filters.

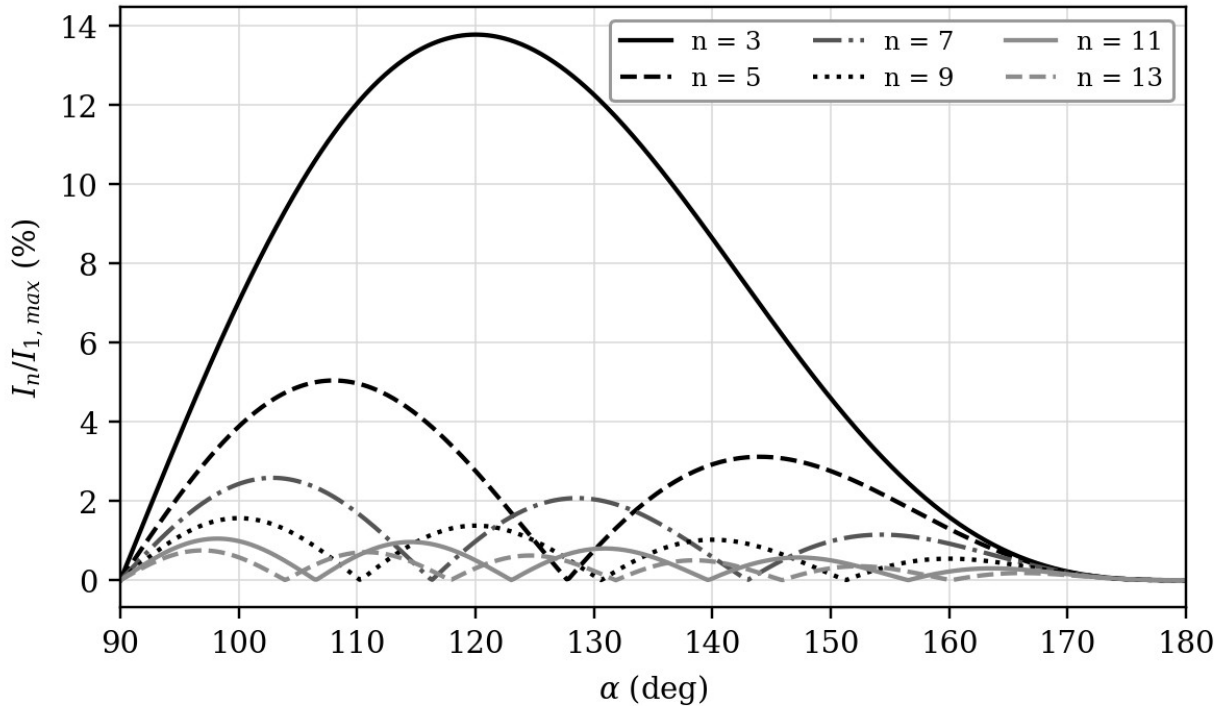


Figure 6.9. Odd harmonic branch currents of a TCR as a percentage of the full-conduction fundamental.

Table 6.3. Maximum amplitude of each TCR harmonic, as a percentage of the full-conduction fundamental, and the firing angle at which it occurs (balanced operation). The values apply to branch and line currents alike, except that triplens are absent from the line current of a delta connection.

n	3	5	7	9	11	13	15	17	19	21	23	25
$I_{n,max} (\%)$	13.78	5.05	2.59	1.57	1.05	0.75	0.57	0.44	0.35	0.29	0.24	0.20
α at max	120.0 °	108.0°	102.9 °	100.0 °	98.2°	96.9°	96.0°	95.3°	94.7°	94.3°	93.9°	93.6°

Example 6.1: Delta-connected TCR: rating, firing angle and harmonics.

A balanced, delta-connected, six-pulse 50 kvar TCR operates from a 415 V, 50 Hz supply. Find (a) the branch reactance and inductance; (b) the branch and line currents at full conduction; (c) the firing angle for 25 kvar; (d) the total RMS branch current and its THD at that angle; (e) the 3rd, 5th and 7th harmonic branch currents; (f) the line-current harmonics; (g) which harmonics reach the supply.

Solution. (a) Each branch sees the line voltage $V = 415$ V and absorbs one third of the rating, so $Q_{3\phi} = 3V^2/X_L$; $X_L = 3(415)^2/50\,000 = 10.3335 \, \Omega$ and $L = X_L/\omega = 32.893$ mH.

(b) $I_b = V/X_L = 415/10.3335 = 40.161$ A; $I_\ell = \sqrt{3}I_b = 69.560$ A.

(c) Half rating requires $B_{TCR}/B_{max} = 0.5$, so from Equation (6.13), $[2(\pi - \alpha) + \sin 2\alpha]/\pi = 0.5$. Solving numerically, $\alpha = 113.83^\circ$ ($\beta = 23.83^\circ$ in the peak-referenced convention).

(d) Use the unrounded solution $\alpha = 113.82677^\circ$. The peak base is $\sqrt{2}(415)/10.3335 = 56.796$ A. Equation (6.17) gives $I_{rms} \approx 20.847$ A. The fundamental is $I_1 = 0.5(40.161) = 20.080$ A, giving

THD $\approx 27.9\%$. Small changes in the last digit result if the printed angle is rounded before substitution.

- (e) From Equation (6.20) with $V_m/\omega L = 56.796$ A (peak), dividing by $\sqrt{2}$: $I_3 = 5.271$ A, $I_5 = 1.786$ A, $I_7 = 0.283$ A RMS (13.12%, 4.45% and 0.70% of the full-conduction fundamental of 40.161 A).
- (f) The 5th and 7th appear in the line multiplied by $\sqrt{3}$: $I_{5\ell} = 3.094$ A and $I_{7\ell} = 0.489$ A. The line fundamental is $\sqrt{3} \times 20.080 = 34.780$ A.
- (g) The third harmonic (5.27 A in each branch) is zero-sequence and circulates in the delta; only $n = 6k \pm 1$ reach the supply under balanced conditions.

Note. This example deliberately combines rating, firing-angle control and harmonic interpretation. The purpose is to connect the branch-level TCR equations with what the supply actually sees through the delta connection. Always distinguish branch harmonics from line-current harmonics.

6.2.6 The three-phase six-pulse TCR

Three single-phase TCRs connected in delta (Figure 6.10) form the standard six-pulse TCR. Each branch is fired with the same α referred to its own line-to-line voltage, so the three firing patterns are displaced by 120° .

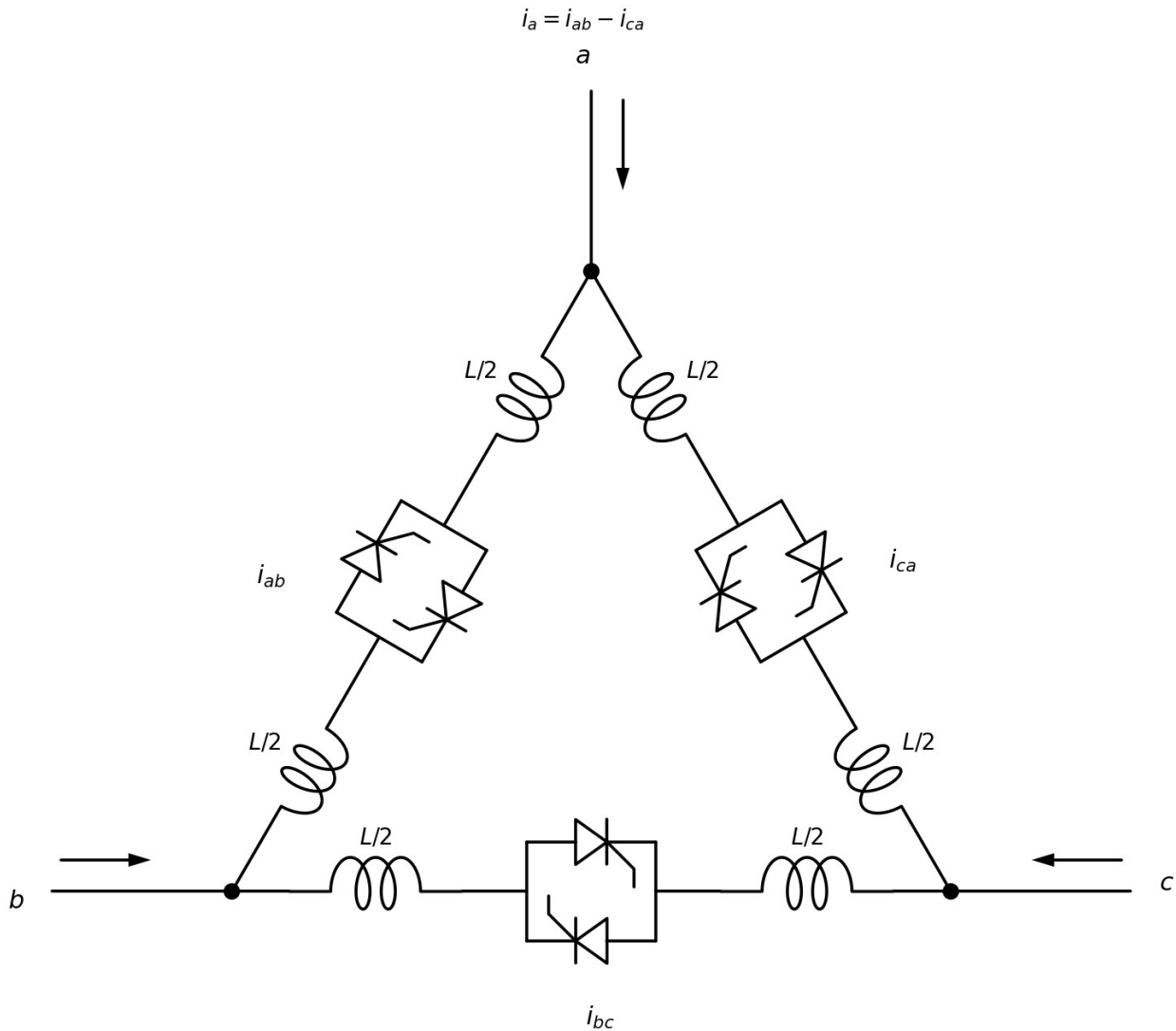


Figure 6.10. Delta-connected six-pulse TCR with split reactors.

Line currents follow from Kirchhoff's current law at each vertex, for example

$$i_a = i_{ab} - i_{ca} \quad (6.21)$$

where i_a is the line current and i_{ab} , i_{ca} the branch currents with arrows $a \rightarrow b$ and $c \rightarrow a$. Applying Equation (6.21) harmonic by harmonic is revealing. The n th harmonic of i_{ca} is displaced from that of i_{ab} by $n \times 120^\circ$. For triplen orders ($n=3, 9, 15, \dots$) this displacement is a multiple of 360° , so the triplens in the three branches are in phase: they form a zero-sequence set that circulates around the delta and cancels in every line current. For $n=6k-1$ (5, 11, 17, ...) the set has negative-sequence rotation; for $n=6k+1$ (7, 13, 19, ...) positive-sequence. Both appear in the line current multiplied by $\sqrt{3}$, exactly as the fundamental does. The line current therefore carries the characteristic harmonics $n=6k \pm 1$ with the same percentages as in Table 6.3, and Figure 6.11 shows its typical waveform.

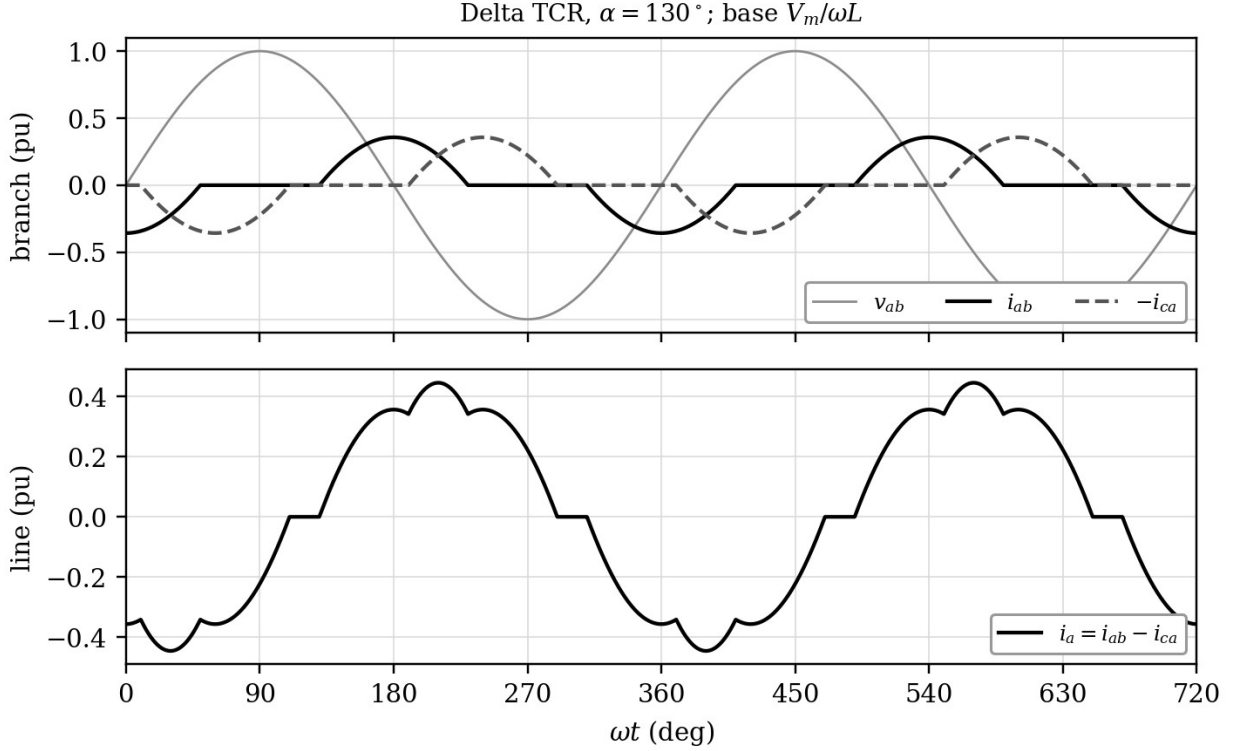


Figure 6.11. Branch and line currents of a balanced delta TCR at $\alpha = 130^\circ$ computed by KCL.

Ideal cancellation assumes identical reactors, balanced sinusoidal voltages and equal firing. In practice, small asymmetries let *non-characteristic* harmonics, triplens included, reach the network, and unequal firing of T_1 and T_2 produces a DC component that can drive the coupling transformer into saturation. Typical reactor quality factors lie between 40 and 100, so a small in-phase current representing losses is also present.

6.2.7 The twelve-pulse TCR

Two six-pulse TCRs of equal rating, supplied from two transformer secondaries whose voltages are displaced by 30° (for example a Y-Y and a Y- Δ winding, Figure 6.12), cancel the 5th, 7th, 17th, 19th, ... harmonics on the primary side. The reason is that a Y- Δ transformer shifts positive-sequence quantities by $+30^\circ$ and negative-sequence quantities by -30° . The group supplied through the Y- Δ winding sees voltages displaced by -30° , so its n th harmonic current is displaced by $-30n^\circ$ relative to the Y-Y group; transferring it back to the primary adds $+30^\circ$ for positive-sequence orders ($n=6k+1$) and -30° for negative-sequence orders ($n=6k-1$). The net displacement is $-30^\circ(n-1)$ or $-30^\circ(n+1)$ respectively, which equals 180° (cancellation) for $n=5, 7, 17, 19, \dots$ and 0° or 360° (addition) for $n=11, 13, 23, 25, \dots$. The primary current therefore contains only $n=12k \pm 1$.

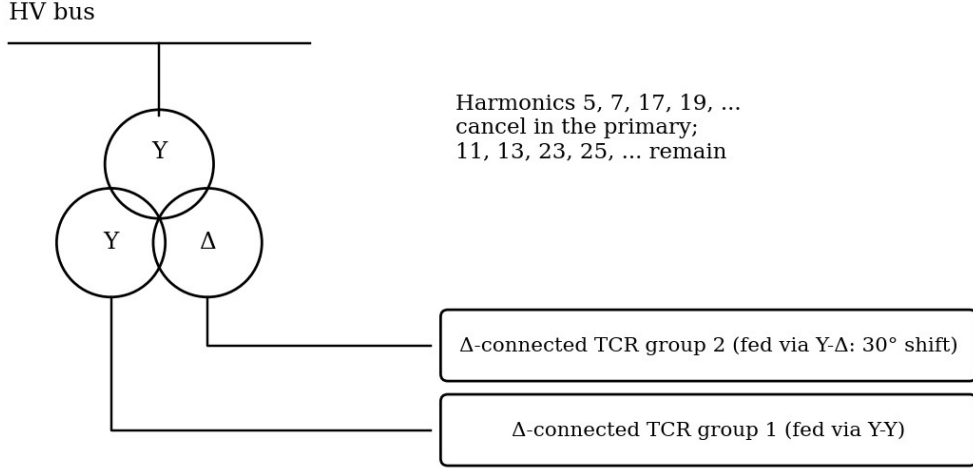
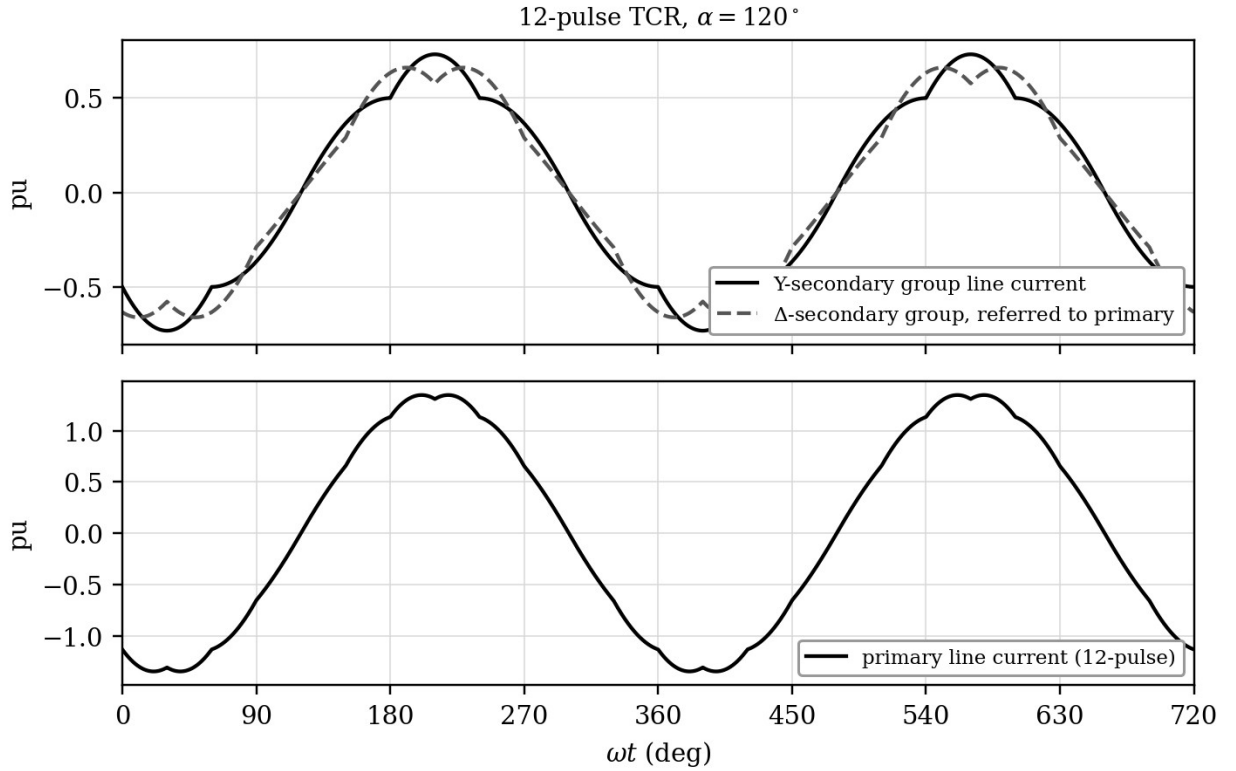


Figure 6.12. Twelve-pulse TCR arrangement with Y-Y and Y-Δ secondaries.

Figures 6.13 and 6.14 illustrate this cancellation. The waveforms use Equation (6.9) for both groups, with the delta-winding current referred through $i_A = (i_{a2} - i_{b2})/\sqrt{3}$ at unity line-voltage ratio. Under ideal balanced conditions the residual characteristic orders are $12k \pm 1$, starting with the 11th and 13th. Selecting tuned, damped or high-pass filters requires the harmonic spectrum, background distortion, network impedance and contingency states; twelve-pulse cancellation alone does not prescribe a high-pass filter. The trade-off is a more complex transformer and two valve groups [19].


 Figure 6.13. Line currents of the two six-pulse groups referred to the primary and their twelve-pulse sum at $\alpha = 120^\circ$.

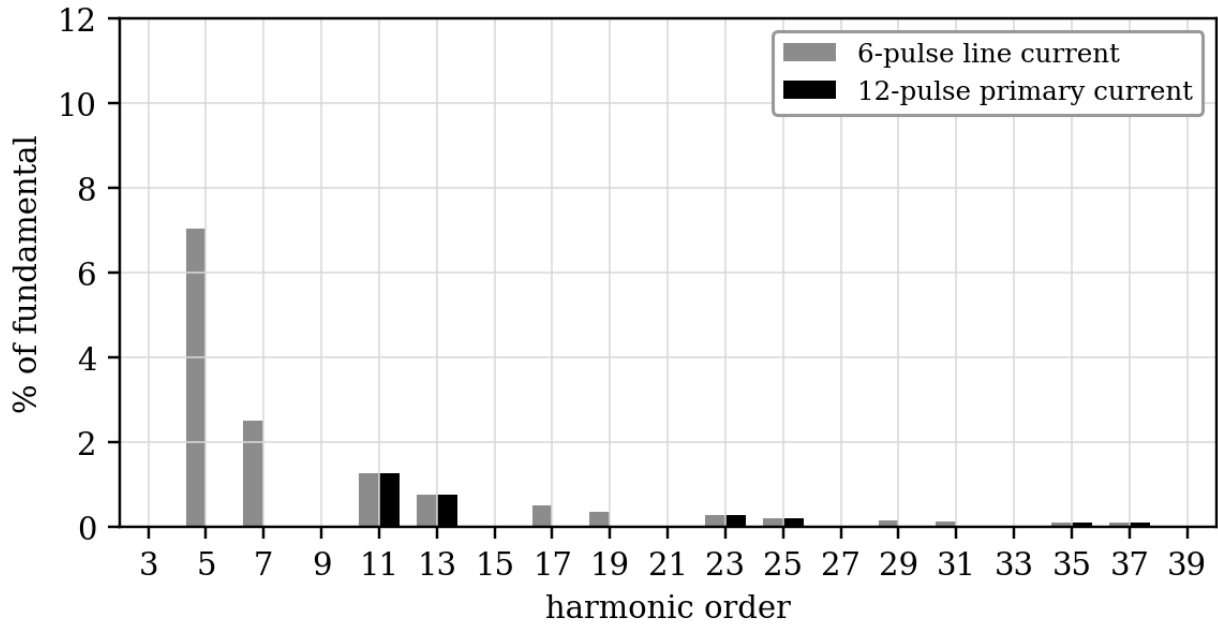


Figure 6.14. Harmonic content of six-pulse line current and twelve-pulse primary current at $\alpha = 120^\circ$.

Example 6.2: Twelve-pulse TCR.

A twelve-pulse TCR is fed from a 415 V primary through a three-winding transformer with Y-Y and Y- Δ secondaries, each 415 V line-to-line. Each secondary supplies a delta TCR of 15 mH per branch, and $\alpha = 120^\circ$. Find (a) the total rating; (b) the branch fundamental and RMS currents; (c) the secondary line fundamental; (d) the primary fundamental; (e) the 5th and 7th harmonics in each secondary line; (f) the 5th and 7th in the primary; (g) the 11th and 13th in the primary.

Solution. (a) $X_L = 4.7124 \Omega$; each group $3 V^2 / X_L = 109.64$ kvar; total 219.28 kvar.

- (b) Base 124.54 A. $I_{1b} = 0.391 \times 124.54 / \sqrt{2} = 34.43$ A; $I_{rms,b} = 36.63$ A.
- (c) $\sqrt{3} \times 34.43 = 59.64$ A in each secondary.
- (d) With 1:1 line-voltage ratios, the two fundamentals add in phase on the primary: $2 \times 59.64 = 119.28$ A. The absorbed power is 85.74 kvar.
- (e) $I_{5\ell} = 4.205$ A and $I_{7\ell} = 1.502$ A in each secondary.
- (f) The Y- Δ group's 5th and 7th are shifted by 180° relative to the Y-Y group's when referred to the primary (Section 6.2.7), so both cancel: zero.
- (g) The 11th and 13th add: $2 \times 0.7645 = 1.529$ A and $2 \times 0.4621 = 0.924$ A.

Note. Twelve-pulse cancellation is a connection-dependent harmonic result, not a statement that the individual six-pulse groups are harmonic-free. Transformer phase shift and balanced operation are essential to the cancellation mechanism.

6.2.8 The segmented (sequentially controlled) TCR

A TCR of total rating Q can be split into m parallel branches of rating Q/m . Only one branch is phase controlled at a time; the others are either fully conducting (acting as TSRs at $\alpha = 90^\circ$) or blocked. Because the controlled branch carries at most $1/m$ of the rated current, every harmonic amplitude, expressed on the total-rating base, is reduced by the factor m . Figure 6.15 shows a four-section TCR delivering 60% of its rating with two sections fully on and the third at $\alpha = 119.5^\circ$. The price is more valves and reactors, which is why segmentation is reserved for installations with tight harmonic limits. Example 6.20 quantifies the benefit.

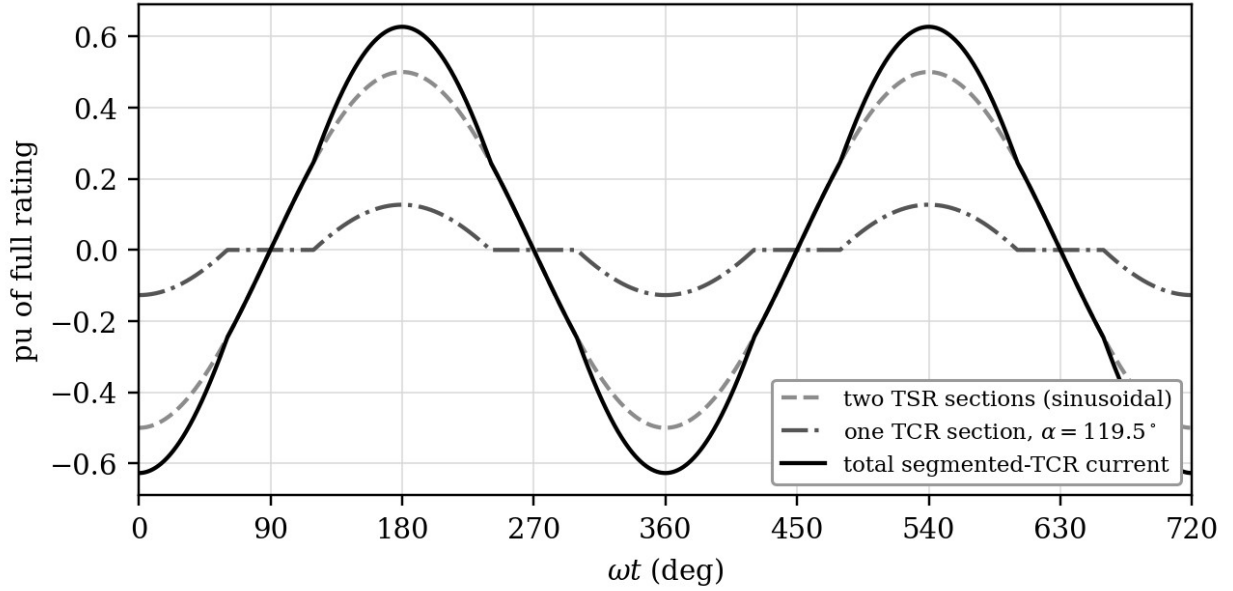


Figure 6.15. Segmented TCR with $m=4$ supplying 60% of rating: two TSR sections plus one phase-controlled section.

6.2.9 Practical design considerations

Valve construction. Each valve consists of parallel strings of series-connected thyristors. The number in series follows from the peak repetitive voltage, including the switching overvoltage factor and a redundancy margin of one or two devices; the number of parallel strings follows from the RMS and surge currents. Each thyristor in a TCR conducts a half-cycle pulse, so at full conduction its RMS current is $I_{pk}/2$ and its average current I_{pk}/π , where I_{pk} is the peak branch current.

Reactor and losses. Air-cored reactors are linear and cannot saturate, which keeps the susceptance law exact, but they offer little overload capability; a TCR must be protected against sustained overvoltage by limiting the conduction angle. Losses are dominated by $I_{rms}^2 R$ in the reactor and the on-state loss of the valve.

A firing-angle command changes the current only at an eligible firing instant; a conducting thyristor cannot be turned off by removing its gate pulse. In balanced TCR operation, eligible instants occur every half-cycle, giving a worst-case command wait of approximately half a cycle and mean wait of a quarter cycle. Natural extinction and the actual controller timing must be included when evaluating the transient response.

Filtering. Six-pulse TCRs are normally accompanied by single-tuned filters at the 5th and 7th harmonics and often a high-pass branch for the 11th and 13th. These filters are capacitive at fundamental frequency, so they double as the fixed capacitive element of the SVC (Section 6.5).

6.2.10 V-I characteristic and operating area

At fundamental frequency a TCR is a susceptance adjustable from zero to B_{max} , so its V-I characteristic is a fan of straight lines through the origin. The operating area (Figure 6.16) is bounded by the $\alpha=90^\circ$ line, by the maximum voltage the valve and reactor can withstand, and by the maximum current. A TSR, by contrast, offers only the discrete lines corresponding to the number of stages in conduction.

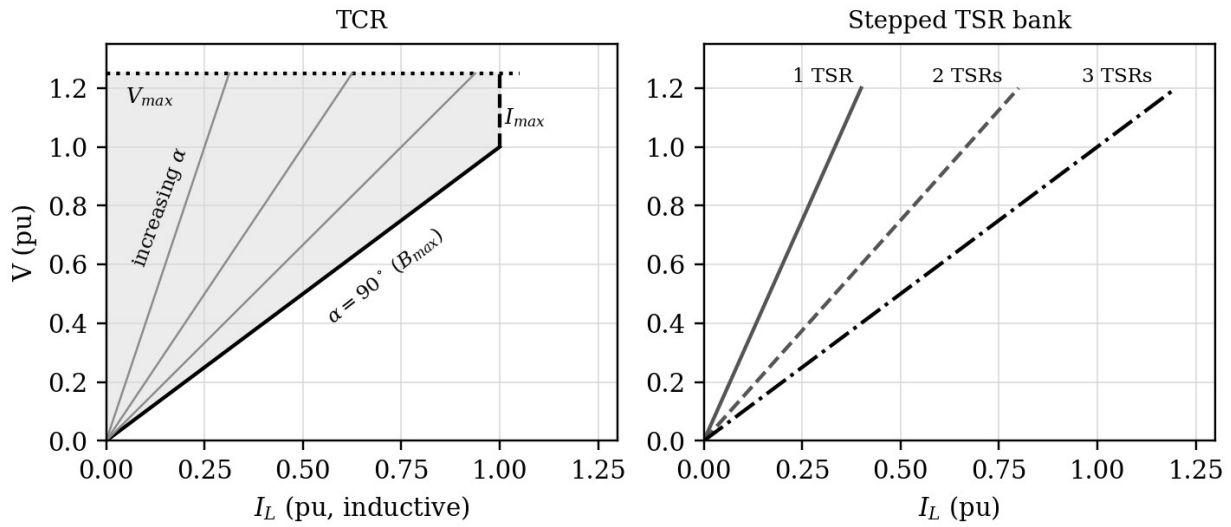


Figure 6.16. Operating area of a TCR (left) and the stepped characteristic of a TSR bank (right).

6.3 The Thyristor-Switched Reactor (TSR)

6.3.1 Principle

If the firing angle of a TCR is fixed at 90° and the valve is either gated continuously or not at all, the branch becomes a thyristor-switched reactor (Figure 6.17a). It offers a fixed inductive admittance whose reactive current is proportional to the applied voltage. Several TSRs in parallel give a susceptance adjustable in steps. Compared with a TCR of the same total rating, the TSR generates no phase-control harmonics, has lower losses and needs no linearizer, but it cannot provide continuous control. It is chosen where the inductive duty is coarse, for example to absorb line charging on light load or to suppress temporary overvoltage after load rejection, often in combination with a smaller TCR for the fine adjustment.

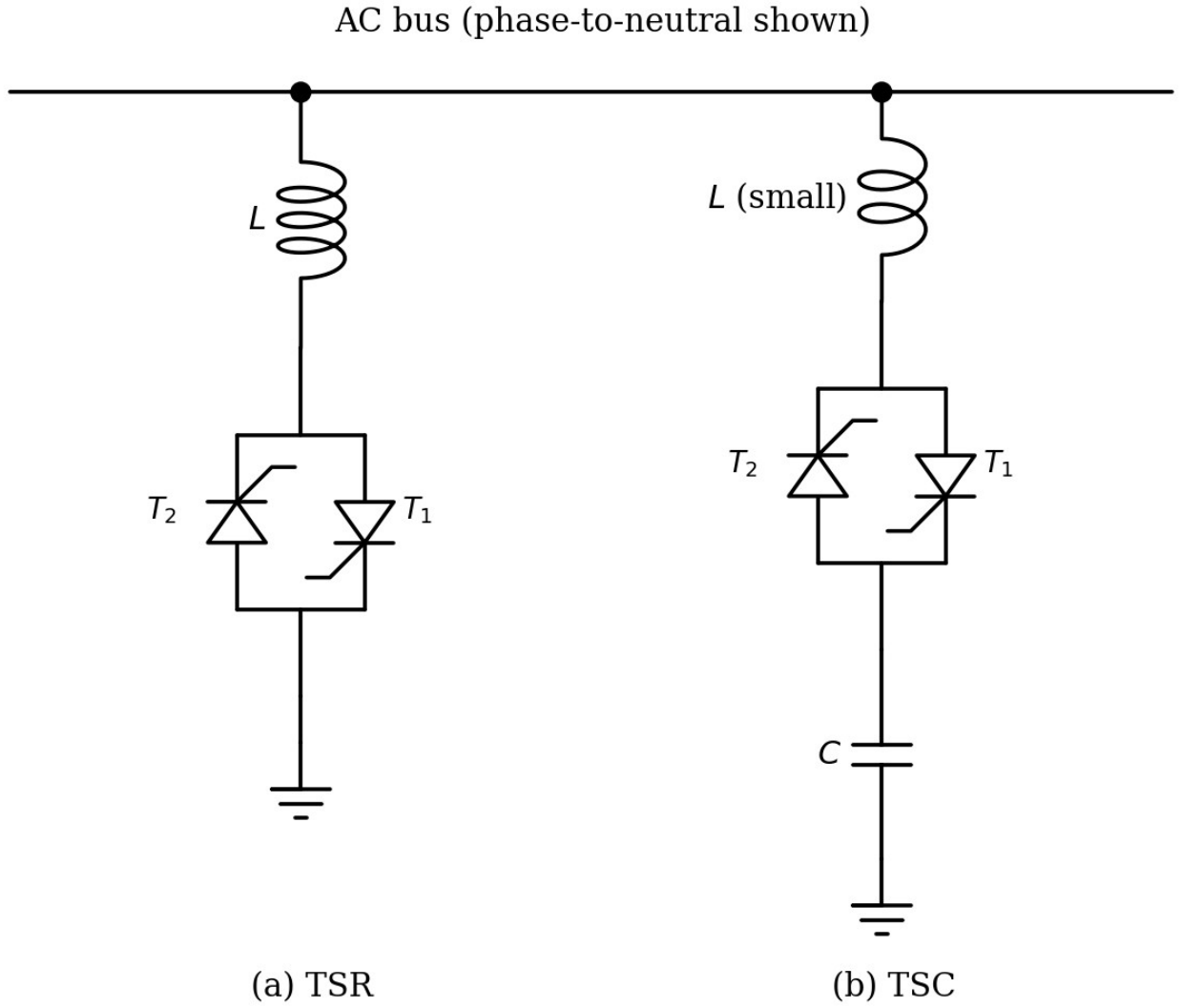


Figure 6.17. Single-phase (a) TSR and (b) TSC branches.

6.3.2 Transient-free connection

Suppose the valve of an initially de-energized reactor is closed at an arbitrary instant $\theta_c = \omega t_c$. Integrating Equation (6.8) from $i(\theta_c) = 0$ while the valve remains gated gives

$$i(\theta) = \frac{V_m}{\omega L} (\cos \theta_c - \cos \theta), \theta \geq \theta_c \quad (6.22)$$

where θ_c is the closing angle measured from the positive-going voltage zero crossing. The second term is the steady sinusoidal current; the first is a DC offset of magnitude $(V_m/\omega L) \cos \theta_c$ that, in a lossless circuit, persists indefinitely and in practice decays with the reactor time constant L/R . The offset vanishes only for $\cos \theta_c = 0$, that is, closure at a voltage peak. Closure at a voltage zero gives a fully offset current whose peak is twice the steady value, as Figure 6.18 shows. The TSR is therefore switched in at $\theta_c = 90^\circ$ (or 270° for the opposite valve) and switched out by removing gate pulses, after which the conducting thyristor commutates naturally at the next current zero.

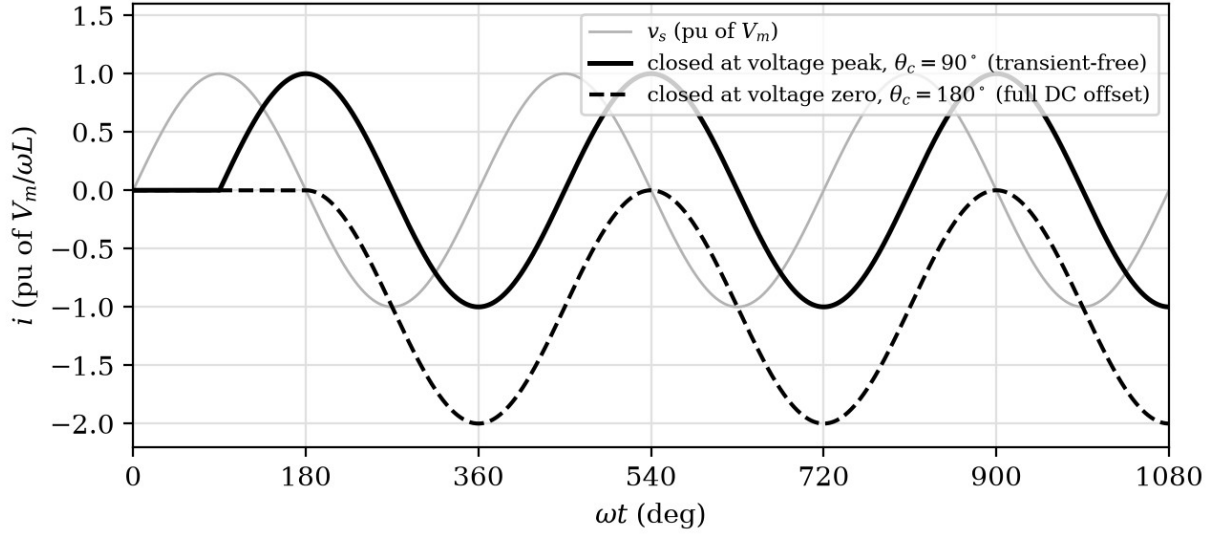


Figure 6.18. TSR current when closed at a voltage peak (transient-free) and at a voltage zero (fully offset).

Example 6.3: Thyristor-switched reactor.

A single-phase 25 kvar TSR is switched at 230 V, 50 Hz. Find (a) reactance and inductance; (b) the transient-free closing instant; (c) the peak and RMS branch currents; (d) the RMS and average current per thyristor; (e) the peak current if closed at a voltage zero; (f) the peak blocking voltage.

Solution. (a) $X = 230^2 / 25\,000 = 2.116\ \Omega$; $L = 6.735\text{ mH}$.

(b) At a voltage peak, $\theta_c = 90^\circ$ (Equation (6.22) with $\cos \theta_c = 0$).

(c) $I_{pk} = 325.27 / 2.116 = 153.72\text{ A}$; $I = 108.70\text{ A}$.

(d) $I_{T,rms} = 76.86\text{ A}$; $I_{T,av} = 48.93\text{ A}$.

(e) At $\theta_c = 0^\circ$ (or 180°) the offset equals the AC peak, so $i_{pk} = 2 \times 153.72 = 307.4\text{ A}$, decaying with L/R .

(f) 325.3 V plus margin.

Note. A TSR is a switched reactor, not a continuously controlled reactor. The numerical result should therefore be interpreted as one discrete susceptance state. Closing near the appropriate voltage crest avoids the large DC offset that would result from an arbitrary switching instant.

6.3.3 Stepped banks

With N branches of binary-weighted susceptance $B, 2B, 4B, \dots, 2^{N-1}B$, the bank provides 2^N states including zero, in uniform steps of B . Three branches of 50, 100 and 200 kvar, for instance, give eight levels from 0 to 350 kvar (Example 6.23). Each thyristor of a TSR carries half-cycle pulses of a full sinusoid, so its RMS current is $I_{pk}/2$, its average current I_{pk}/π , and it must block the full peak branch voltage when off.

6.4 The Thyristor-Switched Capacitor (TSC)

6.4.1 Why capacitor switching is the difficult case

A TSC switches a capacitor bank in or out with a bidirectional valve (Figure 6.17b). Unlike a reactor, the capacitor cannot be phase controlled: partial conduction would produce large current pulses and unacceptable harmonics. The capacitor is therefore switched fully on or fully off, and the entire difficulty lies in choosing the switching instant.

For an ideal capacitor connected directly to an ideal voltage source, a mismatch between residual capacitor voltage and supply voltage requires an impulsive charging current. If the voltages match but $dv_s/dt \neq 0$,

current jumps from zero to $C dv_s/dt$, implying an unbounded idealized rate of rise. At a matched voltage crest, however, $dv_s/dt=0$ and this particular current step is absent; it is an important special case. Real TSCs nevertheless use a series reactor to limit current and di/dt under voltage mismatch, background harmonics and mistimed firing. The reactor also changes resonance and the exact transient-free closing state, so the complete LC problem must be analyzed.

6.4.2 Complete current of the TSC branch

With the valve conducting and resistance neglected, the branch obeys

$$L \frac{di}{dt} + v_C = V_m \sin \omega t, i = C \frac{dv_C}{dt} \quad (6.23)$$

where C is the capacitance, v_C the capacitor voltage and L the series reactor. Two parameters describe the branch. The *tuning ratio* is

$$n = \frac{\omega_n}{\omega} = \frac{1}{\omega \sqrt{LC}} = \sqrt{\frac{X_C}{X_L}} \quad (6.24)$$

where ω_n is the natural resonant frequency of the branch, $X_C = 1/(\omega C)$ and $X_L = \omega L$. The *magnification factor* is

$$k = \frac{X_C}{X_C - X_L} = \frac{n^2}{n^2 - 1} \quad (6.25)$$

The steady-state solution is found by treating the branch as a net capacitive reactance $X_C - X_L$:

$$i_{ss} = I_{AC} \cos \omega t, I_{AC} = \frac{V_m}{X_C - X_L} = k \omega C V_m, v_{C,ss} = k V_m \sin \omega t \quad (6.26)$$

where I_{AC} is the peak steady-state current. Now close the valve at $\theta_c = \omega t_c$ with the capacitor carrying a residual voltage V_{C0} and the reactor current zero. The natural response must make up the mismatch between the initial states and the forced states. Writing $\tau = t - t_c$ and solving Equation (6.23) with these initial conditions gives

$$i(t) = I_{AC} \left[\cos \omega t - \cos \theta_c \cos \omega_n \tau + n \left(\sin \theta_c - \frac{V_{C0}}{k V_m} \right) \sin \omega_n \tau \right] \quad (6.27)$$

The first term is the forced current. The second and third oscillate at ω_n ; they are the switching transient, and without damping they persist. Resistance makes them decay over a few cycles, but their initial amplitude still sets the valve current stress. The amplitude of the oscillatory part, relative to I_{AC} , is

$$\frac{I_{osc}}{I_{AC}} = \sqrt{\cos^2 \theta_c + n^2 \left(\sin \theta_c - \frac{V_{C0}}{k V_m} \right)^2} \quad (6.28)$$

and the worst instantaneous current is bounded by $I_{AC} + I_{osc}$.

6.4.3 Consequences of the series reactor: magnification and the choice of n

Equation (6.26) shows that the reactor raises both the current and the capacitor voltage by the factor k compared with the capacitor alone. Figure 6.19 plots k . For $n \geq 3$ the magnification is modest, 1.125 at $n=3$ and 1.052 at $n=4.5$, but it rises steeply for small n . The capacitor must be voltage rated for $k V_m$, and the bank rating must be computed with the net reactance $X_C - X_L$.

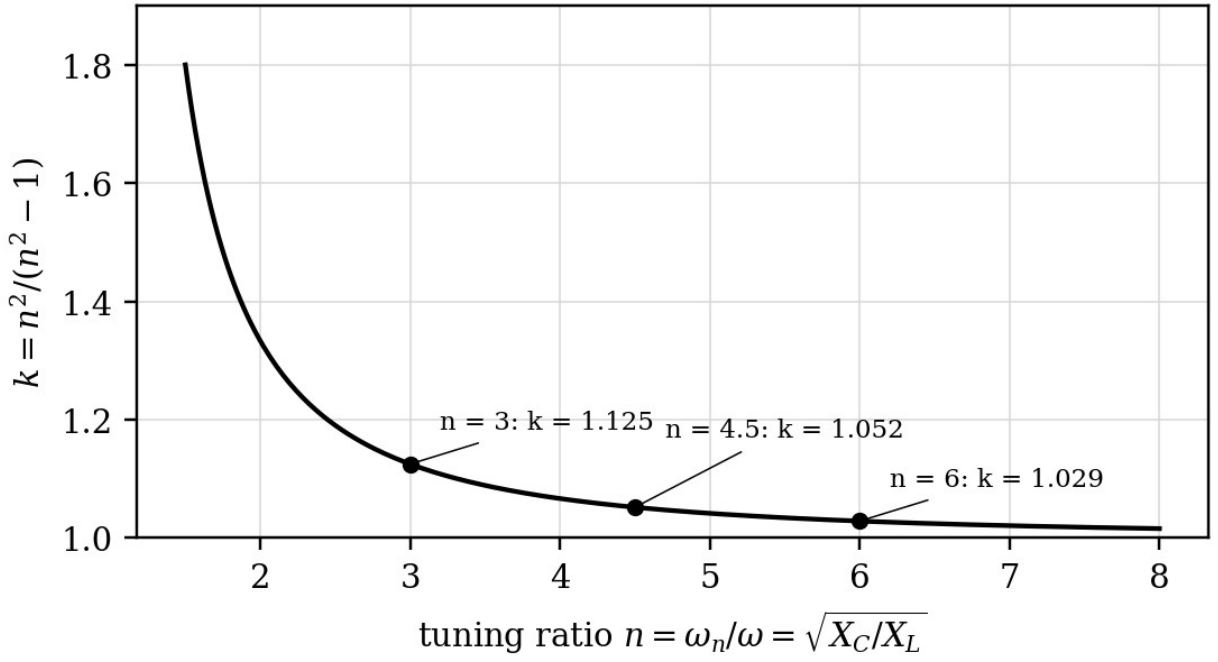


Figure 6.19. Magnification factor $k = n^2 / (n^2 - 1)$ versus tuning ratio.

The series branch is resonant at harmonic order n . At order h its reactance is $X_h = h X_L - X_C / h = X_L (h - n^2 / h)$, capacitive for $h < n$ and inductive for $h > n$. A branch detuned to $n = 4.5$ is therefore capacitive at the fundamental and inductive at the fifth harmonic. Tuning below the fifth, typically between 4.3 and 4.8, has to be coordinated with network resonances, background harmonics, component tolerances and switching duty. Increasing L lowers n and reduces the initial rate of rise for a given voltage mismatch, at the cost of a larger fundamental magnification.

Example 6.4: TSC branch design with detuning reactor.

A single-phase TSC must deliver 10 kvar at 230 V, 50 Hz, with tuning ratio $n = 4.5$. Find (a) k ; (b) X_C and C ; (c) X_L , L and resonant frequency; (d) branch current and capacitor voltage; (e) capacitor var duty; (f) steady-state peak valve voltage after turn-off and a blocking rating with 50% margin.

Solution. (a) $k = 20.25 / 19.25 = 1.05195$.

(b) The net reactance is $V^2 / Q = 5.29 \Omega = X_C - X_L$. With $X_L = X_C / n^2$, $X_C = k \times 5.29 = 5.565 \Omega$ and $C = 572.0 \mu\text{F}$.

(c) $X_L = 0.2748 \Omega$; $L = 0.875 \text{ mH}$; $f_n = 225 \text{ Hz}$.

(d) $I = 43.48 \text{ A RMS}$ (61.49 A peak); $V_C = I X_C = 241.9 \text{ V RMS}$, peak 342.2 V = $k V_m$.

(e) $I^2 X_C = 10.52 \text{ kvar}$: the capacitor must be rated about 5% above the branch output.

(f) $(1 + k) V_m = 667.4 \text{ V}$; with 50% margin, about 1.0 kV.

Note. The series reactor is part of the TSC design, not merely a protective accessory. It changes the branch resonant frequency, capacitor voltage, current magnification and the transient produced by switching.

6.4.4 Conditions for transient-free switching

From Equation (6.27), both oscillatory terms vanish only if, simultaneously,

$$\cos \theta_c = 0 \text{ and } V_{C0} = \pm k V_m \quad (6.29)$$

The first condition requires closure at a positive or negative peak of the supply voltage, where the forced current is zero. The second requires the capacitor to be pre-charged to the magnified steady-state voltage with

the matching polarity. Figure 6.21c shows such a closure: the branch current starts smoothly with no oscillation.

These conditions are naturally satisfied when a bank that was previously in service is reconnected before it discharges: the valve turns off at a current zero, which coincides with the capacitor voltage at $\pm k V_m$, and the next same-polarity peak then provides a transient-free closing instant. Elsewhere they are not met. Distorted bus voltages shift the peaks, residual charge leaks away through the discharge resistors, and maintaining an exact pre-charge would require extra charging equipment and impose a prolonged DC stress that AC capacitors are not designed for. Practical controllers therefore use simpler rules that keep the transient small rather than zero.

6.4.5 Practical firing schemes and switching strategies

Firing scheme 1 (valve-voltage-zero firing). Fire when the instantaneous supply voltage equals the residual capacitor voltage, $V_m \sin \theta_c = V_{C0}$, so that the valve voltage is zero at turn-on and the initial di/dt is zero. Writing $\rho = V_{C0}/V_m$ and substituting $\sin \theta_c = \rho$ into Equation (6.28) gives

$$\frac{I_{osc}}{I_{AC}} = \sqrt{1 - \rho^2 \left(1 - \frac{1}{n^2}\right)}, \theta_c = \sin^{-1} \rho, |\rho| \leq 1 \quad (6.30)$$

Here $\rho = V_{C0}/V_m$ is signed and voltage matching is possible only for $|\rho| \leq 1$. The oscillatory amplitude does not exceed I_{AC} , and is I_{AC} for a discharged capacitor. This rule gives zero initial di/dt and minimizes the oscillatory amplitude in the stated lossless LC model. To verify this, set $x = \sin \theta_c$ and write $A^2 = 1 - x^2 + n^2(x - \rho/k)^2$. Since $n^2/k = n^2 - 1$, differentiation gives $x_{opt} = \rho$. The minimum amplitude is therefore $A_{min} = \sqrt{1 - \rho^2/k}$, exactly the scheme-1 expression: voltage matching does minimize the oscillatory amplitude for $|\rho| \leq 1$. The complete state mismatch still prevents zero transient except at the special crest condition in Equation (6.29).

Firing scheme 2 (crest firing). Fire at the crest of the supply voltage of the same polarity as V_{C0} , where the valve voltage is at its minimum $V_m - V_{C0}$. With $\theta_c = 90^\circ$,

$$\frac{I_{osc}}{I_{AC}} = n \left| \frac{V_{C0}}{k V_m} - 1 \right| = n \left| \left(1 - \frac{1}{n^2}\right) \rho - 1 \right| \quad (6.31)$$

This scheme works for an overcharged capacitor and becomes transient-free as $\rho \rightarrow k$, but its performance deteriorates rapidly as ρ falls: for a discharged capacitor the oscillatory amplitude is n times the steady current. Figure 6.20 compares the two schemes, and Figure 6.21 shows the computed closing transients for four representative cases with $n = 4.5$.

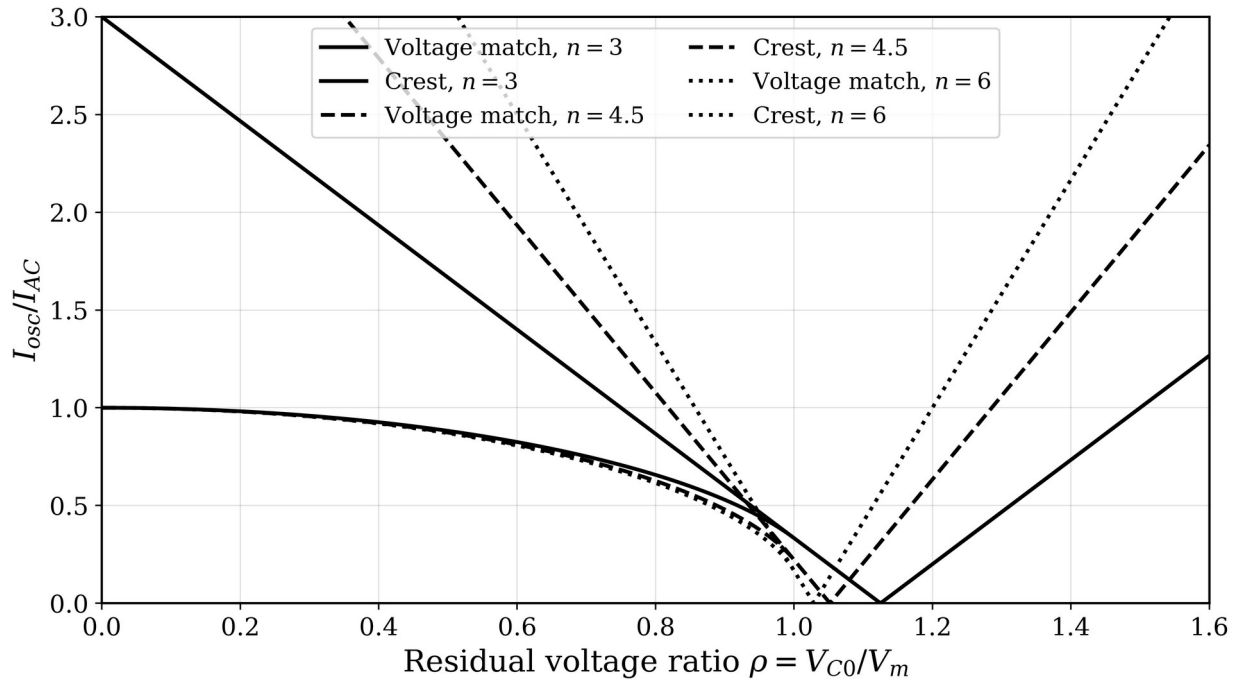


Figure 6.20. Oscillatory current amplitude for voltage-matching and positive-crest firing at three tuning ratios. The curves starting at 1.0 are voltage matching, which exists only for $\rho \leq 1$; the V-shaped lines are crest firing, with zero transient at $\rho = k$.

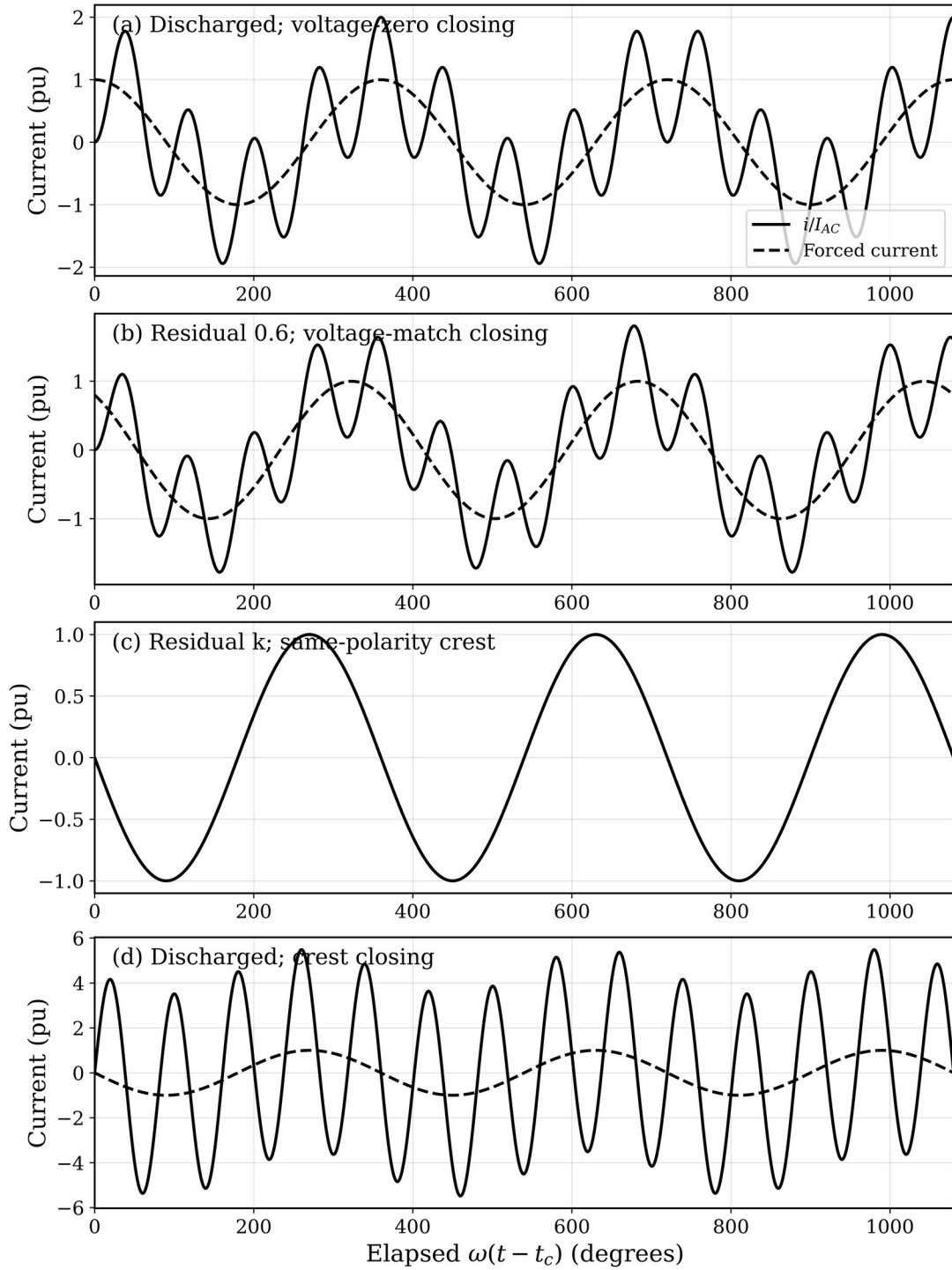


Figure 6.21. Lossless analytical TSC closing currents from Equation (6.27), $n = 4.5$: (a) discharged, voltage-zero firing; (b) residual $0.6V_m$, voltage match; (c) residual kV_m , crest firing; (d) discharged, crest firing. Time starts at closure. Without resistance the natural oscillation does not decay.

One practical rule uses voltage matching for $\angle V_{C0} \angle V_m$ and the same-polarity crest for $\angle V_{C0} \angle V_m$; in the lossless model these choices minimize the oscillatory amplitude over the feasible closing angles. Another arrangement keeps every disconnected capacitor charged near the supply crest and always fires at the crest. At $\angle V_{C0} \angle V_m$ the transient is small but not zero, because the exact steady capacitor crest is kV_m . Which strategy suits a given installation depends on the charging method, leakage, valve polarity, harmonics and capacitor insulation.

Example 6.5: TSC with residual charge: scheme 1.

A 6 kvar, 230 V TSC with $n=5$ has $V_{C0}=150$ V. Find (a) the applicable scheme; (b) the closing angle and delay after the zero crossing; (c) I_{osc}/I_{AC} ; (d) a bound on peak current; (e) I_{osc}/I_{AC} for crest firing; (f) for a discharged capacitor at a voltage zero; (g) the residual voltage for zero transient.

Solution. (a) $\rho = 150/325.27 = 0.4612 < 1$: scheme 1.

(b) $\theta_c = \sin^{-1} 0.4612 = 27.46^\circ$, i.e. 1.53 ms after the positive-going zero.

(c) Equation (6.30): $\sqrt{1 - 0.2127 \times 0.96} = 0.892$.

(d) $I_{AC} = \sqrt{2} \times 26.09 = 36.89$ A (the 6 kvar rating already includes the reactor); bound $36.89 \times 1.892 = 69.8$ A.

(e) Equation (6.31): $5|0.96 \times 0.4612 - 1| = 2.79$, much worse.

(f) 1.0.

(g) $k V_m = 1.04167 \times 325.27 = 338.8$ V, closed at the positive crest.

Note. Matching the instantaneous source voltage to the trapped capacitor voltage reduces the voltage discontinuity, but the complete LC state must also be considered. A minimum-transient switching rule and an exactly transient-free state are not automatically the same condition.

6.4.6 Turn-off, trapped charge and valve voltage stress

In sinusoidal steady state, removing gate pulses turns the valve off at the next current zero, when capacitor voltage is $\pm k V_m$. The capacitor retains this charge initially, neglecting leakage. If turn-off occurs before the switching oscillation has decayed, the trapped voltage can differ from $\pm k V_m$ and must instead be found from the transient state. For the steady-state case, the valve blocks

$$v_T(t) = V_m \sin \omega t - k V_m \quad (6.32)$$

where v_T is the valve voltage for a capacitor left at $+k V_m$. Half a cycle later the supply reaches $-V_m$ and the valve voltage peaks at $-(1+k)V_m$, slightly more than twice the supply peak. Figure 6.22 shows this, together with the slower decline when a discharge resistor is fitted. The initial small step in v_T at turn-off is the voltage across the series reactor, which collapses when the current stops.

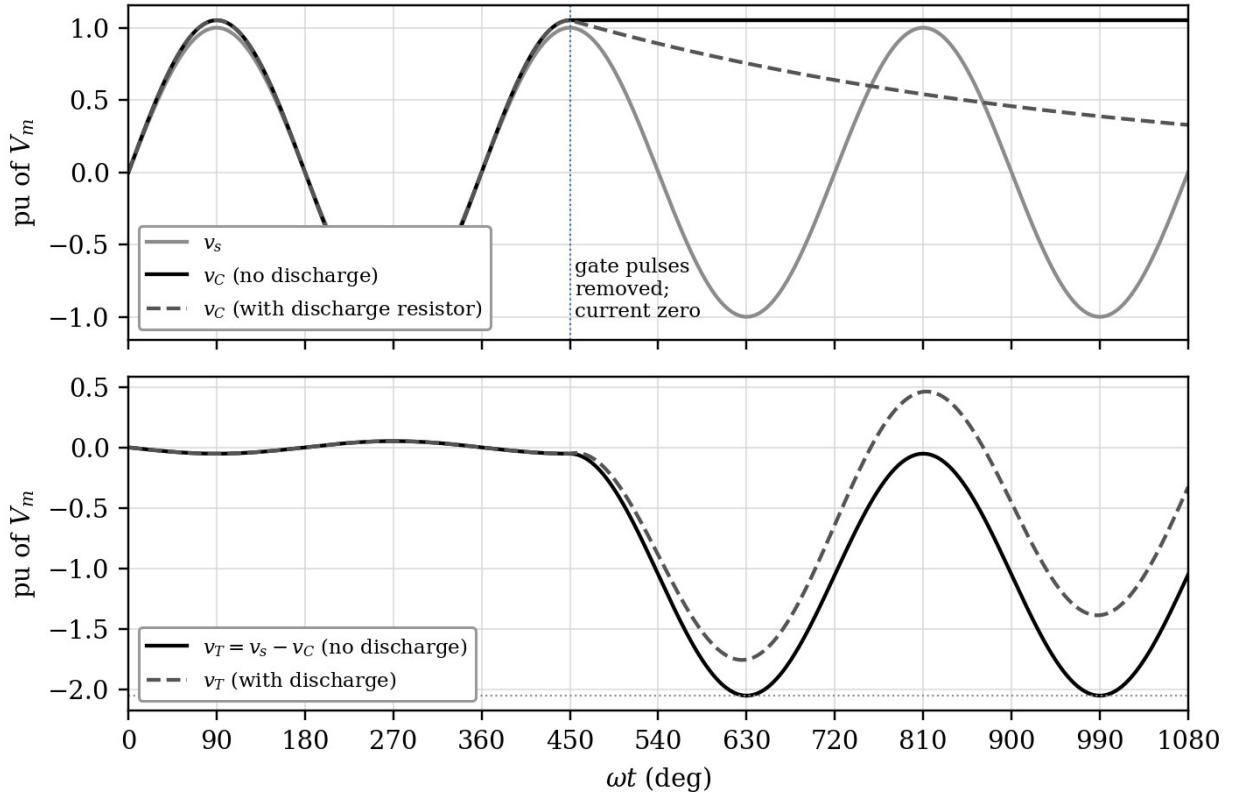


Figure 6.22. TSC turn-off: supply and capacitor voltages (top) and valve voltage (bottom) for non-discharging and discharging capacitors.

In the steady-state comparison, a TSC valve needs approximately twice the blocking-voltage capability of a TCR valve at the same branch voltage. It must also withstand the trapped-charge DC component while disconnected. Transformer ratio, transient overvoltage, redundancy and voltage-sharing design determine the actual valve string. Transmission TSCs commonly use a step-down transformer to place this duty at a suitable medium voltage.

6.4.7 Inrush and di/dt protection

The series reactor must also protect the valve against a mistimed firing, for instance a control fault, a misread synchronizing signal or forward break-over protection firing the valve at an arbitrary instant. At the instant of closure the reactor carries the whole voltage difference, so

$$\left. \frac{di}{dt} \right|_{t_c^+} = \frac{v_s(t_c) - V_{C0}}{L} \quad (6.33)$$

The worst case is closure when the supply is at one peak while the capacitor holds the trapped voltage of opposite polarity, giving $|v_s - V_{C0}| = (1+k)V_m$. The minimum reactor for a valve di/dt capability $(di/dt)_{max}$ is therefore $L_{min} = (1+k)V_m / (di/dt)_{max}$. In a well-designed TSC the reactor chosen for detuning ($n \approx 4.5$) normally satisfies this limit with ample margin; Example 6.25 checks it for a 33 kV bank.

6.4.8 Configurations

Three-phase TSC banks are connected in delta or in ungrounded star. Delta branches see line-to-line voltage and can be switched independently. In an ungrounded star the neutral shifts during switching and couples the phases, so the firing sequence has to be derived from the actual residual and line-to-line voltages rather than from a fixed quarter-cycle delay. Some low-voltage banks replace one thyristor of each valve by a diode, which keeps the capacitor charged to the crest and permits crest firing; the harmonic, polarity and insulation duties of that circuit then need separate checking.

With binary branch weights $B, 2B, \dots, 2^{N-1}B$, a bank provides $2^N - 1$ nonzero susceptance levels. Repeated weights such as $C, C, C/2$ give five nonzero levels, although several switch combinations yield the same output. Repeated assemblies can simplify spares and redundancy. Apply all weights to the net branch susceptance including the series reactor; Example 6.26 treats a binary bank.

6.4.9 V-I characteristic and hysteresis

Each TSC stage contributes a fixed capacitive susceptance, so a bank of N stages has the discrete characteristic of Figure 6.23: one straight line through the origin for each number of stages in service. To regulate voltage, a stage is switched in when the bus voltage falls below $V_{ref} - \Delta V/2$ and switched out when it rises above $V_{ref} + \Delta V/2$. The hysteresis band ΔV must exceed the voltage change caused by one step, $\Delta V_{step} \approx X_{th} B_{stage} V$, otherwise the controller hunts. Because the response is quantized, a TSC alone regulates voltage only within this band. Continuous control requires a TCR or a converter alongside it.

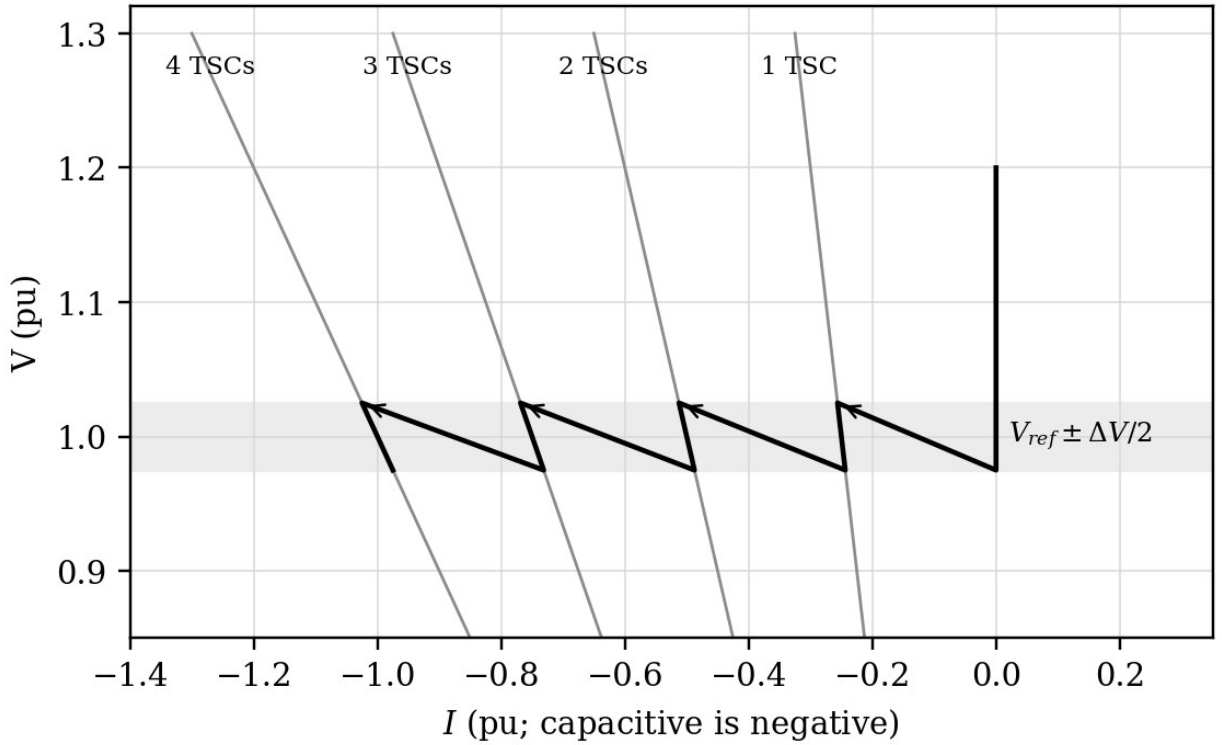


Figure 6.23. Discrete V-I characteristic of a four-stage TSC bank with hysteresis band $V_{ref} \pm \Delta V/2$.

A TSC also has a non-zero response delay. After a command to switch in, the valve must wait for the next permissible firing instant, which with trapped charge of a given polarity can be up to one full cycle. Switching out takes effect at the next current zero, within half a cycle. These delays are small compared with typical voltage-control time constants, but they matter in flicker compensation (Section 6.5.10).

6.5 The Static Var Compensator (SVC)

6.5.1 Definition and objectives

The IEEE and CIGRE definition of a static var compensator is a shunt-connected static var generator or absorber whose output is adjusted to exchange capacitive or inductive current so as to maintain or control specific parameters of the power system, typically bus voltage [17], [23]. “Static” distinguishes it from the rotating synchronous condenser: an SVC has no moving parts other than cooling equipment. Physically it is an assembly of the branches already studied (TCRs, TSRs, TSCs, fixed capacitors and harmonic filters),

connected to a medium-voltage busbar that is usually fed from the transmission bus through a step-down transformer (Figure 6.24).

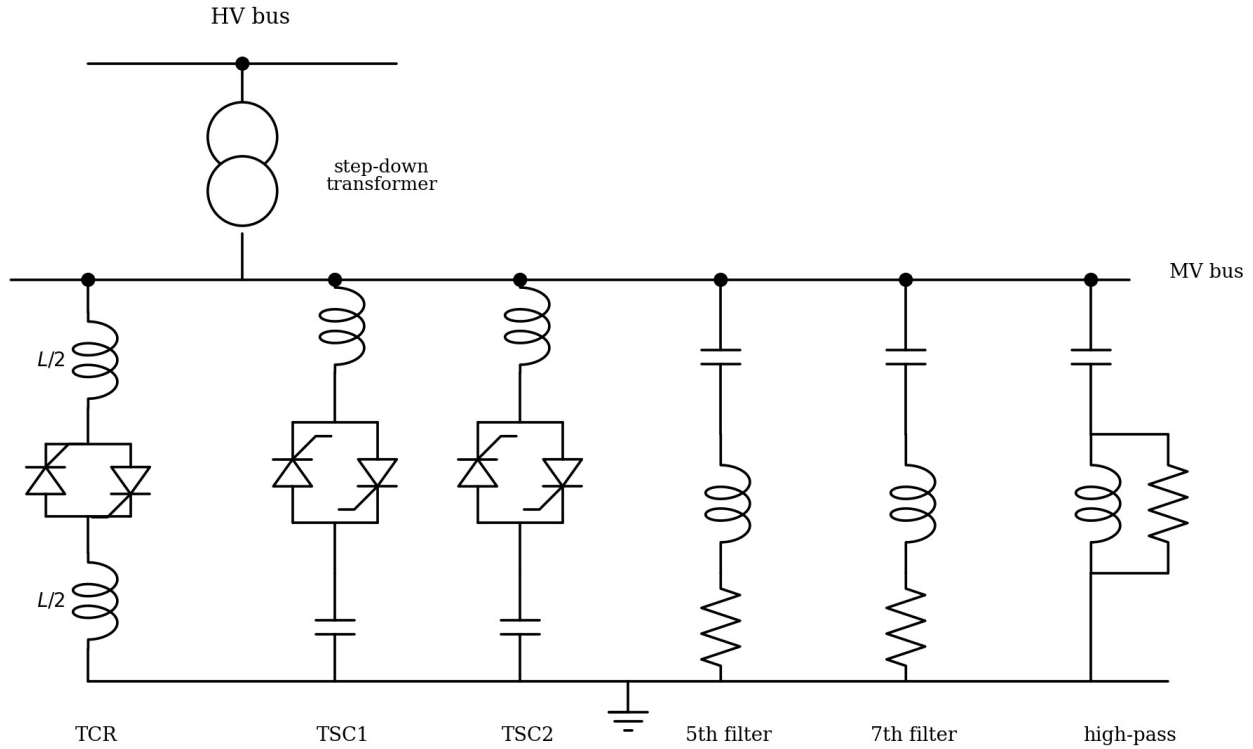


Figure 6.24. Single-line diagram of a transformer-coupled TSC-TCR SVC with 5th, 7th and high-pass filters.

Two broad families exist. *Transmission SVCs* regulate the voltage at a substation or along a long corridor. They raise transfer limits, improve transient stability, damp electromechanical oscillations (and occasionally subsynchronous torsional modes), and limit dynamic overvoltages after load rejection. *Industrial SVCs* sit close to arc furnaces, rolling mills, mine hoists and traction substations. There they reduce voltage flicker, balance single-phase loads and correct power factor, which requires very fast, phase-by-phase action.

6.5.2 V-I characteristic: ideal and practical

An ideal SVC would hold its bus voltage at V_{ref} for any current, capacitive or inductive, without losses, harmonics or delay: a horizontal line on the V-I plane. A practical SVC approaches this inside its control range and departs from it at the limits. Its characteristic is built by combining the characteristics of its branches, as Figure 6.25 shows for an FC-TCR. The fixed capacitor contributes a straight line of slope $-1/B_C$ on the capacitive side, and the TCR a fan of lines on the inductive side. The controller selects, at each voltage, the TCR susceptance that places the operating point on the regulating line.

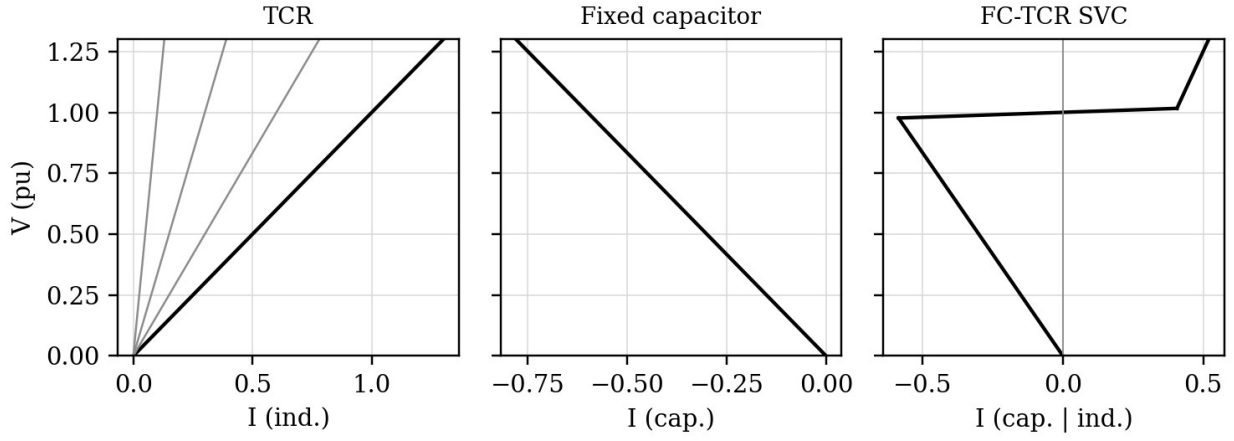


Figure 6.25. Building the SVC characteristic: TCR fan, fixed-capacitor line, and the resulting FC-TCR characteristic.

Inside the control range the SVC follows

$$V = V_{ref} + X_{SL} I_{SVC} \quad (6.34)$$

where V is the regulated bus voltage, V_{ref} the voltage at zero SVC current, X_{SL} the slope (droop) reactance, typically 1 to 5% on the SVC rating, and I_{SVC} the SVC reactive current, positive when inductive. Outside the range the SVC behaves as a fixed susceptance:

$$I_{SVC} = -B_{C,max} V \text{ (capacitive limit)}, I_{SVC} = B_{L,max} V \text{ (inductive limit)} \quad (6.35)$$

where $B_{C,max}$ and $B_{L,max}$ are the maximum net capacitive and inductive susceptances. The limits are lines through the origin, not vertical lines. At the capacitive limit the SVC current, and with it the support it can give, collapses in proportion to the voltage.

6.5.3 Operating point and the role of the slope

The steady operating point is the intersection of Equation (6.34) with the system load line, Equation (6.1). Solving the pair,

$$I_{SVC} = \frac{E_{th} - V_{ref}}{X_{th} + X_{SL}}, V = \frac{X_{SL} E_{th} + X_{th} V_{ref}}{X_{th} + X_{SL}} \quad (6.36)$$

Figure 6.26 shows how the operating point moves as the Thevenin voltage varies. When the system voltage would otherwise be high, the SVC absorbs reactive power, and when it would be low, the SVC injects it.

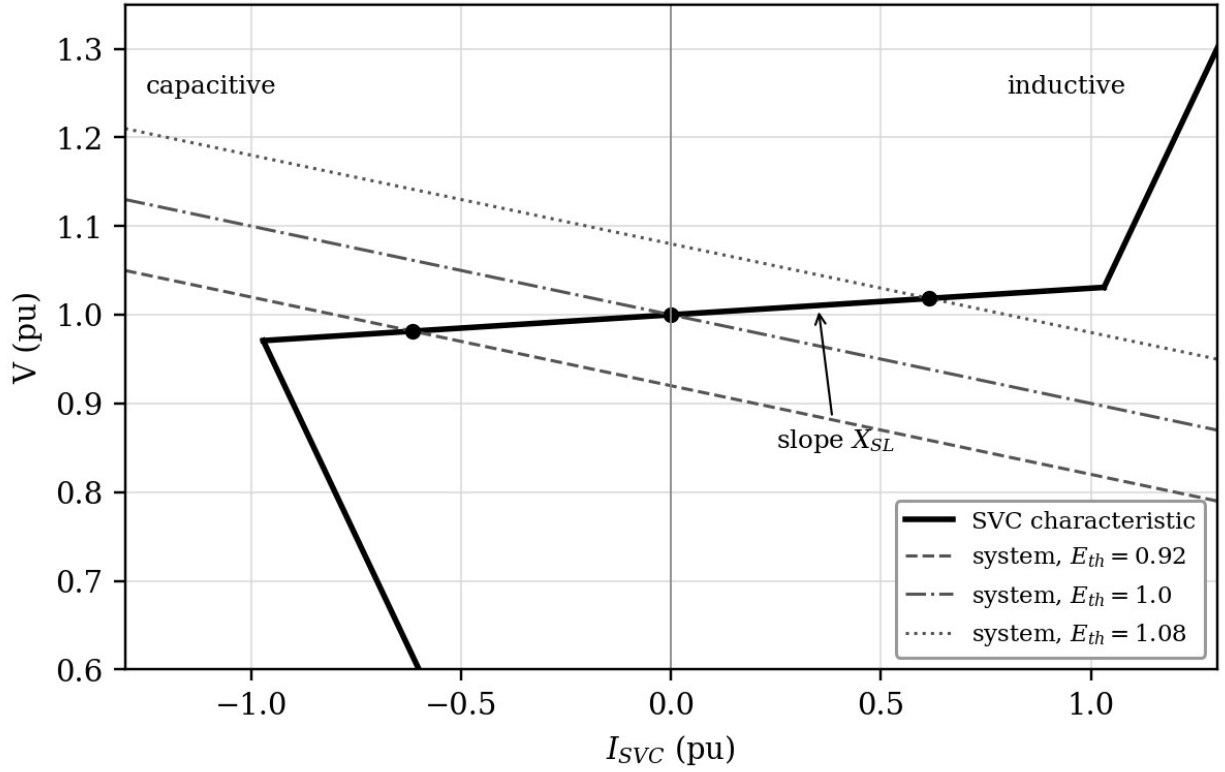


Figure 6.26. SVC operating points as the intersection of the SVC characteristic with system load lines for three Thevenin voltages.

Why not set $X_{SL}=0$ for perfect regulation? Equation (6.36) answers part of the question: the current needed to absorb a given excursion ΔE_{th} is $\Delta E_{th}/(X_{th}+X_{SL})$, so a non-zero slope reduces the required rating (Figure 6.27). For a stiff bus ($X_{th}=0.05$ pu) a 3% slope cuts the rating by 37.5%. There are three further reasons. The slope lets several SVCs, or an SVC and nearby generators, share the regulating duty in a stable way instead of fighting each other. It keeps the SVC away from its limits for ordinary voltage variations, preserving dynamic reserve for disturbances. And it makes the voltage loop gain less sensitive to network strength (Section 6.5.7). The voltage regulation obtained is $\Delta V/\Delta E_{th} = X_{SL}/(X_{th}+X_{SL})$.

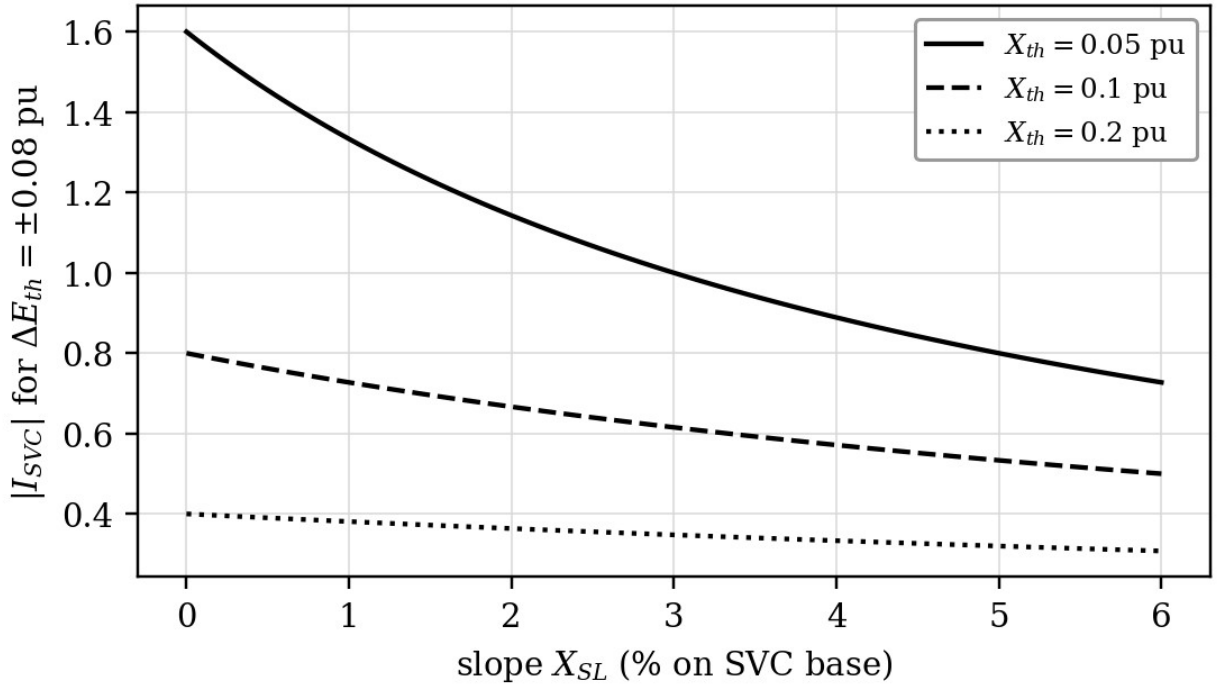


Figure 6.27. SVC current required to absorb a ± 0.08 pu excursion of the Thevenin voltage, as a function of slope, for three network strengths.

Example 6.6: SVC operating point, slope and saturation.

The Thevenin voltage at a bus varies between 0.92 and 1.08 pu with $X_{th}=0.06$ pu. An SVC with $V_{ref}=1.0$ pu is installed. Find (a) the SVC current range for $X_{SL}=0$; (b) the range for $X_{SL}=0.03$ pu; (c) the bus voltage band in case (b); (d) the reactive power at the extremes; (e) the capacitive susceptance needed at $E_{th}=0.92$; (f) the actual operating point if $B_{C,max}=0.8$ pu.

Solution. (a) Equation (6.36) with $X_{SL}=0$: $I=\pm 0.08/0.06=\pm 1.333$ pu, and $V=1.0$ pu exactly.

(b) $I=\pm 0.08/0.09=\pm 0.889$ pu, a 33% reduction in rating.

(c) $V=1\pm 0.03\times 0.889$, i.e. 0.9733 to 1.0267 pu.

(d) $Q=VI$: -0.865 pu (capacitive) at $E_{th}=0.92$ and $+0.913$ pu (inductive) at $E_{th}=1.08$.

(e) $B=0.889/0.9733=0.913$ pu.

(f) The SVC saturates and behaves as $B=0.8$ pu. With $I=-BV$ in Equation (6.1), $V=E_{th}/(1-X_{th}B)=0.92/0.952=0.9664$ pu, $I=-0.773$ pu and $Q=BV^2=0.747$ pu. The voltage is below the droop line, and the support is 14% less than the regulating characteristic would have demanded.

Note. The droop equation determines the commanded operating line, while the capacitive and inductive susceptance limits determine whether that command is physically achievable. A calculated operating point outside the limits must be clipped to the appropriate boundary.

6.5.4 SVC configurations

Fixed capacitor-thyristor controlled reactor (FC-TCR)

The FC-TCR (Figure 6.28) extends the purely inductive TCR into the capacitive region by adding a permanently connected capacitor, which in practice is split into tuned filter branches. Its net susceptance is

$$B_{SVC}(\alpha)=B_C-B_{TCR}(\alpha) \quad (6.37)$$

where B_C is the total fundamental capacitive susceptance of all fixed branches and filters, and B_{TCR} is given by Equation (6.13). Maximum capacitive output occurs with the TCR blocked ($\alpha=180^\circ$), and maximum inductive output at full conduction. To deliver a range from $+Q_{cap}$ to $-Q_{ind}$ at rated voltage, the ratings must be

$$Q_C = Q_{cap}, Q_{TCR} = Q_{cap} + Q_{ind} \quad (6.38)$$

where Q_C is the fixed-capacitor rating and Q_{TCR} the TCR rating at full conduction. The TCR is therefore always larger than the SVC's inductive range, because it must first cancel the capacitor.

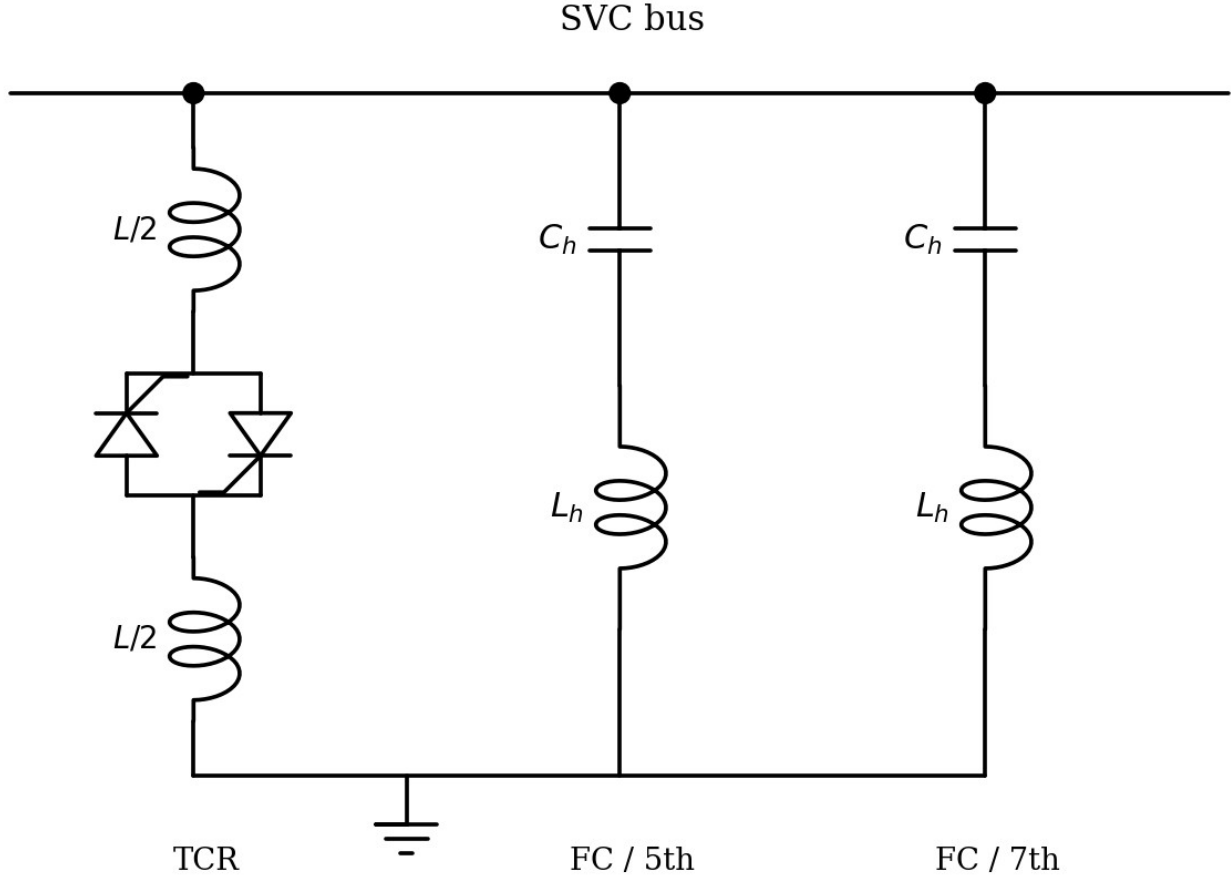


Figure 6.28. FC-TCR SVC with the fixed capacitance provided by 5th and 7th harmonic filters.

Figure 6.29 shows the operating area and the net susceptance versus firing angle. The fixed capacitance stays connected and the TCR varies its fundamental current continuously, so there are no capacitor-switching transients. The TCR still generates harmonics, which the filters reduce but do not eliminate. The weakness is loss: at zero net output the TCR must cancel the full capacitor current, as the simplified comparison of Figure 6.31 shows. Industrial users often accept this for the continuous response, whereas transmission SVCs usually prefer switched stages that avoid circulating current at idle.

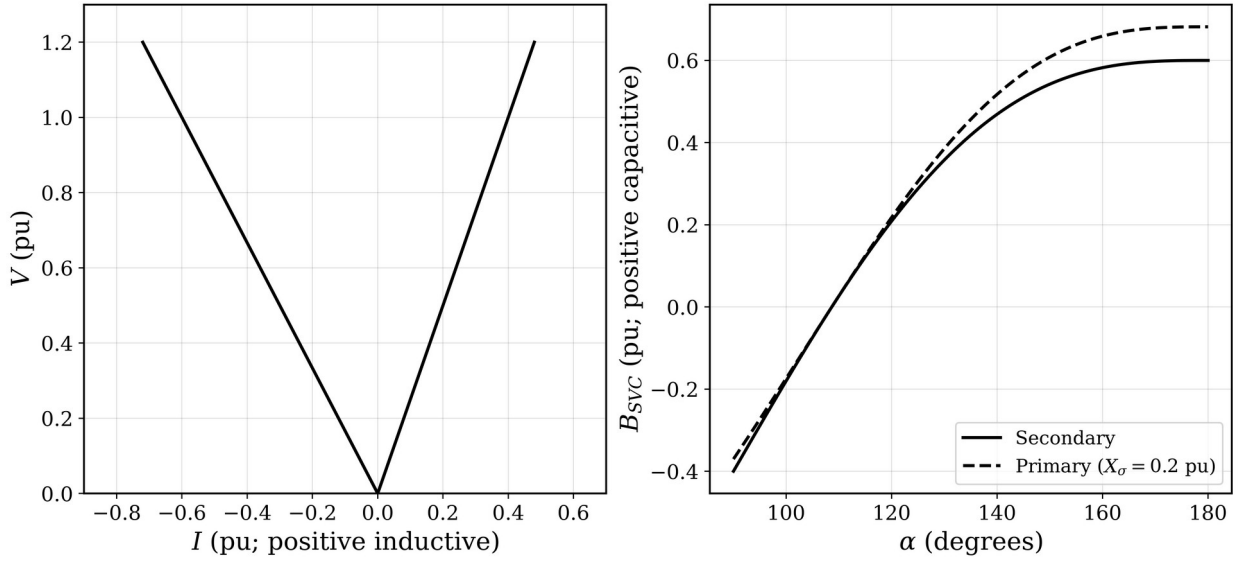


Figure 6.29. FC-TCR limiting V-I envelope and net susceptance versus firing angle, with $BC = 0.6$ pu, $B_{TCR,max} = 1.0$ pu and $X_\sigma = 0.2$ pu. The full voltage-control characteristic also includes its droop segment.

The FC-TCR controller synchronizes firing to branch-voltage zero crossings, determines the required reactor absorption, converts it through the inverse firing law and distributes gate pulses. With delivered reactive-power reference Q_{ref} , its ideal rated-voltage balance is $Q_{TCR} = Q_C - Q_{ref}$. With positive capacitive injection-current reference $I_{Q,ref}$, the corresponding balance is $I_{TCR} = I_C - I_{Q,ref}$. The chosen current sign must be retained throughout the regulator and distribution logic.

Thyristor switched capacitor-thyristor controlled reactor (TSC-TCR)

The TSC-TCR (Figure 6.24) replaces the fixed capacitor by N switched capacitor stages, each of rating Q_{TSC} , and retains one TCR for continuous fine control. For a capacitive demand Q , the controller switches in

$$N_{on} = \lceil \frac{Q}{Q_{TSC}} \rceil, Q_{TCR}(\alpha) = N_{on} Q_{TSC} - Q \quad (6.39)$$

For Equation (6.39), Q and the stage ratings refer to the same operating voltage, and $0 \leq Q \leq N Q_{TSC}$ is assumed. At other voltages use $Q_{stage}(V) = B_{stage} V^2$ or dispatch directly in susceptance. Negative commands require zero or an appropriate number of capacitor stages and the specified inductive TCR range. A minimum-stage dispatch keeps residual TCR absorption below one capacitor step for positive demand; hysteresis may retain an extra stage temporarily. At zero output with all switched stages off, branch losses are small, although auxiliaries and transformers still consume power.

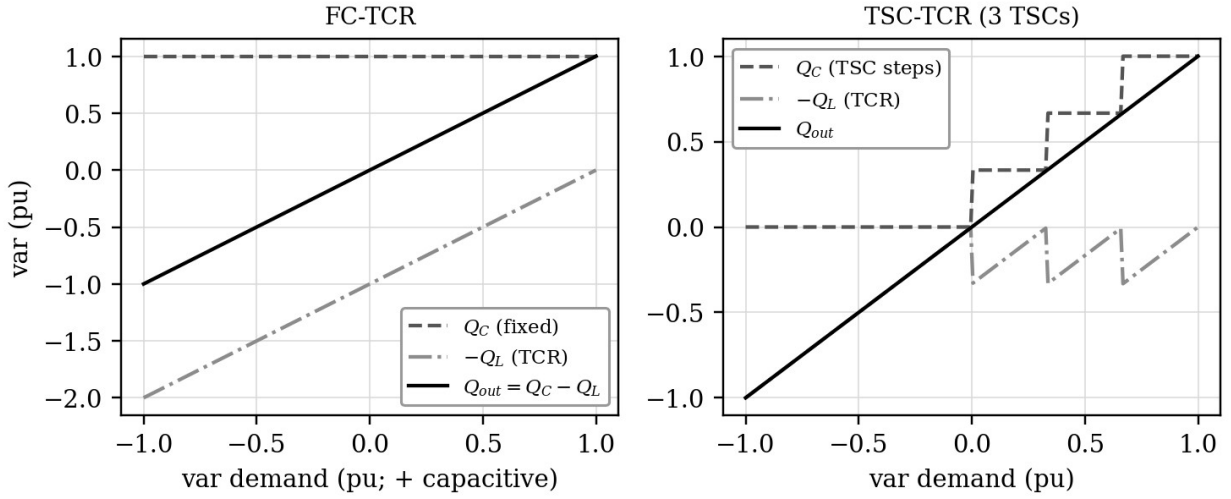


Figure 6.30. Var output versus var demand: FC-TCR (left) and TSC-TCR with three stages (right).

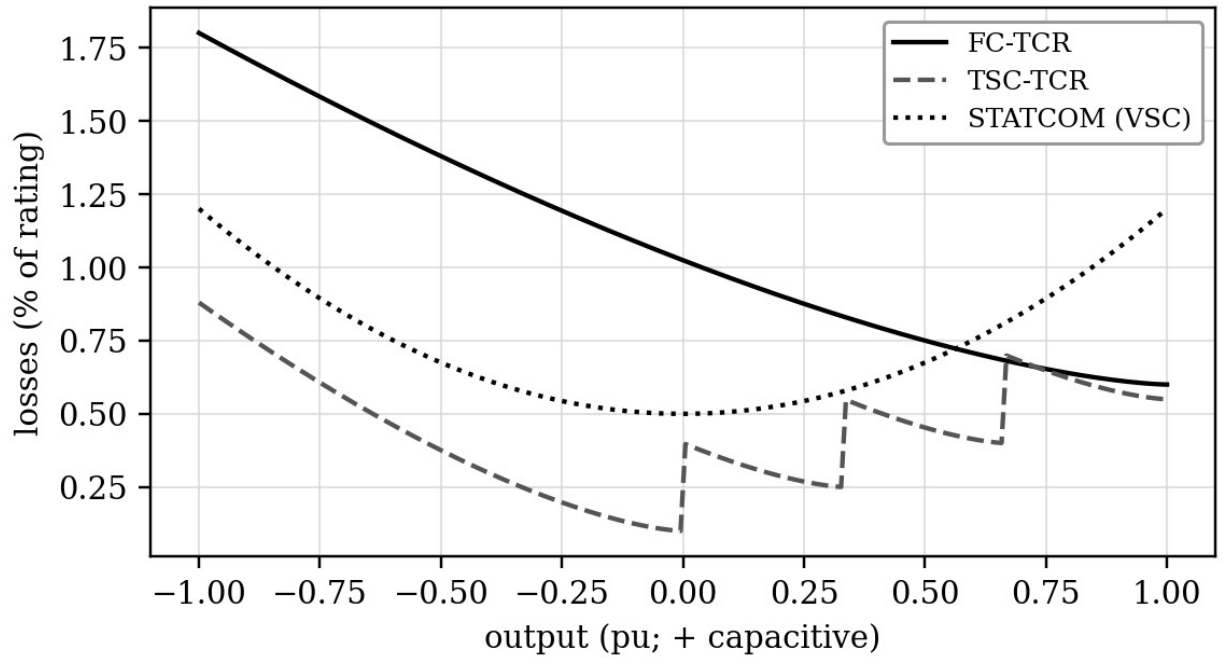


Figure 6.31. Illustrative loss characteristics of FC-TCR, TSC-TCR and VSC-based STATCOM of equal rating (simplified loss models; actual values depend on design).

For continuous coverage the TCR range must at least bridge one TSC step, and in practice exceed it slightly so that the switching-in and switching-out thresholds can be separated by a hysteresis margin:

$$B_{TCR,max} \geq B_{TSC}(1+\epsilon) \quad (6.40)$$

where B_{TSC} is the susceptance of one TSC stage and ϵ a margin of a few percent. If this condition is violated the characteristic develops dead zones, ranges of susceptance that no combination of stages and firing angle can produce (Figure 6.32). Inside such a gap the SVC can only alternate between neighbouring stages, which defeats continuous control and causes repeated switching.

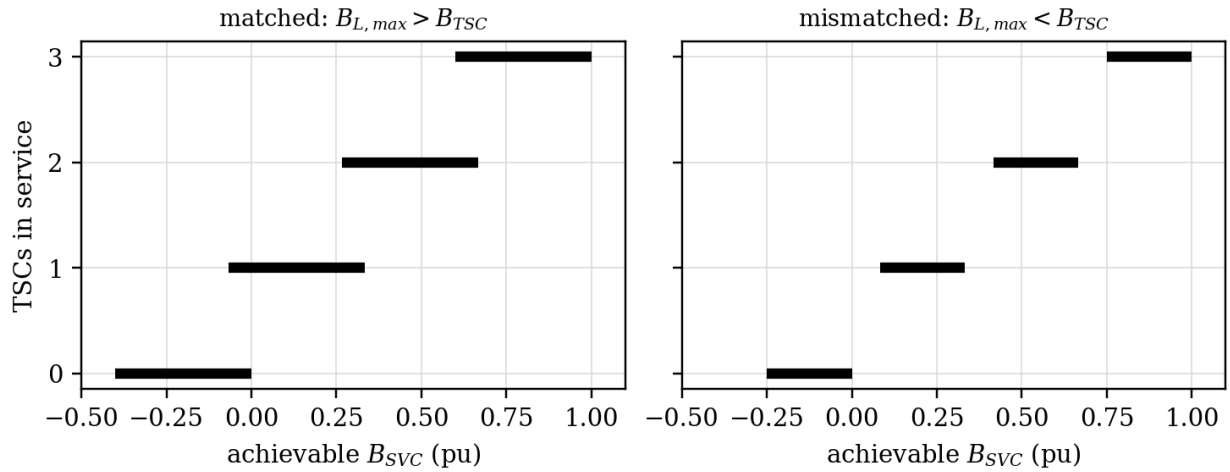


Figure 6.32. Achievable net susceptance of a three-stage TSC-TCR: matched TCR (left) and undersized TCR producing dead zones (right).

Other configurations

Table 6.4 summarizes common configurations. An MSC-TCR combines slow mechanical steps with continuous TCR trim. A TSC-TSR arrangement avoids phase-control harmonics in ideal sinusoidal steady state, but background harmonics, resonance and switching transients can still require damping or filtering. Saturated-reactor compensators illustrate the historical development of static compensation.

Table 6.4. Comparison of SVC configurations.

Configuration	Control	Harmonics	Losses at zero output	Typical use
FC-TCR	Continuous	TCR harmonics, absorbed by FC filters	High (TCR cancels FC)	Arc furnaces, rolling mills
TSC-TCR	Continuous	Small TCR (one stage) harmonics	Low	Transmission voltage control
TSC-TSR	Stepped	Negligible	Low	Slow voltage control, low cost
MSC-TCR	Continuous within TCR range, slow steps	TCR harmonics	Medium	Subtransmission, slow load cycles
Saturated reactor + FC	Inherent, fixed slope	Low (with slope correction)	High	Historical flicker compensators

6.5.5 SVC with a coupling transformer

Most transmission SVCs connect through a step-down transformer of leakage reactance X_σ , which places the thyristor valves at a convenient medium voltage. Seen from the high-voltage bus, the branch susceptance B on the secondary (positive when capacitive) appears in series with X_σ . The equivalent admittance of the series combination is $j/(1/B - X_\sigma)$, giving

$$B_{eq} = \frac{B}{1 - X_\sigma B} = \frac{B_\sigma B}{B_\sigma - B}, B_\sigma = \frac{1}{X_\sigma} \quad (6.41)$$

where B_{eq} is the susceptance seen from the primary and B_σ the transformer leakage susceptance magnitude. The transformer *enlarges* the capacitive end of the range ($B > 0$, denominator below one) and *shrinks* the inductive end ($B < 0$, denominator above one). The secondary voltage is raised on capacitive output and lowered on inductive output:

$$V_2 = \frac{V}{1 - X_o B} \quad (6.42)$$

where V_2 is the secondary bus voltage referred to the primary. Two practical consequences follow. Secondary overvoltage at full capacitive output sets the voltage rating of the capacitors and valves. And because B_{eq} is a nonlinear function of B , the linearizer must include the transformer, or the regulator gain will vary with operating point. For $X_o B \ll 1$, Equation (6.41) linearizes to $B_{eq} \approx B(1 + X_o B)$, which is adequate for preliminary sizing (Example 6.7).

Example 6.7: SVC behind a coupling transformer.

An FC-TCR has $B_C = 0.8$ pu and $B_{L,max} = 0.9$ pu on the secondary, connected through a transformer with $X_o = 0.2$ pu ($B_o = 5$ pu). Find (a) the net secondary susceptance range; (b) the exact primary range; (c) the linearized values and their errors; (d) the secondary voltages at the limits with 1.0 pu on the primary; (e) the practical consequence.

Solution. (a) $+0.8$ pu (TCR blocked) to $0.8 - 0.9 = -0.1$ pu (full conduction).

(b) Equation (6.41): $0.8/(1 - 0.16) = 0.9524$ pu and $-0.1/(1 + 0.02) = -0.0980$ pu.

(c) $B(1 + X_o B)$: 0.928 pu (-2.6%) and -0.0980 pu (-0.04%). The linear form is poor at the capacitive end.

(d) Equation (6.42): $V_2 = 1/(1 - 0.16) = 1.190$ pu at full capacitive output and $1/1.02 = 0.980$ pu at full inductive output.

(e) The transformer raises the capacitive range by 19%, but the capacitors and TSC valves must be rated for 1.19 pu secondary voltage. Writing the series combination as $B_o B/(B_o + B)$ would give a *reduced* capacitive range of 0.69 pu; that form has the wrong sign for a capacitive branch.

Note. Referring an SVC through its coupling transformer is not a simple turns-ratio exercise. Transformer leakage reactance modifies the effective susceptance and therefore the attainable capacitive and inductive ranges seen from the transmission bus.

6.5.6 Harmonic filters and resonance

A single-tuned filter branch is a series R-L-C circuit tuned to harmonic order h . At fundamental frequency it is capacitive, and its net reactance must deliver the rated filter output Q_F :

$$X_C - X_L = \frac{V^2}{Q_F}, X_L = \frac{X_C}{h^2} \Rightarrow X_C = \frac{h^2}{h^2 - 1} \cdot \frac{V^2}{Q_F} \quad (6.43)$$

where V is the voltage across the branch (phase voltage for star connection with per-phase Q_F , or equivalently line voltage with three-phase Q_F), X_C and X_L are the fundamental-frequency reactances of the filter capacitor and inductor, and h the tuning order. The resistance follows from the chosen quality factor Q_f as $R = h X_L / Q_f$. Filters are tuned a few percent below the nominal harmonic, for example $h = 4.9$ for the 5th, so that a loss of capacitance (ageing or failed elements, which raises the tuned frequency) or a fall in system frequency (which lowers the harmonic frequency) does not leave them tuned above the harmonic. The parallel resonance with the network lies just below the tuned frequency and would then move onto the harmonic. Figure 6.33 shows the impedance of typical 5th and 7th branches.

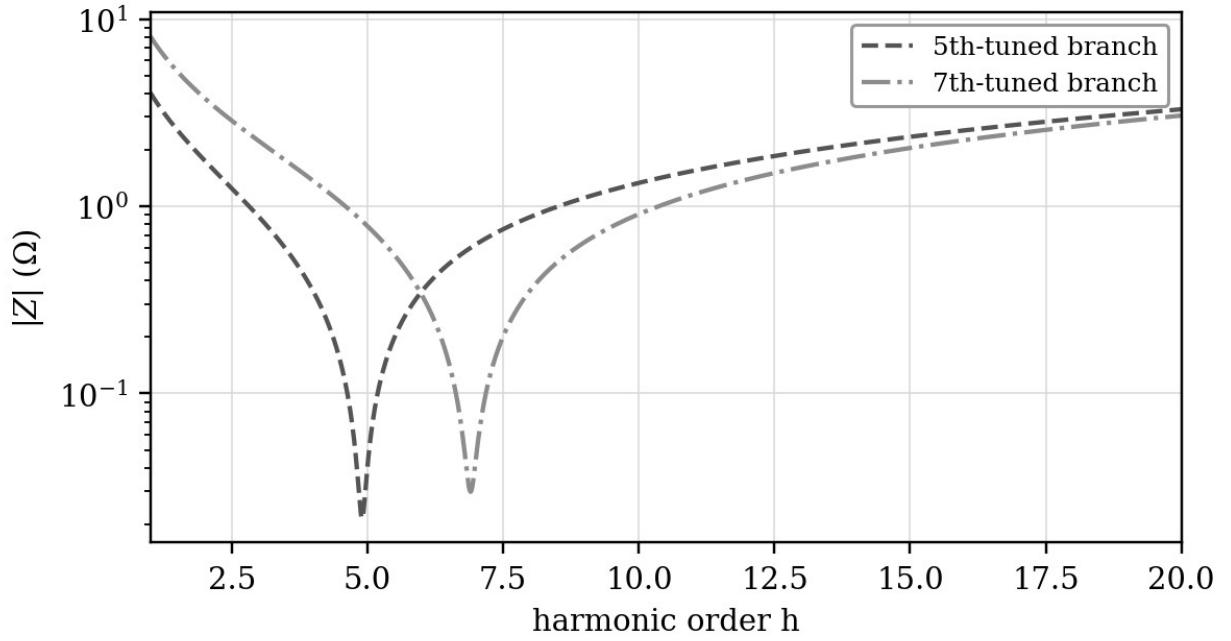


Figure 6.33. Impedance magnitude of 5th- and 7th-tuned filter branches ($Q_f = 40$) versus harmonic order.

Every capacitive branch also forms a parallel resonance with the network inductance. Below its tuned frequency a filter is capacitive, and with the system reactance X_{th} it resonates at approximately $h_p \approx \sqrt{X_C / (X_{th} + X_L)}$, where X_C and X_L are the branch reactances at fundamental frequency. Filter design must ensure that h_p does not coincide with a harmonic the TCR or the network can excite, under all switching states of the SVC and of the surrounding network.

6.5.7 SVC control

Control structure

Figure 6.34 uses I_{SVC} positive for inductive absorption and B_{ref} positive for capacitive injection. To implement Equation (6.34), the voltage error is $e_V = V_{ref} + X_{SL} I_{SVC} - V_m$. Thus the current droop term enters with a plus sign. A positive error must command more capacitive susceptance. The PI regulator output, plus a limited supplementary damping signal, passes through susceptance limits and stage distribution. Switching logic selects TSC/TSR states, and the TCR linearizer supplies the firing angle for the residual reactor absorption. A PLL provides synchronized firing pulses. Anti-windup must track the realizable susceptance after both clipping and stage dispatch.

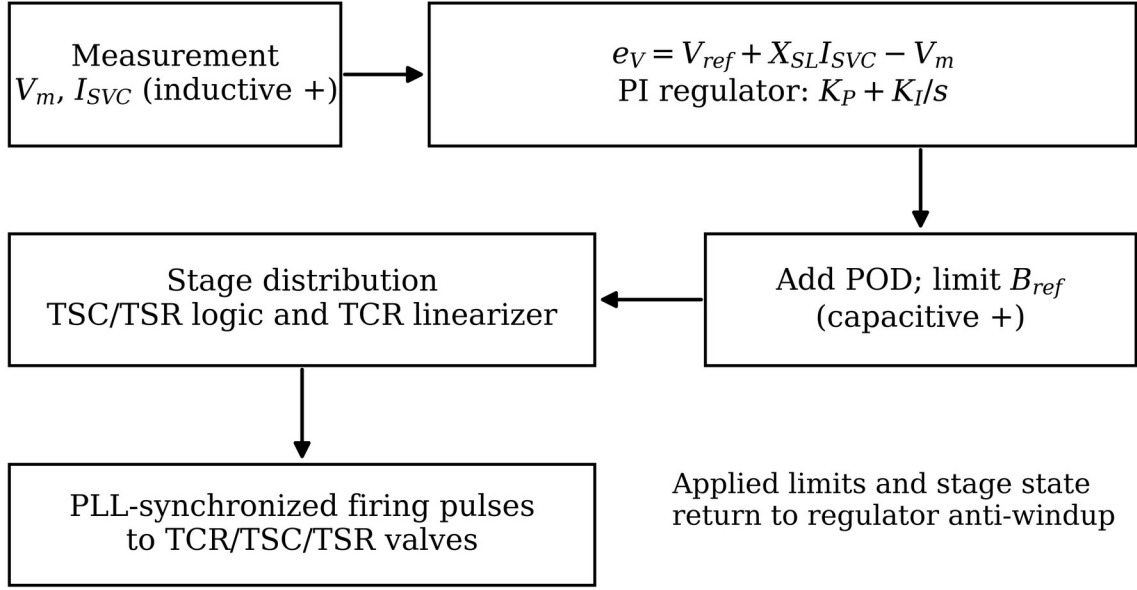


Figure 6.34. General SVC voltage control structure.

Regulator design against network strength

The closed-loop behaviour is dominated by the gain from SVC susceptance to bus voltage. Linearizing Equation (6.1) around $V \approx 1$ with $I_{SVC} = -B_{SVC}V$, a change ΔB in SVC susceptance (on the SVC rating) produces

$$\Delta V \approx K_N \Delta B, K_N = X_{th} = \frac{1}{E S C R} \quad (6.44)$$

Here K_N is the network gain and $E S C R$ the effective short-circuit ratio at the SVC bus, expressed on the SVC rating. The approximation comes from the lossless steady relation $V = E_{th}/(1 - X_{th}B)$, whose sensitivity at an operating point B_0 is $K_N = \partial V / \partial B = E_{th} X_{th} / (1 - X_{th} B_0)^2$, evaluated at $B = B_0$; it reduces to X_{th} for $V \approx E_{th} \approx 1$ pu and small $X_{th} B_0$. With consistent uncompensated short-circuit and MVA bases $X_{th} = 1/SC R$; using $1/E S C R$ instead is an effective-strength approximation, and the fixed capacitance must then not be counted a second time as an explicit controlled branch.

The thyristor actuator is modelled as a transport delay of mean value $T_b \approx T/4$ (5 ms at 50 Hz), approximated by $1/(1 + sT_b)$. The slope feedback contributes a further gain K_{SL} around the loop. With the regulator

$$G_R(s) = K_C \frac{1 + sT_C}{sT_C}, T_C = T_b \quad (6.45)$$

the zero cancels the actuator lag and the open-loop transfer function reduces to $K_C(K_N + K_{SL})/(sT_b)$. The closed loop is then first order with time constant

$$T_\omega = \frac{T_b}{K_C(K_N + K_{SL})} \quad (6.46)$$

For this preliminary lag-and-droop model, choose $K_C = 1/[2(K_{SL} + K_{N,max})]$ using the largest expected voltage sensitivity; the crossover then lies near $1/(2T_b)$ at that operating point. This is a starting value, and

the final design must include the true firing delay, measurement lags, the current-to-susceptance conversion and the expected topology changes. Within the simplified model,

$$T_{\omega} = 2T_b \frac{K_{SL} + K_{N,max}}{K_{SL} + K_N}, t_{95} \approx 3T_{\omega}, \Delta V = \frac{K_N}{K_N + K_{SL}} \Delta V_{ref} \quad (6.47)$$

Here t_{95} is the time to 95% of the final voltage change. The reduced model predicts slower response in a stronger network (Figure 6.35). Increasing gain to recover speed can reduce stability margins after network weakening. Gain scheduling or supervision may address this trade-off, but its switching logic and limits must also be validated.

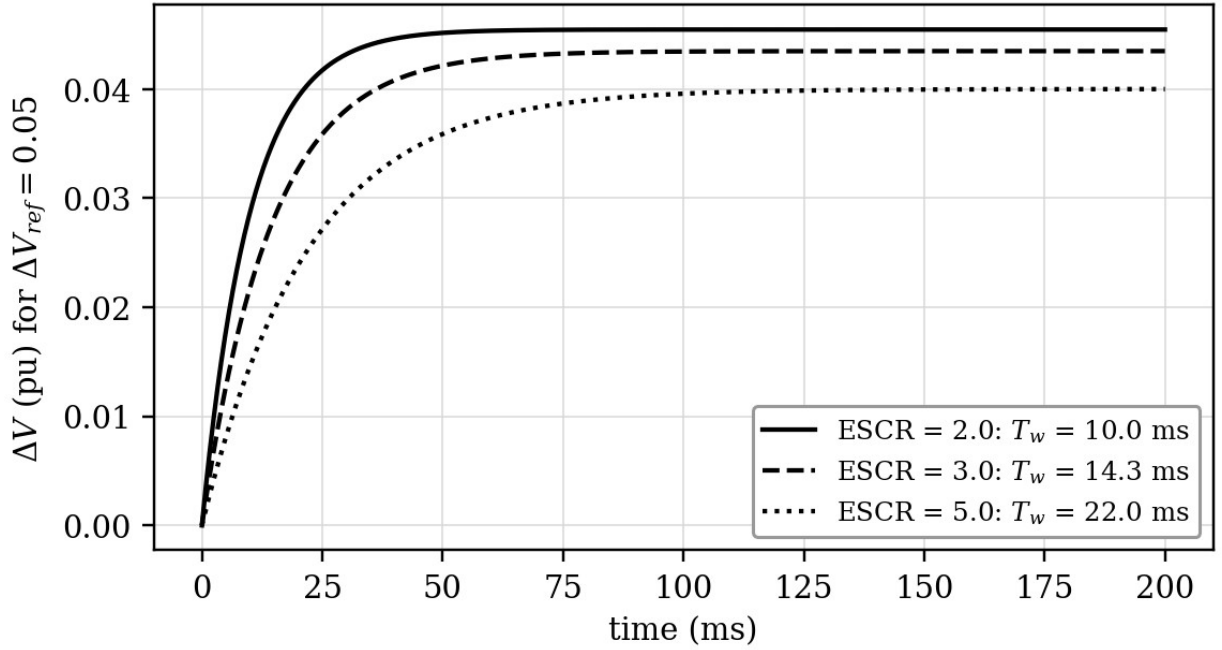


Figure 6.35. Closed-loop voltage response to a 0.05 pu reference step for a regulator designed at ESCR = 2, operating at ESCR = 2, 3 and 5.

Standard models and supplementary controls

For power-system studies the IEEE and WECC define generic SVC models. The common structure contains a measurement lag, a voltage regulator (proportional with lag, or PI), a slope feedback, a thyristor-susceptance block with delay T_d and lag T_b , and susceptance limits. Some models also include a slow susceptance regulator that drifts the operating point back towards a preset susceptance after a disturbance, keeping the fast range in reserve for the next event.

Supplementary signals improve damping. Candidate inputs include line current, line active or reactive power, bus voltage angle, bus frequency, and the speed or accelerating power of a nearby generator. Line power and locally synthesized frequency are preferred because they are available at the SVC site and are sensitive to inter-area modes. The supplementary controller uses a washout filter, lead-lag compensation for the phase of the target mode, and limits; its design is developed in Chapter 11. A well-known side effect deserves mention here. An SVC regulating voltage at a generator bus near a series-compensated line can interact with torsional modes, and the same supplementary channel, designed with the torsional frequency in mind, can add subsynchronous damping.

Example 6.8: Voltage regulator design against ESCR.

An SVC regulator is designed for $ESCR_{min} = 2$ with $K_{SL} = 0.05$ pu at 50 Hz. Find (a) $K_{N,max}$; (b) K_C ; (c) $G_R(s)$; (d) T_{ω} , t_{95} and ΔV for a 0.05 pu reference step at ESCR = 5; (e) the same at ESCR = 2; (f) the engineering conclusion.

Solution. (a) $K_{N,max} = 1/2 = 0.5$ pu.

(b) $K_C = 1/[2(0.05 + 0.5)] = 0.909$ pu.

(c) $T_b = T/4 = 5$ ms; $G_R(s) = 0.909(1 + 0.005s)/(0.005s)$.

(d) $K_N = 0.2$; $T_\omega = 2(5)(0.55/0.25) = 22$ ms; $t_{95} = 66$ ms; $\Delta V = 0.2 \times 0.05/0.25 = 0.040$ pu.

(e) $T_\omega = 10$ ms; $t_{95} = 30$ ms; $\Delta V = 0.0455$ pu.

(f) The simplified response slows by a factor of 2.2 in the strong network. Gain scheduling or gain supervision is an option if one fixed tuning cannot satisfy the required response and stability margins across the specified strengths.

Note. Regulator tuning cannot be separated from network strength. The same controller gains can produce quite different closed-loop behaviour as the Thevenin reactance changes, which is why weak-grid and contingency conditions must be included in an SVC control study.

6.5.8 Midpoint SVC on a long transmission line

For long lines the lumped-reactance result of Section 6.1.2 must be replaced by the distributed-parameter solution. A lossless line of surge impedance Z_c , phase constant β and length l has electrical length $\theta = \beta l$. With equal end voltages V and power angle δ , the uncompensated power and the open-circuit midpoint voltage are

$$P_u = \frac{V^2}{Z_c \sin \theta} \sin \delta = \frac{P_0}{\sin \theta} \sin \delta, V_{m0} = V \frac{\cos(\delta/2)}{\cos(\theta/2)} \quad (6.48)$$

where $P_0 = V^2/Z_c$ is the surge impedance loading. With the midpoint held at V , each half behaves as a line of electrical length $\theta/2$ carrying angle $\delta/2$:

$$P_c = \frac{P_0}{\sin(\theta/2)} \sin \frac{\delta}{2}, \frac{P_{c,max}}{P_{u,max}} = \frac{\sin \theta}{\sin(\theta/2)} = 2 \cos \frac{\theta}{2} \quad (6.49)$$

For short lines $\cos(\theta/2) \rightarrow 1$ and the improvement approaches the factor of two found earlier; for a 500 km line at 50 Hz ($\theta = 30^\circ$) it is 1.93. The reactive power the midpoint device must supply follows from the receiving-end reactive power of each half-line:

$$Q_{dev} = \frac{2P_0[\cos(\theta/2) - \cos(\delta/2)]}{\sin(\theta/2)} \quad (6.50)$$

which is negative (absorption) for $\delta < \theta$, when line charging exceeds the series reactive loss, and positive (injection) beyond. The SVC must therefore be able to absorb on light load and inject on heavy load, which is exactly the duty of the TSC-TCR. Seen from the midpoint, the network has the Thevenin reactance $X_{th} = (Z_c/2) \tan(\theta/2)$, the parallel combination of the two half-lines; this value is used in Equation (6.36) when the SVC has a finite slope (Example 6.30).

6.5.9 Load balancing and power-factor correction

An SVC with independent phase control can balance currents and correct power factor under a balanced positive-sequence supply. Write each delta load admittance as $Y_{ab}^L = G_{ab} + jB_{ab}^L$, with positive susceptance capacitive, and let Y denote load plus compensation. Two simultaneous conditions are required: zero negative-sequence source current, represented by $a^2 Y_{ab} + Y_{bc} + a Y_{ca} = 0$ with $a = e^{j2\pi/3}$; and zero total reactive power, $B_{ab,total} + B_{bc,total} + B_{ca,total} = 0$. Individual branch susceptances need not vanish. Separating the complex current condition into real and imaginary parts and using the reactive-power condition gives

$$B_{ab}^c = -B_{ab}^L + \frac{G_{ca} - G_{bc}}{\sqrt{3}}, B_{bc}^c = -B_{bc}^L + \frac{G_{ab} - G_{ca}}{\sqrt{3}}, B_{ca}^c = -B_{ca}^L + \frac{G_{bc} - G_{ab}}{\sqrt{3}} \quad (6.51)$$

For a single-phase load $G + jB^L$ connected between a and b this reduces to the Steinmetz circuit: a capacitive branch $G/\sqrt{3}$ across b - c , an inductive branch $G/\sqrt{3}$ across c - a , and a branch $-B^L$ across a - b that cancels the load's own reactive power. The supply then sees a balanced resistive load drawing $P/(\sqrt{3}V_{LL})$ in each line. A TCR-FC per phase realizes these susceptances dynamically. This is the principle of traction and arc-furnace balancing compensators [18] (Example 6.31).

6.5.10 Industrial SVCs and field experience

Arc furnaces draw erratic, unbalanced reactive power that produces visible flicker at frequencies of a few hertz. Flicker reduction requires the compensator to track the furnace reactive power within roughly a half-cycle, phase by phase. FC-TCRs with open-loop (feedforward) control computed from the measured load current achieve flicker reduction factors of about two. Converter-based compensators, whose bandwidth is much higher, can do substantially better.

Table 6.5 gives historical installations from the Siemens reference list [41], with the Devers data from [40], and the Hitachi Energy Gerdau case [43]. Years are in-service years, which can differ by a year from the contract years in supplier lists. Positive range entries denote capacitive output and negative entries inductive absorption. Gerdau is a STATCOM installation, not a TSC-only bank; its 32 MVA converter and 0 to 64 Mvar station range illustrate the distinction between converter and complete-station ratings.

Table 6.5. Selected historical shunt-compensation installations (SVC and STATCOM).

Installation	Year	Voltage	Range	Configuration
Radsted, Denmark	2007	132 kV	+80.2 to -65 Mvar	2 TCRs, 2 filters; 12-pulse
Greenbank and South Pine, Australia	2008	275 kV	+250 to -100 Mvar; +350 Mvar one-hour overload	1 TCR, 2 TSCs, 3 filters
Devers, California, USA	2006	525 kV equipment; nominal 500 kV system, 60 Hz	SVC +330 to -110 Mvar; +440 Mvar including 110 Mvar MSC	2 TCRs, 3 TSCs, 2 filters; separate MSC
Strathmore, Australia	2007	275 kV	+180 to -80 Mvar; +260 Mvar overload in supplier list	SVC; detailed branch count not stated here
Bom Jesus da Lapa II, Brazil	2002	500 kV	+250 to -250 Mvar	2 TCRs, 1 TSC, 2 filters
Islington, New Zealand	2010	220 kV	+150 to -75 Mvar	1 TCR, 1 TSC, 1 filter
Gerdau, Charlotte, USA	2006	13.2 kV furnace bus	0 to +64 Mvar	STATCOM; 32 MVA three-level NPC VSC with passive compensation

6.6 The Static Synchronous Compensator (STATCOM)

6.6.1 Principle: a synchronous condenser without rotating mass

A STATCOM controls the AC voltage synthesized by a voltage-source converter behind a coupling reactance. It can exchange reactive power without a large switched AC capacitor or reactor bank, but it still requires coupling inductance, DC capacitors and, depending on its topology, filters (Figure 6.36). Its fundamental-frequency voltage-source model resembles a synchronous condenser: raising converter voltage above the bus voltage supplies capacitive vars, and lowering it absorbs vars. The dynamic behavior, stored energy and fault-current capability differ substantially from a rotating condenser.

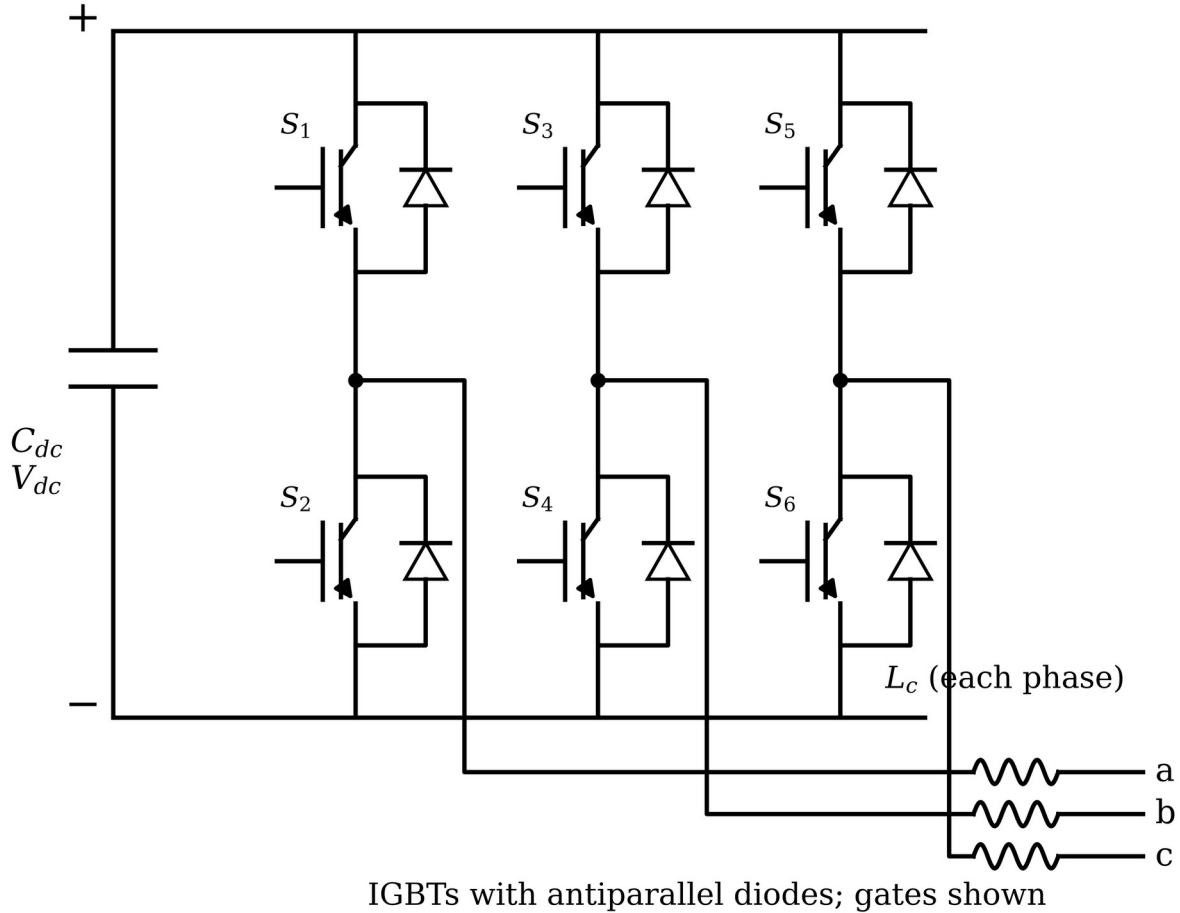


Figure 6.36. Two-level three-phase IGBT STATCOM bridge with a common DC capacitor, antiparallel diodes and one coupling inductance in each phase. Gate connections and phase outputs are shown; crossings without dots are not connections.

A static synchronous generator (SSG) can exchange sustained active and reactive power when its DC side includes an energy source or store. A capacitor-only STATCOM supplies reactive current and draws active power to cover losses; brief real-power exchange is possible within its capacitor energy window [22]. Voltage-source topologies suit available turn-off devices and capacitor-based DC voltage support, whereas current-source topologies require different reverse-blocking and energy-storage arrangements. Neither topology is inherently lossless. A charged DC capacitor establishes the voltage reference but does not replace surge arresters, valve protection, pre-charge or fault handling.

6.6.2 Current and power exchange

Refer the converter and bus phasors to the same side of the transformer and take the current positive from converter to bus. With $V = V \angle 0$, $V_c = V_c \angle \delta_c$ and a lossless coupling reactance X ,

$$I = \frac{V_c \angle \delta_c - V}{jX} \quad (6.52)$$

where I is the current phasor, V and V_c the RMS bus and converter phase voltages and δ_c the angle by which the converter voltage leads the bus voltage. The complex power delivered to the bus, $S = VI^*$, gives

$$P = \frac{VV_c}{X} \sin \delta_c, Q = \frac{V(V_c \cos \delta_c - V)}{X} \quad (6.53)$$

where P and Q are the active and reactive power delivered to the bus, per phase in SI units or three-phase in per unit. For $\delta_c = 0$ the exchange is purely reactive:

$$I_Q = \frac{V_c - V}{X}, Q = \frac{V(V_c - V)}{X} = \left(\frac{V_c}{V} - 1 \right) \frac{V^2}{X} \quad (6.54)$$

where I_Q is the reactive current, positive for capacitive injection. If $V_c > V$ the converter supplies capacitive vars and the current it delivers lags the bus voltage by 90° ; seen from the bus it draws a leading current, exactly like a capacitor. If $V_c < V$ it absorbs vars like a reactor (Figure 6.37). Because X is small, typically 0.1 to 0.15 pu, a change of only 10 to 15% in V_c swings the output across its full range. A small, deliberate phase shift δ_c makes the converter absorb active power from the bus ($\delta_c < 0$), which charges the DC capacitor and covers the converter losses.

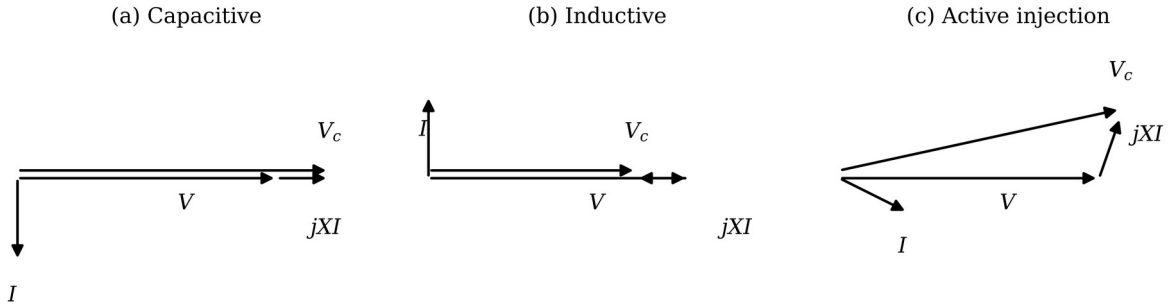


Figure 6.37. Converter-to-bus phasor relation $V_c = V + jXI$: (a) capacitive injection; (b) inductive absorption; (c) nonzero phase angle with active injection. Current vectors use a separate illustrative scale.

Example 6.9: STATCOM reactive exchange.

A STATCOM is connected to a 132 kV bus through $X = 0.15$ pu on 100 MVA. Find (a) I_Q and Q for $V_c = 1.08$ pu, $V = 1.0$ pu; (b) the current in amperes; (c) the mode of operation; (d) the unconstrained response if V falls to 0.92 pu with V_c unchanged; (e) the actual response with $I_{max} = 1.0$ pu; (f) the converter voltage for 0.333 pu inductive; (g) the angle that supplies 1% losses.

Solution. (a) Equation (6.54): $I_Q = 0.08/0.15 = 0.533$ pu; $Q = 0.533$ pu = 53.3 Mvar.

(b) $I_B = 100/(\sqrt{3} \times 132) = 437.4$ A; $I = 233.3$ A.

(c) $V_c > V$: capacitive. Writing $Q = -53.3$ Mvar from $(1 - V_c/V)$ gives the power absorbed by the converter, not the power delivered to the bus.

(d) $I_Q = 1.067$ pu, $Q = 0.981$ pu.

- (e) The limiter holds 1.0 pu: $V_c = 0.92 + 0.15 = 1.07$ pu, $Q = 0.92$ pu = 92 Mvar.
- (f) $V_c = 1 - 0.333 \times 0.15 = 0.95$ pu.
- (g) $\sin \delta_c = 0.01 \times 0.15 / 1.08$, so $\delta_c = -0.080^\circ$ (converter lagging, absorbing).

Note. The sign of reactive power follows the voltage difference across the coupling reactance. Before substituting numbers, state the current direction and the sign convention for injected reactive power. This prevents the most common STATCOM phasor-model error.

6.6.3 Instantaneous power balance and the role of the DC capacitor

For balanced sinusoidal converter-terminal voltages and currents exactly in quadrature with those voltages, $p = v_{ca}i_a + v_{cb}i_b + v_{cc}i_c = 0$ at every instant. The individual phase powers are nonzero and sum to zero; energy is exchanged through the three-phase circuit. A common DC capacitor is therefore not required to buffer the full balanced fundamental reactive-power rating. It supports DC voltage and buffers switching ripple, losses, unbalance and transient real-power mismatch. During a transient, energy also changes in the coupling inductance. In chain-link topologies, individual floating capacitors still experience phase/arm ripple even when aggregate three-phase power is zero (Section 6.7.2). A nominal ± 100 Mvar dynamic range describes 200 Mvar of variation; AC voltage, current and overload limits still set the equipment rating.

6.6.4 V-I characteristic and operating area

Under voltage control the STATCOM follows the same droop law as the SVC, Equation (6.34), within its range. At the limits the difference appears (Figure 6.38). The converter current is bounded by device ratings, not by a susceptance, so the limits are *vertical* lines on the V-I plane: the STATCOM can deliver its full capacitive current down to very low bus voltages, limited eventually by its ability to stay synchronized and keep its DC voltage. An SVC at its limit is a fixed susceptance whose current falls in proportion to voltage.

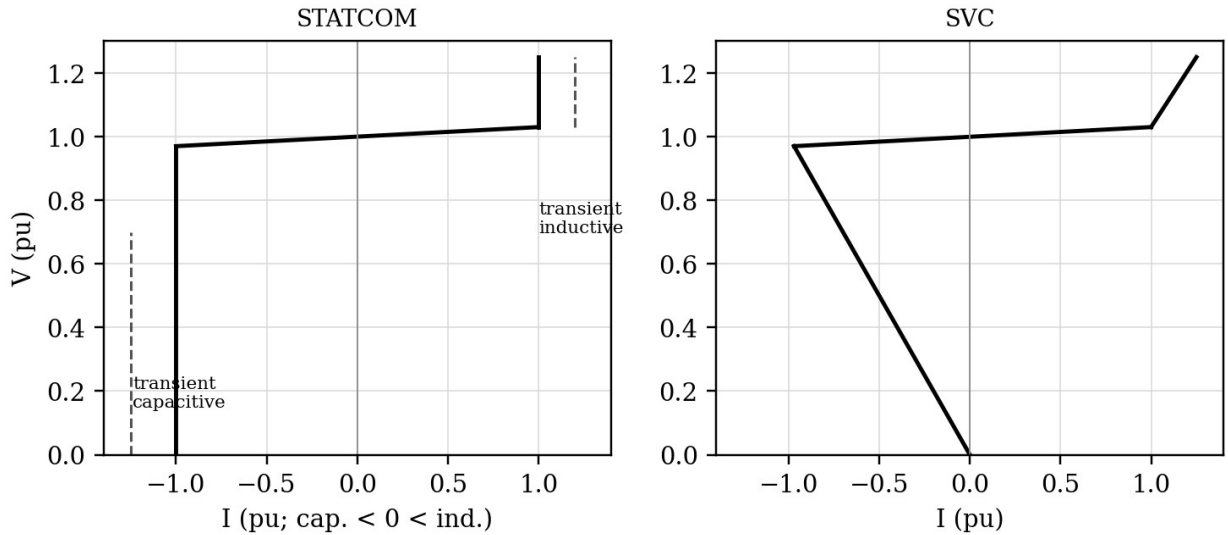


Figure 6.38. V-I characteristics: STATCOM with constant-current limits and transient ratings (left); SVC with susceptance limits (right).

Short-time capacitive and inductive overload limits depend on topology, switching pattern, transistor and diode duty, junction temperature, cooling and protection. Turn-off current capability made some earlier converters stronger in one direction than the other, but modern PWM and multilevel STATCOMs show no general asymmetry by the sign of Q , and the manufacturer's complete V-I envelope should be used. Under purely reactive operation, the ideal current-limited comparison is

$$Q_{cap,max} = V I_{max}, Q_{SVC,max} = B_{C,max} V^2 \quad (6.55)$$

where I_{max} is the converter current limit. Figure 6.40 compares the two laws on equal nominal ratings. At $V=0.7$ pu the STATCOM supplies 70% of rated vars and the SVC 49%; at 0.5 pu the ratio is two to one. The comparison is made at the respective capacitive limits. It does not mean that a STATCOM delivers constant Mvar during a voltage collapse: its current stays at the limit, so its output falls in proportion to V , while the SVC output falls with V^2 . The current limit also implies a circular P-Q capability, $P^2 + Q^2 \leq (V I_{max})^2$, but sustained operation away from the Q axis requires an energy source: a capacitor-only STATCOM lives on the narrow locus $P \approx -P_{loss}$. Figure 6.39 shows how the circle contracts with voltage. When a battery or supercapacitor is added, active and reactive services compete for the same semiconductor current, so a controller needs a priority rule during disturbances: voltage-support priority, active-power priority or a weighted allocation (Section 6.7.3).

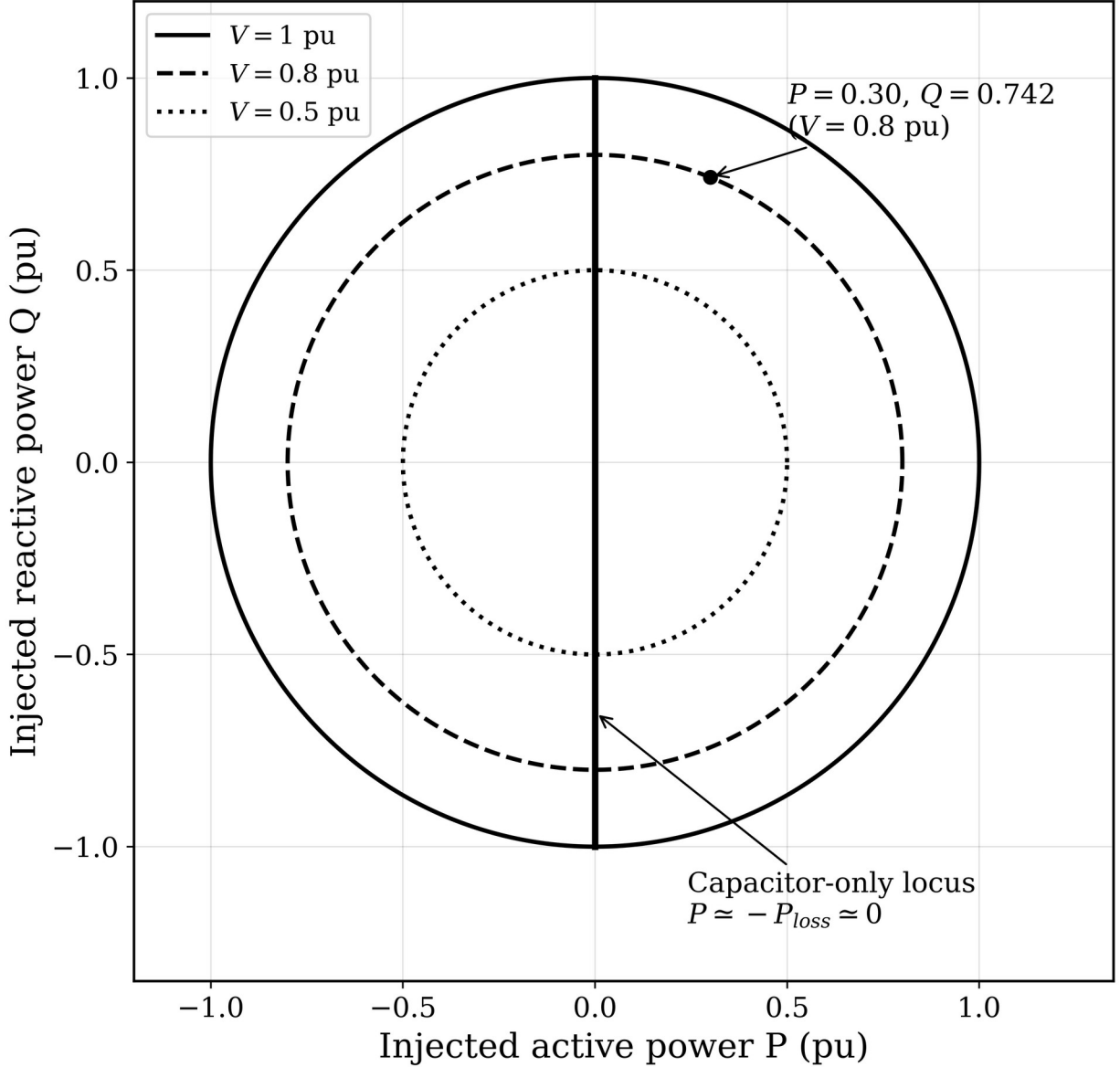


Figure 6.39. Current-limited P-Q capability of a VSC-based STATCOM at three bus voltages. The thick vertical line is the operating locus of a capacitor-only STATCOM.

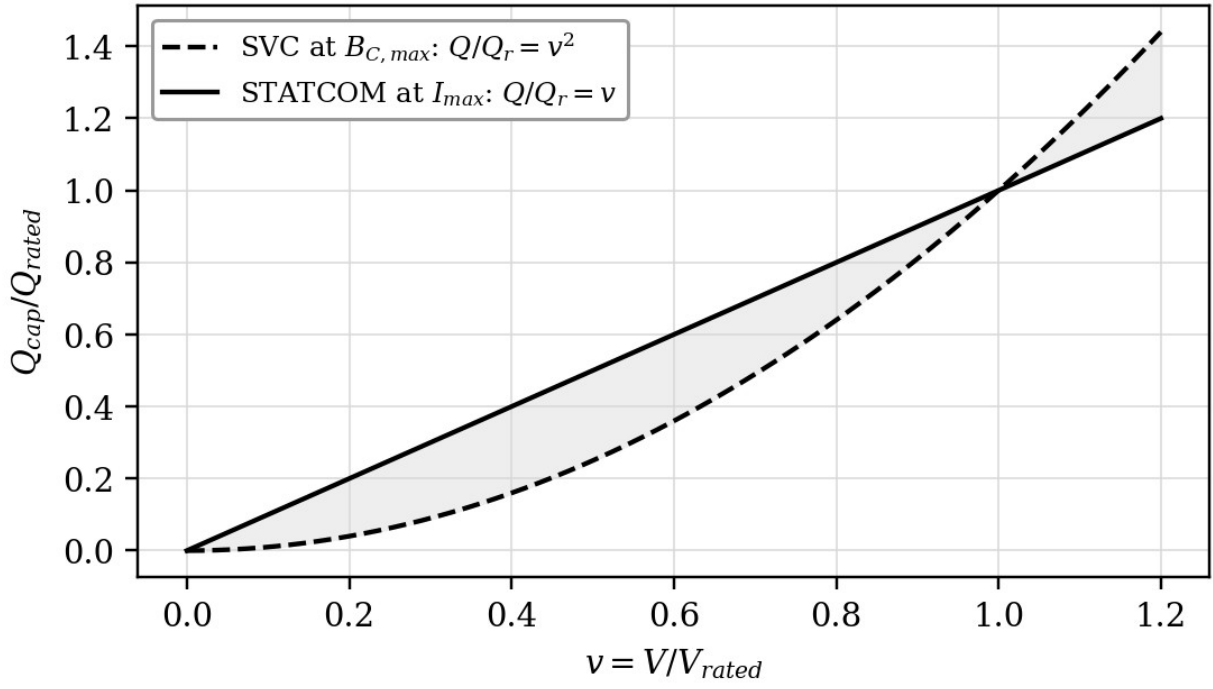


Figure 6.40. Maximum capacitive output versus bus voltage for SVC ($\propto V^2$) and STATCOM ($\propto V$) of equal rating.

6.6.5 Converter configurations

Two-level six-pulse converter

Operated with one switching per cycle (square-wave or six-step operation), each pole of the two-level converter connects its phase alternately to the positive and negative DC rails for 180° . The resulting phase-to-neutral voltage of the three-phase bridge is the six-step wave, whose Fourier series is

$$v_{cn}(t) = \frac{2V_{dc}}{\pi} \left(\sin \omega t + \frac{1}{5} \sin 5 \omega t + \frac{1}{7} \sin 7 \omega t + \frac{1}{11} \sin 11 \omega t + \dots \right) \quad (6.56)$$

where V_{dc} is the DC voltage and the harmonics present are $n=6k \pm 1$ with amplitudes V_1/n . The fundamental is fixed by V_{dc} , so in six-step operation the AC voltage, and with it the reactive output, can only be changed by changing V_{dc} . That is the indirect control method of Section 6.6.6.

Multi-pulse converters

Combining $p/6$ six-pulse converters whose outputs are displaced by $360^\circ/p$ through phase-shifting transformers cancels all harmonics except $n=kp \pm 1$, with amplitudes still V_1/n . The voltage THD is therefore

$$THD_v = \sqrt{\sum_{k=1}^{\infty} \left[\frac{1}{(kp-1)^2} + \frac{1}{(kp+1)^2} \right]} \quad (6.57)$$

Equation (6.57) gives 31.1%, 15.2%, 7.6% and 3.8% for 6, 12, 24 and 48 pulses. Each doubling approximately halves this ideal voltage THD. Retaining only two harmonic pairs underestimates it by about 12%. For a sinusoidal bus and inductive coupling with harmonic reactance nX , $I_n = V_n/(nX) = V_1/(n^2 X)$. The examples use $V_1=1$ pu and rated fundamental current as the current base. Actual current harmonics depend on converter voltage, background bus harmonics and the frequency-dependent coupling. Early transmission STATCOMs used multi-pulse transformers; Table 6.6 contrasts them with multilevel designs.

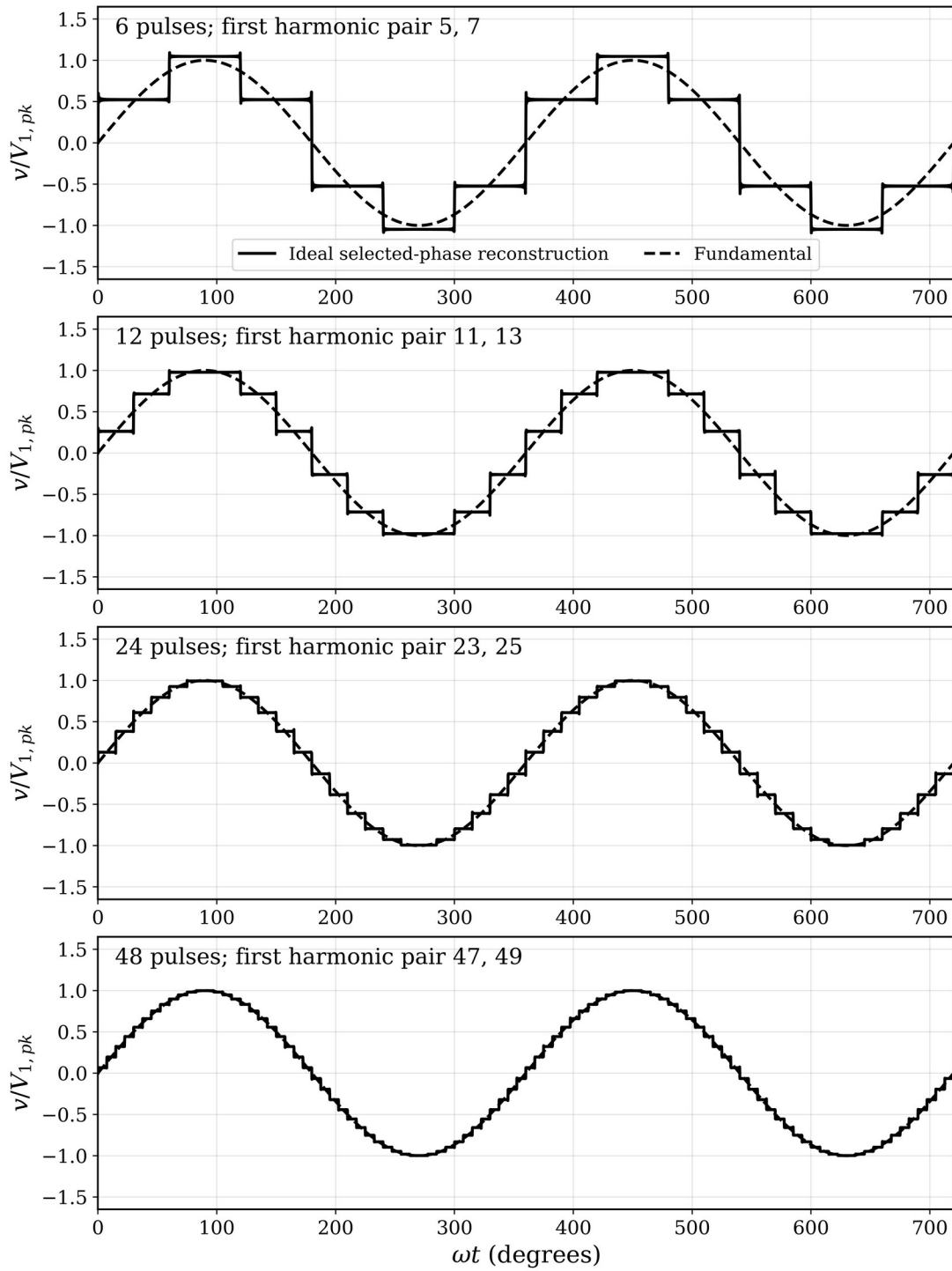


Figure 6.41. Analytical reconstructions of ideal 6-, 12-, 24- and 48-pulse voltage spectra. Harmonic magnitudes are V_1/n for $n = kp \pm 1$; positive sine phases are selected and 99 pairs retained. These illustrate spectral cancellation, not a unique transformer or gate implementation. Dashed curve is the fundamental.

Three-level, PWM and multilevel converters

A three-level neutral-point-clamped (NPC) converter adds a zero level to each pole voltage. Holding the zero level for an interval 2β around each zero crossing gives the quasi-square wave of Figure 6.42, whose fundamental amplitude

$$V_1 = \frac{4}{\pi} \cdot \frac{V_{dc}}{2} \cos \beta \quad (6.58)$$

can be varied through β at constant DC voltage. This makes direct control possible without PWM. Pulse-width modulation achieves the same with a two-level bridge: the fundamental equals $m_a V_{dc}/2$ per pole in the linear range ($m_a \leq 1$), and harmonics are pushed to sidebands of the carrier frequency, at the cost of switching losses.

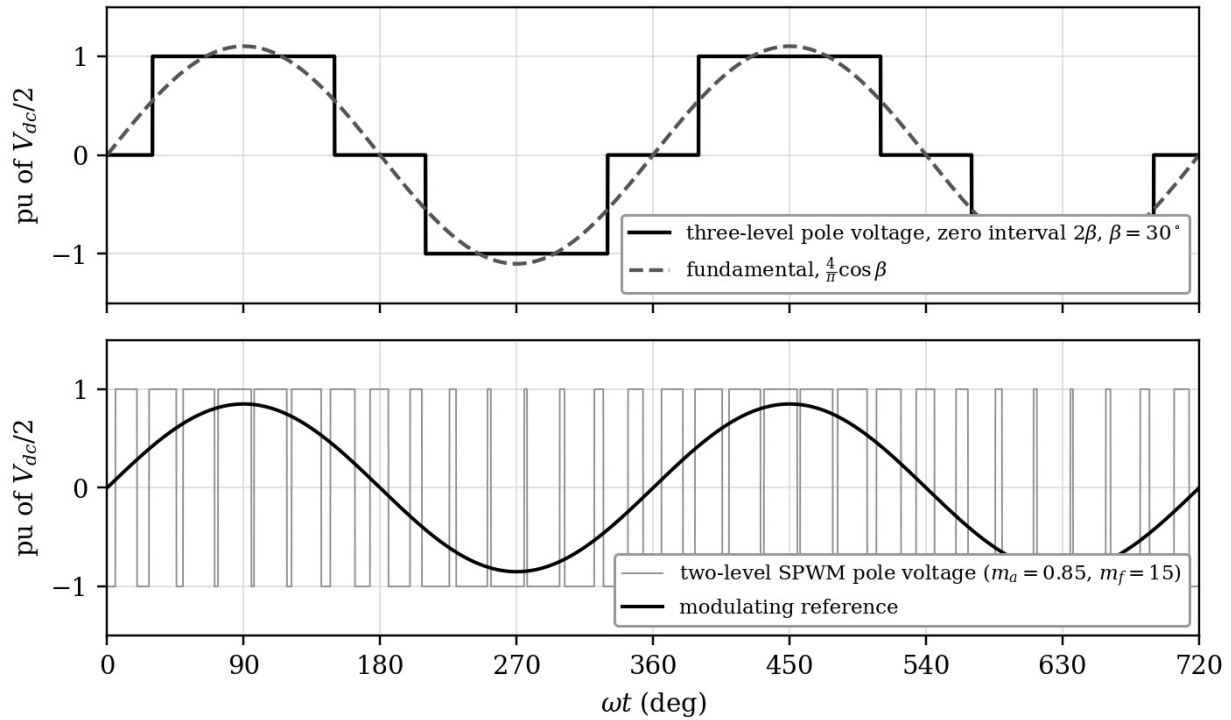
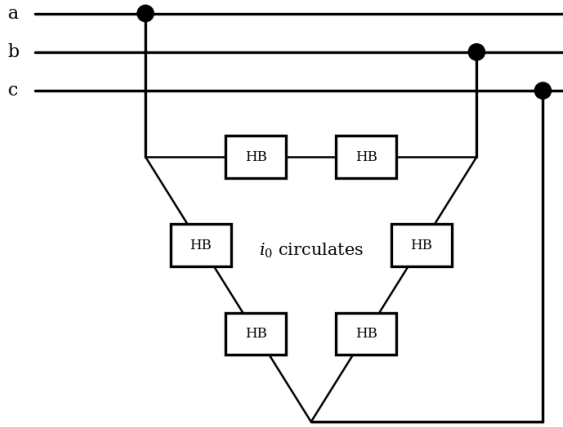
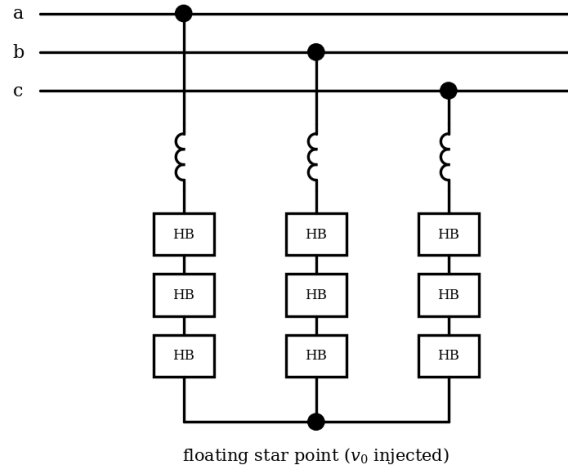


Figure 6.42. Three-level pole voltage with zero interval 2β (top) and two-level sinusoidal PWM (bottom).

Cascaded H-bridge (CHB) and modular multilevel designs, developed in Chapter 5, are widely used in high-power STATCOMs. CHB clusters are series chains of H-bridge cells with floating capacitors; double-star MMCs have upper and lower arms and require a different internal-energy model. Do not treat these arrangements as identical. Akagi's MMCC classification [14] distinguishes single-star bridge-cell (SSBC) and single-delta bridge-cell (SDBC) arrangements without an external common DC link. The delta path permits balancing circulating current under unbalanced duty; the star uses a floating-neutral voltage. The external voltage-source model remains useful, but cell and cluster energies are additional internal states (Section 6.7.2). Multilevel conversion can reduce filter duty; residual harmonics and resonance must still be assessed.



(a) SDBC: single-delta chain-link



(b) SSBC: single-star chain-link

Figure 6.43. Chain-link STATCOM clusters without a common DC link: (a) single-delta bridge-cell (SDBC), (b) single-star bridge-cell (SSBC). HB denotes an H-bridge cell with its own capacitor; cluster inductors are omitted in (a).

Table 6.6. STATCOM converter topologies.

Topology	Switching	Lowest harmonics	Control of V_c	Remarks
Two-level, six-step	Fundamental	5, 7	Indirect (V_{dc})	Needs heavy filtering; basis of multi-pulse
Multi-pulse (12 to 48)	Fundamental	$k p \pm 1$	Indirect	Complex magnetics; early transmission STATCOMs
Three-level NPC	Fundamental or PWM	Depends on β or carrier	Direct (β or m_a)	Neutral-point balancing needed
Two-level PWM	Carrier (1 to 3 kHz)	Carrier sidebands	Direct (m_a)	Low-voltage and distribution STATCOMs
CHB / MMC	Low per cell, high effective	Depends on modulation and cell count; assess complete spectrum	Direct	Modular, scalable, redundant cells; cell balancing control

Example 6.10: Multi-pulse converter harmonics.

For ideal fundamental-frequency-switched converters with harmonic magnitudes $V_n = V_1/n$, $V_1 = 1.0$ pu and lossless coupling reactance $X = 0.2$ pu at fundamental frequency, find for 6, 12, 24 and 48 pulses (a) the lowest orders; (b) voltage THD; (c) THD retaining two pairs; (d) lowest current harmonics as percentages of rated current; (e) harmonic RMS current as a percentage of rated fundamental current. Assume a sinusoidal bus and $X_n = n X$.

Solution. (a) 5, 7; 11, 13; 23, 25; 47, 49.

(b) Equation (6.57): 31.08%, 15.22%, 7.57%, 3.78%.

(c) 27.31%, 13.29%, 6.60%, 3.30%; truncation underestimates by about 12%.

(d) $I_n = 1/(n^2 X)$: 20.0% and 10.2%; 4.13% and 2.96%; 0.95% and 0.80%; 0.23% and 0.21%.

(e) 23.2%, 5.28%, 1.29%, 0.32%.

Note. Multi-pulse conversion reduces characteristic low-order harmonics by phase-shift cancellation, but it requires transformer complexity. Modern MMC-based transmission STATCOMs achieve waveform quality through a different mechanism, using many voltage levels and distributed submodule capacitors.

6.6.6 Control of the STATCOM

Internal and external control

The control system has two layers. The *external* (system) control decides what the STATCOM should do: regulate bus voltage along a droop, hold a reactive reference, damp oscillations or balance phases. It produces a reactive current reference $I_{Q,ref}$. The *internal* control makes the converter produce that current. It synchronizes with the AC system through a PLL, computes the required magnitude and angle of the converter voltage, and generates the gating pattern for the switches while keeping devices within their voltage and current limits. With its internal control in place, the converter behaves to the outside world as a controllable sinusoidal voltage behind the coupling reactance.

Indirect control (Type 2 converter)

When the converter operates with fixed modulation, as in six-step and multi-pulse designs, $V_c = k V_{dc}$ with k constant. The only handle is the angle δ_c . To increase capacitive output, the controller momentarily makes δ_c negative. By Equation (6.53) the converter then absorbs active power, the DC capacitor charges, V_{dc} and hence V_c rise, and Q increases. Once the target is reached, δ_c returns to the small negative value that just covers losses. The reverse sequence lowers V_{dc} . Figure 6.44 simulates this behaviour. The response is limited by the energy that must flow into or out of the capacitor (Example 6.33), and it is typically a few cycles.

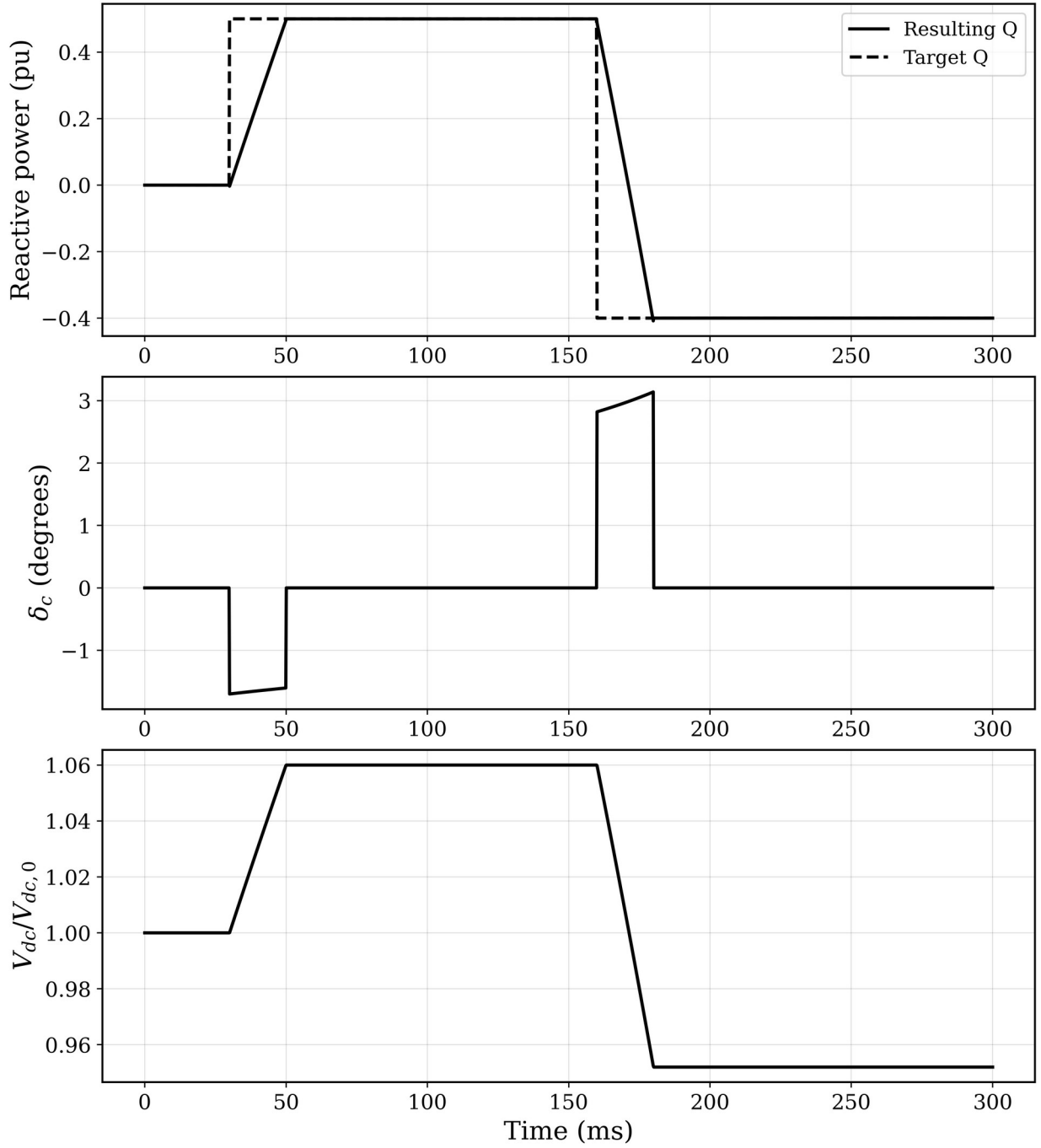


Figure 6.44. Illustrative indirect-control energy trajectory: initial stored energy 4 MJ on 100 MVA, $X = 0.12$ pu, prescribed 20 ms charging and discharging ramps in capacitor energy. Converter magnitude follows $\sqrt{(W/W_0)}$; angle follows AC power needed for dW/dt . Losses and inductor transients are neglected. This is an analytical illustration, not a measured controller response.

Direct control (Type 1 converter)

With PWM, three-level or multilevel converters, the magnitude of V_c can be varied directly at constant V_{dc} , through m_a or the zero interval β . A separate DC-voltage loop holds V_{dc} by adjusting δ_c , and therefore the small active current, while the reactive loop acts on the magnitude. Response is limited only by the current-control bandwidth, often within one cycle.

dq model and decoupled current control

In a synchronously rotating frame aligned with the positive-sequence bus voltage ($v_q=0$), with resistance R and inductance L of the coupling and the amplitude-invariant transformation, the converter current obeys

$$L \frac{di_d}{dt} = v_{cd} - v_d - Ri_d + \omega Li_q, L \frac{di_q}{dt} = v_{cq} - v_q - Ri_q - \omega Li_d \quad (6.59)$$

where i_d, i_q are the d and q components of the converter-to-bus current, v_{cd}, v_{cq} those of the converter voltage and v_d, v_q those of the bus voltage. The cross terms $\pm \omega Li$ couple the axes (Figure 6.45). The power delivered to the bus is

$$P = \frac{3}{2}(v_d i_d + v_q i_q), Q = \frac{3}{2}(v_q i_d - v_d i_q) \quad (6.60)$$

so with $v_q = 0$, i_d controls active power and i_q reactive power, and capacitive injection corresponds to negative i_q .

Converter-to-bus current; voltage-aligned rotating frame

$$L \dot{i}_d = v_{cd} - v_d - Ri_d + \omega Li_q$$

$$L \dot{i}_q = v_{cq} - v_q - Ri_q - \omega Li_d$$

Figure 6.45. Signed dq coupling model for converter-to-bus current. The $\pm \omega Li$ terms arise from frame rotation; they are not additional physical inductors.

The decoupled controller (Figure 6.46) computes

$$v_{cd}^* = v_d + Ri_d - \omega Li_q + u_d, v_{cq}^* = v_q + Ri_q + \omega Li_d + u_q \quad (6.61)$$

The PI outputs u_d and u_q act on $\dot{i}_d - i_d$ and $\dot{i}_q - i_q$. With the full resistance and speed feedforward in Equation (6.61), the decoupled plants are $L di_d/dt = u_d$ and $L di_q/dt = u_q$. They are integrators, rather than stable first-order lag plants. If resistance feedforward is omitted, each plant is $1/(Ls + R)$. For the ideal integrator and $u = K_p e + K_i \int e dt$, the closed-loop characteristic is $Ls^2 + K_p s + K_i$; a useful starting point is $K_p = 2\zeta \omega_n L$, $K_i = \omega_n^2 L$, before adding delay and sampling. The AC-voltage loop must produce negative $\dot{i}_q$ for a positive capacitive-support error. With converter-to-bus current positive, a DC-energy deficit must produce negative $\dot{i}_d$ to absorb charging power. The limiter is connected before both current regulators, enforces $\dot{i}_d^2 + \dot{i}_q^2 \leq I_{max}^2$ and returns the applied limits to outer-loop anti-windup. Active or reactive priority depends on the duty and grid requirements; DC-energy protection must always remain effective. Figure 6.46 makes these signs explicit. The instantaneous power theory [15] provides related stationary-frame tools for harmonics and unbalance.

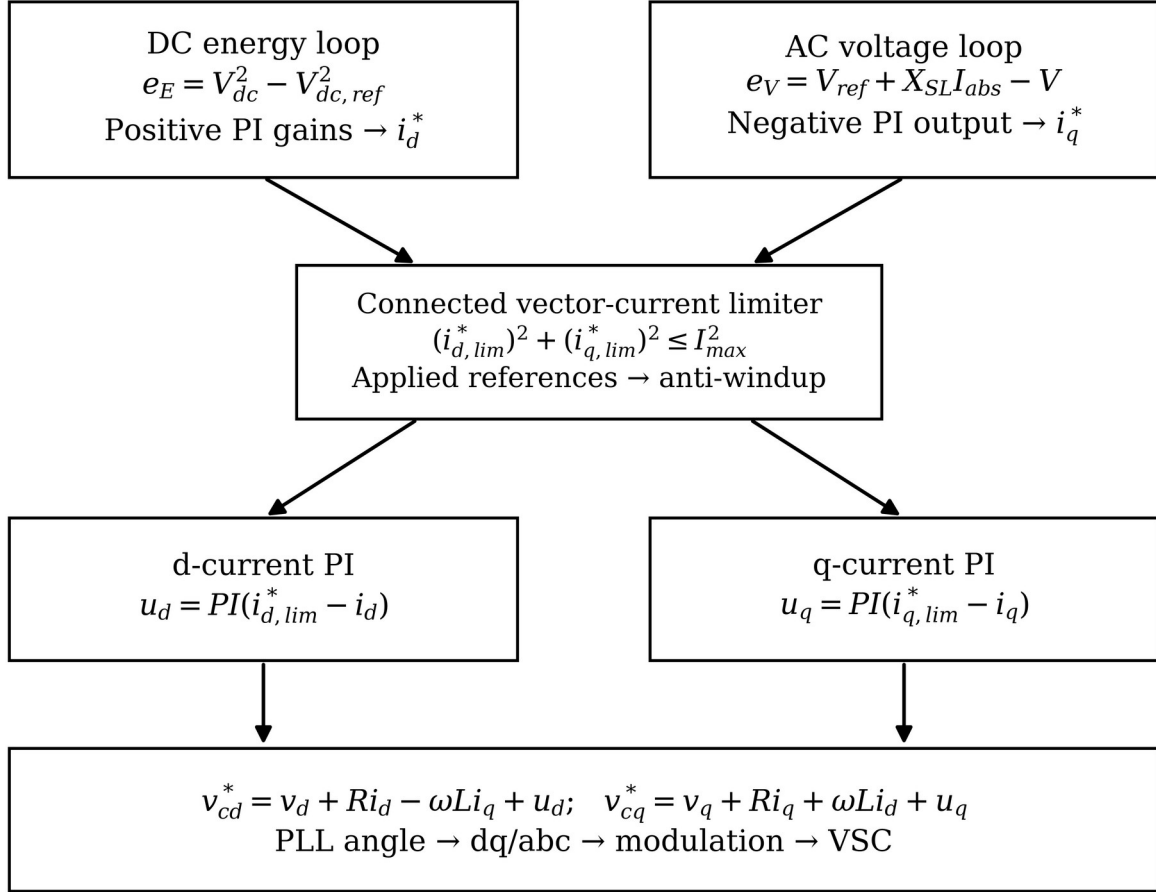


Figure 6.46. Connected current-limited dq control with explicit DC-energy and AC-voltage signs. Positive PI gains are used in the current loops; the voltage-loop output is negated to command capacitive i_q . Applied limits must feed anti-windup.

Example 6.11: dq quantities and converter voltage.

In per unit (amplitude-invariant dq, with consistent peak voltage and current bases so that $P = v_d i_d$ and $Q = -v_d i_q$ at $v_q = 0$), a converter has $v_d = 1.0$, $i_d = 0.02$, $i_q = -0.8$, $R = 0.005$ and $X = 0.12$. The secondary is 6.6 kV line-to-line. Treat the electrical current state as steady; an external DC energy source supports positive active-power delivery. A capacitor-only STATCOM could maintain it only transiently. Find (a) P and Q; (b) operating mode; (c) converter dq voltages; (d) converter magnitude and angle; (e) minimum DC voltage for two-level SPWM; (f) for ideal third-harmonic injection.

Solution. (a) $P = 0.02$ pu delivered to the bus and $Q = 0.8$ pu. (A STATCOM drawing its own losses from the bus would show a small negative i_d .)

(b) $i_q < 0$: capacitive.

(c) From Equation (6.59) in steady state: $v_{cd} = 1 + 0.005(0.02) - 0.12(-0.8) = 1.0961$; $v_{cq} = 0.005(-0.8) + 0.12(0.02) = -0.0016$.

(d) 1.0961 pu at -0.08° .

(e) $V_{pk} = 6.6\sqrt{2}/\sqrt{3} = 5.389$ kV; converter peak 5.907 kV; $V_{dc} \geq 2 \times 5.907 = 11.81$ kV.

(f) $V_{dc} \geq \sqrt{3} \times 5.907 = 10.23$ kV.

Note. The dq equations are most useful when each term is given a physical meaning. The cross-coupling terms arise from the rotating reference frame and should not be mistaken for additional physical inductors. Feedforward decoupling improves current-loop behaviour but does not remove converter limits.

6.6.7 Sizing the DC capacitor

The DC capacitor is sized from *real*-power imbalance and ripple, not from the reactive rating. The energy balance of the converter is

$$\frac{d}{dt} \left(\frac{1}{2} C V_{dc}^2 \right) = p_{ac} - P_{loss} \quad (6.62)$$

where C is the DC capacitance, p_{ac} the active power entering the converter from the AC side and P_{loss} the converter losses. Two duties determine C .

Transient deficit. If a net discharge power ΔP persists for Δt (a loss surge, a delayed DC-voltage loop, or a deliberate brief active-power exchange) while V_{dc} is allowed to fall from $V_{dc,max}$ to $V_{dc,min}$,

$$C = \frac{2 \Delta P \Delta t}{V_{dc,max}^2 - V_{dc,min}^2} \quad (6.63)$$

Unbalance ripple. With a negative-sequence bus voltage V^- and positive-sequence current I^+ , the instantaneous power contains a component at twice the line frequency of amplitude $\hat{P}_2 \approx 3 V^- I^+$ (RMS phase quantities), that is, $\hat{P}_2 \approx (V^- / V^+) S$ for a converter operating at apparent power S . Integrating it in Equation (6.62) gives a peak-to-peak ripple

$$\Delta V_{dc,pp} \approx \frac{\hat{P}_2}{\omega C V_{dc0}} \quad (6.64)$$

Equation (6.64) assumes small voltage ripple. The exact relation for sinusoidal energy exchange is $V_{dc,max}^2 - V_{dc,min}^2 = 2 P_2 / (\omega C)$, where P_2 is the peak twice-frequency power, and uncompensated DC ripple also produces AC sidebands. Sizing the capacitor from a supposed balanced three-phase reactive energy has no physical basis; for the numbers of Example 6.12 that rule gives 1.125 mF, 2.25 times the transient-deficit value.

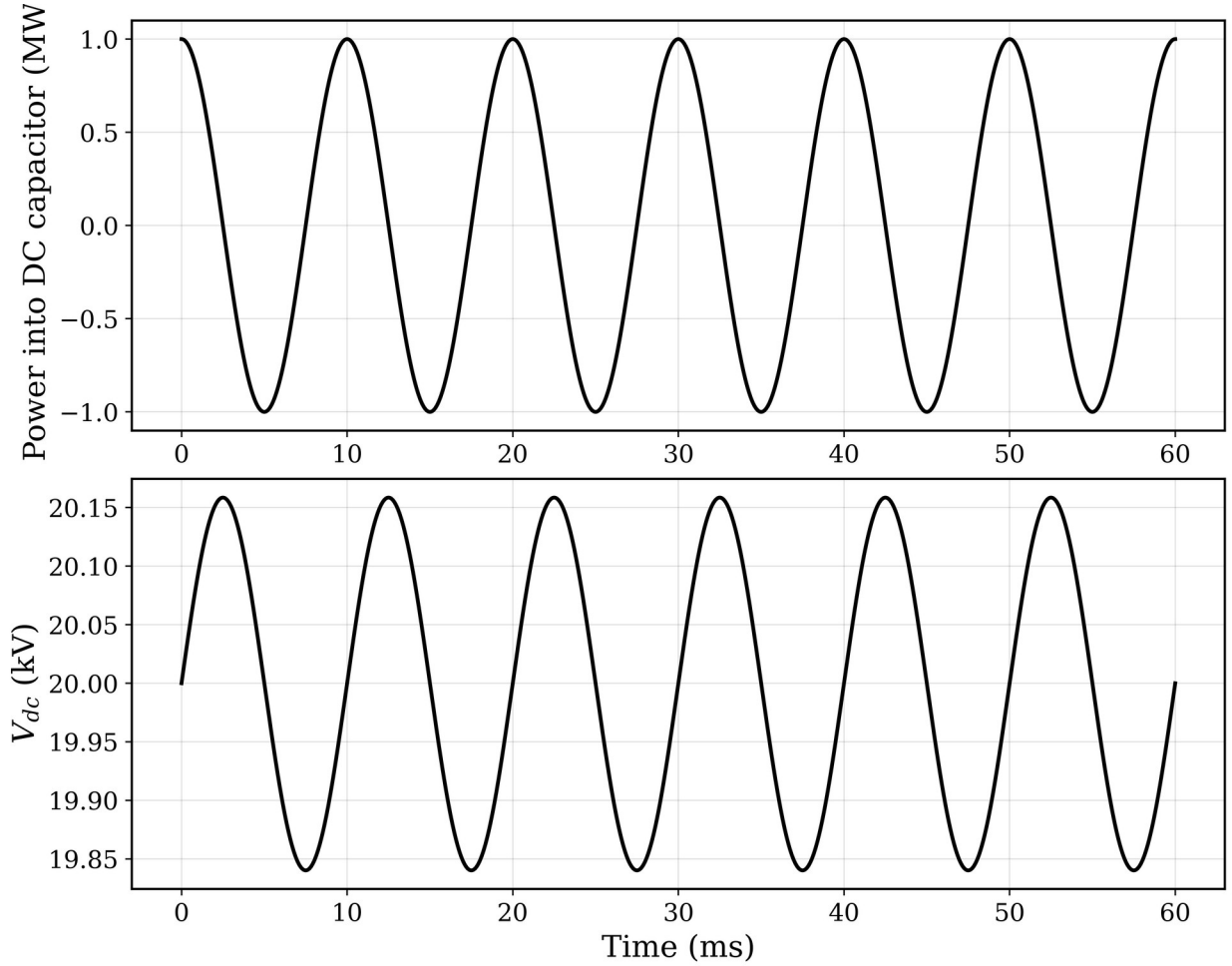


Figure 6.47. Exact squared-voltage integration of 1 MW peak twice-frequency capacitor power at 50 Hz with $C = 0.50$ mF and nominal $V_{dc} = 20$ kV (Example 6.12). The resulting ripple is approximately 318 V peak-to-peak.

For an MMC STATCOM the aggregate result is necessary but not sufficient. In a chain-link cluster each cell capacitor carries a second-harmonic energy ripple even at purely reactive output; the arms of a double-star MMC also carry a fundamental-frequency component. Cell capacitance is therefore sized from cluster or arm energy ripple (Chapter 5 and Section 6.7.2).

Example 6.12: DC capacitor sizing.

A 20 Mvar STATCOM has nominal DC voltage 20 kV. Find (a) C for a net deficit of 2 MW for 10 ms within 21 to 19 kV; (b) stored energy and its equivalent duration at 20 MW; (c) ripple caused by 5% negative-sequence voltage at rated positive-sequence current; (d) capacitance needed for ripple alone within the same window; (e) the numerical result and limitation of a hypothetical reactive-energy sizing rule.

Solution. (a) Equation (6.63): $C = 2(2 \times 10^6)(0.01)/(21^2 - 19^2) \times 10^{-6} = 0.50$ mF.

(b) $W = \frac{1}{2} C V_{dc}^2 = 100$ kJ, equivalent to 5 ms of rated output.

(c) $\hat{P}_2 \approx 0.05 \times 20 = 1$ MW. Equation (6.64): $\Delta V_{pp} = 10^6 / (314.16 \times 0.5 \times 10^{-3} \times 20000) = 318$ V, or 1.6%.

(d) $C \geq 2 \hat{P}_2 / [\omega (V_{max}^2 - V_{min}^2)] = 0.080$ mF; the transient deficit governs.

(e) Balanced fundamental reactive power does not cycle energy through the DC link. If the hypothetical energy $\sqrt{2}Q/(2\omega) = 45.016$ kJ were nevertheless treated as a usable energy excursion, then

$C = 2(45.016 \text{ kJ}) / [(21 \text{ kV})^2 - (19 \text{ kV})^2] = 1.125 \text{ mF}$, 2.25 times the 0.50 mF of part (a). The ratio is specific to these numbers; the rule itself is not a valid way to size a balanced common DC link.

Note. A STATCOM DC capacitor is not sized by multiplying reactive power by time. In balanced steady-state reactive operation the average real power into the DC link is approximately the converter loss. Energy sizing follows real-power imbalance, voltage ripple, modulation headroom and, for MMCs, internal arm and submodule energy requirements.

6.6.8 Hybrid STATCOMs and loss characteristics

The losses of a STATCOM are roughly symmetric about zero output and, at zero output, are higher than those of a TSC-TCR (Figure 6.31), because the converter switches continuously. Where the duty is asymmetric, a smaller converter can be combined with passive elements (Figure 6.48). A fixed capacitor in parallel shifts the range towards capacitive output: a $\pm 50 \text{ Mvar}$ STATCOM with a 30 Mvar capacitor spans $+80$ to -20 Mvar at rated voltage. A fixed reactor shifts it towards inductive output. A STATCOM combined with TSCs and a TCR provides a wide range with a small converter rating. The price is that the passive part re-introduces the V^2 characteristic, so the low-voltage support of the hybrid lies between that of a pure STATCOM and a pure SVC. Commercial “hybrid SVC” and “SVC with STATCOM” designs use exactly this trade-off [37]–[39].

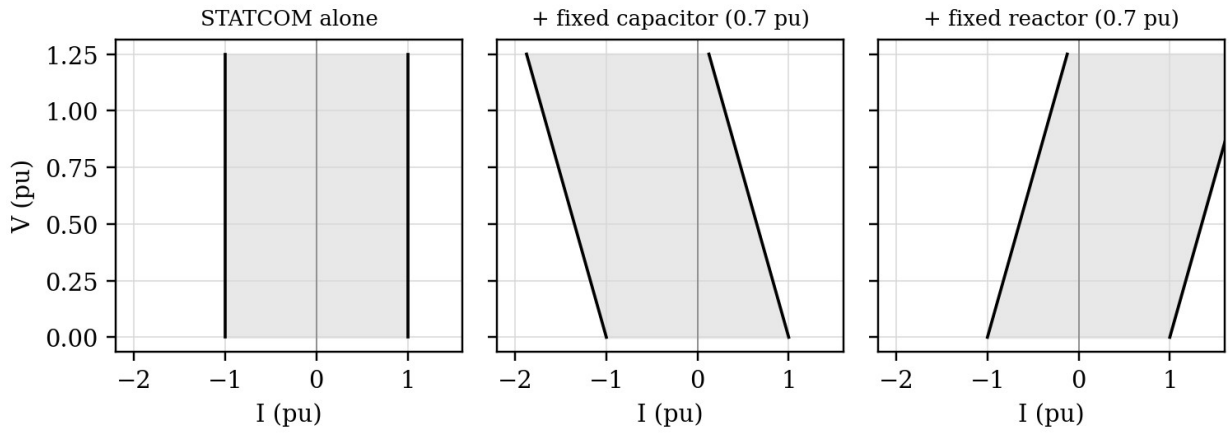


Figure 6.48. V-I operating areas of a STATCOM alone and hybridized with a fixed capacitor or fixed reactor.

6.6.9 Midpoint STATCOM with a current limit

Return to the two-machine system of Section 6.1.2 and let the midpoint compensator be a current source of fixed capacitive current I_Q , as a STATCOM is once it reaches its limit. The Thevenin equivalent at the midpoint is $V \cos(\delta/2)$ behind $X/4$ (the two half-lines in parallel), so the midpoint voltage becomes $V_m = V \cos(\delta/2) + I_Q X/4$. Substituting into the half-line power equation $P = (2 V V_m / X) \sin(\delta/2)$,

$$P = \frac{V^2}{X} \sin \delta + \frac{V I_Q}{2} \sin \frac{\delta}{2} \quad (6.65)$$

where I_Q is the STATCOM’s capacitive current. The STATCOM adds a $\sin(\delta/2)$ term to the uncompensated curve. By contrast, an SVC at its capacitive limit B_C is a fixed capacitor at the midpoint, and the same Thevenin argument gives

$$P = \frac{V^2}{X(1 - X B_C/4)} \sin \delta \quad (6.66)$$

which only scales the uncompensated curve and still peaks at $\delta = 90^\circ$. With equal ratings ($I_Q = B_C = 1 \text{ pu}$) on a line of $X = 0.5 \text{ pu}$, the STATCOM curve peaks at 2.36 pu near 95° and the SVC curve at 2.29 pu at 90° ; more importantly, the STATCOM curve remains higher at large angles, which enlarges the decelerating area available in a transient (Figure 6.3, Example 6.34).

6.6.10 Applications and control functions

Voltage regulation and support. The STATCOM's principal duty, executed through the droop law with a response limited by its current loop. Its capacitive output survives deep voltage sags far better than an SVC's.

Transient stability. During the first swing after a fault, temporarily raising V_{ref} , or simply forcing maximum capacitive current, maximizes the midpoint voltage and the decelerating area. The reference is then returned to normal.

Power oscillation damping. The reference voltage, or alternatively the reference current, is modulated in proportion to a signal derived from local frequency or line power, so that the STATCOM raises the transmitted power when the sending machine accelerates and lowers it when it decelerates. Current modulation avoids interference with the voltage regulator. With an energy store, active-power modulation gives much stronger damping, which is the motivation for E-STATCOMs (Section 6.7).

Var reserve control. A slow outer loop trims V_{ref} so that, in steady state, the STATCOM sits near zero output and the mechanically switched capacitors and reactors in the area carry the slowly varying base load. The full dynamic range then remains in reserve for disturbances.

Flicker and unbalance. Railway traction loads, which are large, single-phase and fluctuating, are a major application in Europe and Asia. STATCOMs with negative-sequence current control restore voltage balance and suppress flicker far more effectively than SVCs because of their bandwidth.

Case history: TVA Sullivan. The ± 100 MVA STATCOM commissioned in 1995 at the Sullivan substation of the Tennessee Valley Authority, designed and built by Westinghouse under EPRI sponsorship, used GTO-based 48-pulse converters with indirect control. It regulated the 161 kV bus, reducing the duty of transformer tap changers in the area, and its operating record established the credibility of the technology [6].

6.6.11 STATCOM versus SVC

Figure 6.49 compares typical dynamic responses, and Table 6.7 collects the comparison. Response times must always be specified with their metric: valve delay, current-loop rise time and full voltage recovery are different quantities.

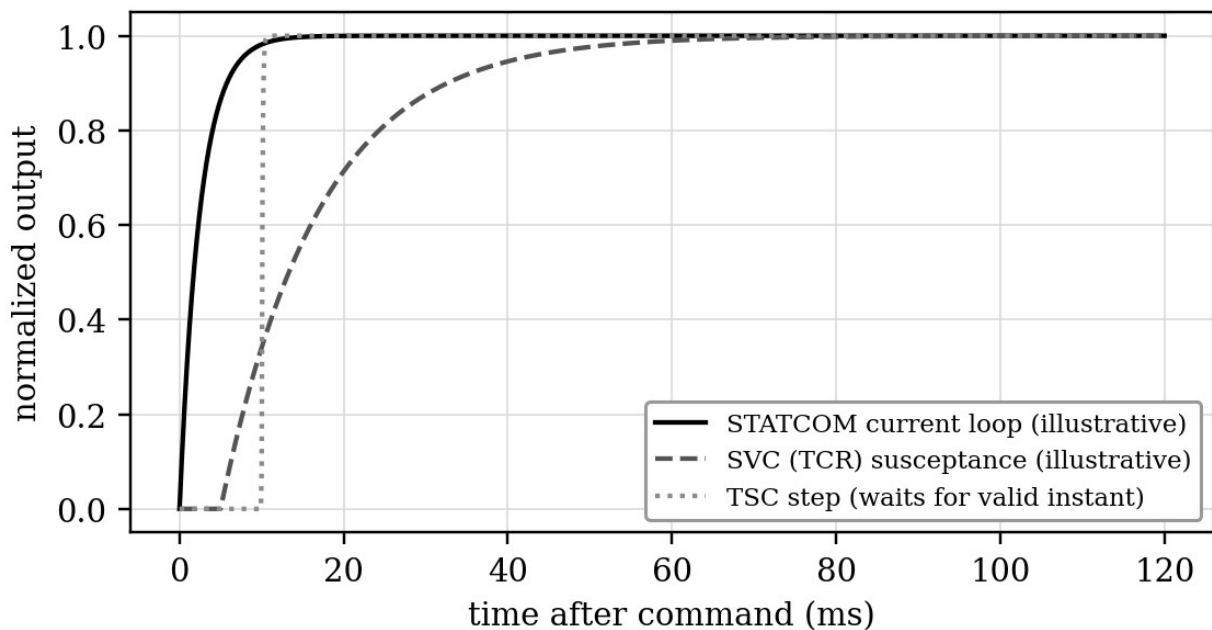


Figure 6.49. Illustrative response to a step in reactive reference: STATCOM current loop, TCR susceptance and TSC switching (stated assumptions; not a measured comparison).

Table 6.7. Comparison of SVC and STATCOM.

Attribute	SVC (TSC-TCR)	STATCOM (VSC/MMC)
Principle	Controlled susceptance	Controlled voltage source behind reactance
Output at limit	$Q \propto V^2$	$Q \propto V$ (constant current)
Low-voltage support	Collapses with voltage	Current support within synchronization, DC and modulation limits
Response	TCR/TSC firing delay plus regulator; turn-on may wait up to one cycle	Current loop within about one cycle
Harmonics	TCR harmonics, filters required	Often reduced filter duty with multilevel design; network study required
Losses at zero output	Low (TSC-TCR), high (FC-TCR)	Moderate (switching losses)
Losses at full output	Moderate	Moderate to high
Footprint	Large (capacitors, reactors, filters)	Considerably smaller for the same dynamic range
Active power capability	None	Possible with DC energy storage
Overload	Limited by capacitor/reactor ratings	Limited by semiconductor ratings, short-time only
Cost (per Mvar)	Often economical for bulk vars; project-dependent	Project-dependent; compare complete station and required duty

6.7 Recent Advances, Research Gaps and Emerging Directions

The TCR, TSC, SVC and STATCOM theory of this chapter is settled, and none of it is overturned by recent work. What has changed is the setting. Transmission systems now host large amounts of wind, solar and battery generation connected through converters, synchronous machines are being retired, and the short-circuit strength at many buses is falling. A modern shunt compensator is therefore asked to do more than hold a voltage: it may have to ride through faults in a weak grid, supply real power for a few seconds, behave as a voltage source rather than a current source, and avoid unstable interaction with neighbouring converters. This section shows, for a research student, which new states, constraints and equations enter the problem when the classical STATCOM model of Section 6.6 is extended in each of these directions. Each subsection ends with open questions; Table 6.9 at the end collects them.

Advanced reading path. Sections 6.7.2 to 6.7.6 assume symmetrical components, capacitor energy balance, the swing equation and basic feedback control. A rating study follows the same order: AC current and modulation headroom first, then stored-energy limits, then synchronization and dynamic stability. Readers meeting the material for the first time can start from the worked examples; research students should follow the internal-state derivations and the validation steps.

6.7.1 Why the problem has changed: system strength

The strength of a bus is measured by its short-circuit ratio,

$$SCR = \frac{S_{sc}}{S_{rated}}, ESCR = \frac{S_{sc} - Q_C}{S_{rated}} \quad (6.67)$$

S_{sc} is the three-phase short-circuit MVA of the network, S_{rated} the compensator or plant rating, and Q_C the fixed capacitive compensation subtracted in the conventional ESCR. A shunt capacitor changes voltage sensitivity and resonance but adds no short-circuit strength. SCR and ESCR are screening indices rather than stability criteria, and they are least reliable when several converters interact. A PLL-based grid-following STATCOM both supports and perturbs the voltage it synchronizes to, so at low system strength its PLL, outer

loops, the network impedance and nearby converters have to be analysed together; no single SCR threshold settles the question.

6.7.2 Modular multilevel STATCOMs: internal energy as a design variable

Section 6.6.5 introduced the chain-link (cascaded H-bridge) clusters used in modern STATCOMs, and Figure 6.43 showed the two arrangements without a common DC link. Each cluster is a series string of H-bridge cells with floating capacitors. A cluster is a single-phase element, so even at purely reactive output its instantaneous power pulsates. With cluster voltage $\sqrt{2}V \cos \omega t$ and current $\sqrt{2}I \sin \omega t$, the cluster power is $VI \sin 2\omega t$ and its energy swings by VI/ω peak to peak. If the cell voltages may ripple by a fraction ε peak to peak about their mean, the stored energy summed over the three clusters must satisfy

$$\frac{E_{stored}}{S} \geq \frac{1}{2\omega\varepsilon} \quad (6.68)$$

Here $S=3VI$ is the balanced reactive rating on consistent cluster bases, and $\varepsilon = \Delta V_{cell,pp}/V_{cell,mean}$. Because capacitor energy varies with voltage squared, a fractional peak-to-peak swing ε corresponds to an energy window of about $2\varepsilon E_{stored}$. Equating that window to the total cluster ripple gives Equation (6.68), whose ideal lower bound at 50 Hz and $\varepsilon=10\%$ is 15.9 kJ/MVA. Installed energy must also cover overload, negative-sequence duty, bypassed cells, tolerances and control headroom.

Cluster energy balancing under negative-sequence current. When a STATCOM supplies negative-sequence current I^- , for example to balance a traction or arc-furnace load (Section 6.5.9), the average power of each cluster is no longer zero, even though the total is. The three cluster capacitors would drift apart in voltage unless a zero-sequence component is added that redistributes power among clusters without reaching the grid. The two arrangements do this in different ways [14], [34], [35].

Positive-sequence phasors form a balanced set in the supply phase order; negative sequence has the opposite phase order. A zero-sequence quantity is identical in the three clusters. In a closed delta it can circulate internally; in a floating star its voltage shifts the neutral without changing line-to-line voltage. The superscripts $+$ and $-$ identify sequences; an asterisk in the following balance equations means complex conjugation, not a reference command. Complex conjugates arise because average power is $\Re(VI^*)$. Equations (6.69) and (6.70) assume balanced positive-sequence cluster voltage, no negative-sequence voltage, negligible losses, and zero total active-power command. Sequence currents must be expressed on the relevant cluster bases.

In the delta (SDBC) a zero-sequence current I_0 circulates inside the delta. With line-to-line cluster voltage phasor V_Δ and negative-sequence cluster current I_Δ^- , the three cluster powers balance when

$$I_0 = -\frac{V_\Delta I_\Delta^{-*}}{V_\Delta^*}, |I_0| = |I_\Delta^-| \quad (6.69)$$

Under these assumptions the magnitude result is exact for every current ratio, and the worst-case cluster current over all negative-sequence angles is $I^+ \vee I^- \vee I_0$. When negative-sequence voltage is also present, the complete cluster-power equations have to be solved and their solvability checked.

In the star (SSBC) the neutral floats, so a zero-sequence voltage V_0 is injected instead. The balance condition is

$$V_0 I^+ + V_0 I^- = -V^+ I^{-*} \frac{r}{1+r} \quad (6.70)$$

The envelope follows from the triangle inequality: the left-hand side has magnitude between $V_0 \vee I^+ \vee I^-$ and $V_0 \vee (I^+ \vee I^-)$. The upper bound $r/(1+r)$ holds for $0 \leq r < 1$, the range plotted in Figure 6.50a. At $r=1$ the balance equation is singular: some phase relations have no solution and

others have infinitely many, so the required voltage does not diverge for every phase as $r \rightarrow 1$. If $I^{+i=0i}$ and $I^- \neq 0$, the equation gives $iV_0 \vee iV^{+i \vee i}$. Severe unbalance can exceed the available cell-voltage headroom, but whether it does has to be checked from the actual phasors, cluster voltages and semiconductor limits [42].

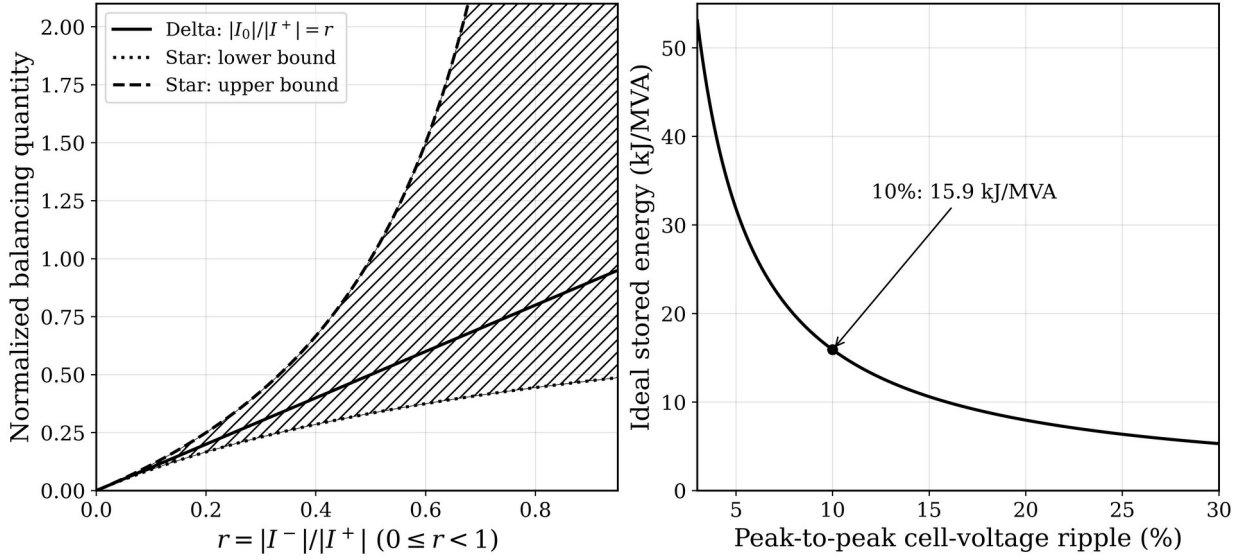


Figure 6.50. (a) Delta circulating-current magnitude and star zero-sequence voltage envelope under balanced positive-sequence voltage; the star envelope is plotted only for $0 \leq r < 1$. (b) Ideal chain-link stored-energy lower bound versus permitted peak-to-peak cell-voltage ripple at 50 Hz. Design margins are additional.

Open questions. Joint optimization of cell capacitance, losses and negative-sequence capability; energy-ripple shaping without extra semiconductor stress; fault-tolerant operation with bypassed cells while keeping cluster balance; lifetime models that couple capacitor and semiconductor ageing to the actual duty cycle.

6.7.3 Energy-storage STATCOMs: separating power and energy

Adding a battery, supercapacitor or other store to the DC side turns the STATCOM into a static synchronous generator in the sense of Section 6.6.1. Supercapacitor-equipped, grid-forming STATCOMs are now offered by several manufacturers [28], [37], [38], and Narula et al. showed the stability benefit of one at an offshore-wind connection [27]. CIGRE Technical Brochure 935 surveys the integration options, models and services [20]. Two constraints must be written separately, because they are set by different hardware:

$$P^2 + Q^2 \leq (V I_{max})^2, E(t) = E(0) - \int_0^t P_{ES}(\tau) d\tau, E_{min} \leq E(t) \leq E_{max}. \quad (6.71)$$

Positive P_{ES} withdraws energy from the store and negative P_{ES} charges it, within $-P_{ch,max} \leq P_{ES} \leq P_{dis,max}$. The current limit is written with AC-terminal P and Q on common per-unit bases, whereas the energy equation uses the actual stored energy. If P_{ES} is measured at the external storage port rather than at the stored-energy boundary, use $dE/dt = -P_{ES}/\eta_{dis}$ for discharge and $dE/dt = -\eta_{ch}P_{ES}$ for charge, with auxiliaries added separately. For a supercapacitor $E_{usable} = 1/2 C (V_{max}^2 - V_{min}^2)$, so a 2:1 voltage window uses 75% of the maximum stored energy. Power and energy ratings are separate design variables (Figure 6.51a).

Inertia-like and frequency support. If the converter injects power in proportion to the rate of change of frequency, it emulates a machine of inertia constant H on its rating S :

$$P_{inj} = -\frac{2HS}{f_0} \frac{df}{dt}, E_{del} = \frac{2HS}{f_0} |\Delta f| \text{ for a monotonic frequency fall.} \quad (6.72)$$

Positive P_{inj} is delivered to the grid. A falling frequency has $df/dt < 0$ and therefore commands positive injection; a rising frequency commands absorption. Here H is in seconds, S is the rated power base, and f_0 is nominal frequency. Equation (6.72) is a small-excursion inertia-emulation law; delivered energy is the integral of signed power. The absolute-excursion formula applies to a monotonic fall, not to an arbitrary oscillatory trajectory. Measurement filtering, activation delay, current saturation and stored-energy headroom determine achievable support. Figure 6.51b shows magnitudes.

Damping. Section 6.6.10 noted that reactive-power modulation damps oscillations only through the voltage sensitivity of loads and line power. Modulating real power acts on the swing equation directly, so a small store gives much stronger damping per MVA. Beza and Bongiorno showed an adaptive controller that extracts the oscillatory component and commands both P and Q [16]; a fair comparison must charge the damping benefit against the converter current and energy it consumes.

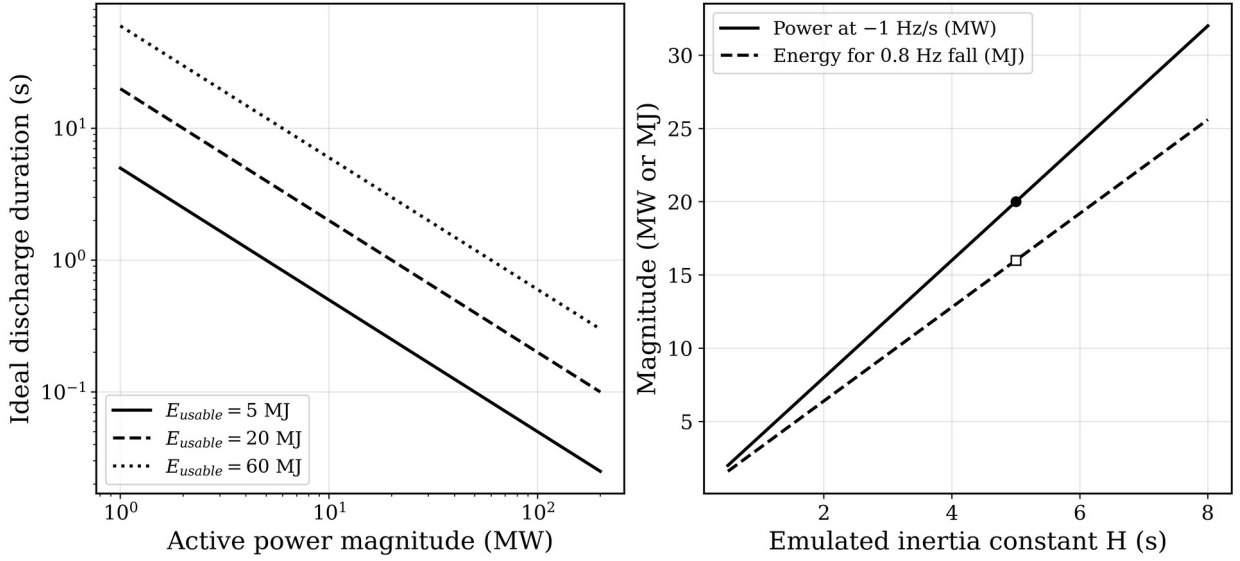


Figure 6.51. (a) Ideal duration E_{usable}/P neglecting losses and power limits. (b) Power-injection magnitude for $df/dt = -1$ Hz/s and delivered energy for a monotonic 0.8 Hz frequency fall, on a 100 MVA, 50 Hz base.

Open questions. Unified P-Q priority rules during faults that respect both constraints of (6.71); state-of-charge management that keeps reserve for the next event; degradation-aware dispatch; and metrics such as damping ratio gained per MJ of storage, which allow a fair comparison with synchronous condensers.

6.7.4 Grid-forming STATCOMs and current limiting

A grid-following controller commonly obtains its angle from a PLL and controls terminal current. A grid-forming controller maintains an internal voltage angle and magnitude and synchronizes through its interaction with the network, while respecting current and energy limits. Two common ideal laws are droop control [32],

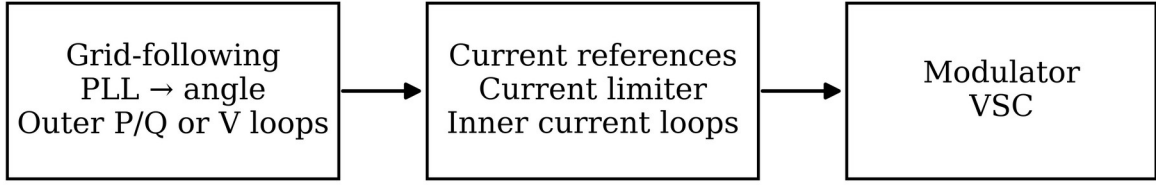
$$\omega = \omega_0 + m_p(P^* - P), E = E_0 + m_q(Q^* - Q) \quad (6.73)$$

and the virtual synchronous machine (VSM) [31], which adds inertia and damping:

$$2H \frac{d\omega_{pu}}{dt} = P_{pu}^* - P_{pu} - D(\omega_{pu} - 1), \frac{d\theta}{dt} = \omega_b \omega_{pu}, \omega_b = 2\pi f_0. \quad (6.74)$$

Equation (6.73) uses angular frequency in rad/s and compatible droop-gain units. In Equation (6.74), powers are per unit on a common S_b , $\omega_{pu} = \omega/\omega_b$, H is in seconds, D is the per-unit damping coefficient and θ is in electrical radians. For an angle relative to a nominal synchronous frame, $d\delta/dt = \omega_b(\omega_{pu} - 1)$. The constant-coefficient swing law is the usual nominal-frequency approximation. Power-synchronization control [36] and virtual-oscillator control are other approaches. With a lossless reactance, $P = EV_g \sin \delta / X$ describes synchronization only while the voltage-source model remains feasible. Grid-forming control can improve

weak-grid operation by avoiding a PLL in the primary synchronization law, but stability still depends on impedance, gains, limiting, stored energy and other devices [25], [26], [33].



Current, voltage and stored-energy limits constrain both structures.

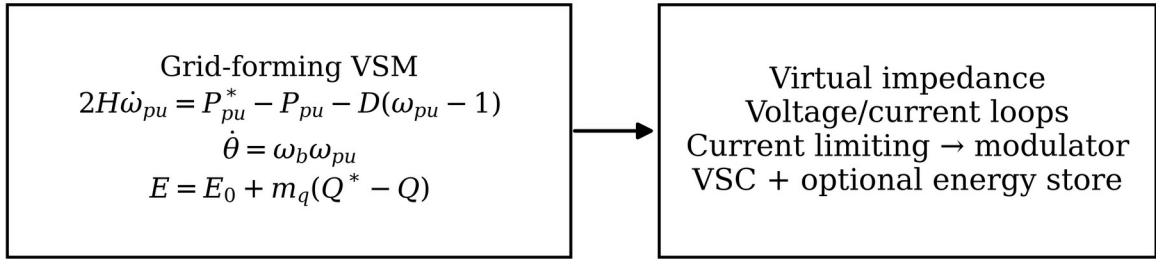


Figure 6.52. Grid-following and grid-forming STATCOM control structures. In both cases the current limiter and, for real-power services, the stored energy bound the response.

A capacitor-only STATCOM can be grid-forming for voltage and reactive power, but it cannot hold an internal frequency against the grid for long, because any sustained real-power exchange would discharge its DC link. Sustained frequency or inertia-like support therefore needs storage, which is why grid-forming and energy-storage STATCOMs are usually discussed together [20], [28].

Current limiting is the central difficulty. A voltage source behind a reactance X of 0.15 pu would deliver $(E - V_g)/X$, more than 6 pu for a bolted fault. Semiconductors allow perhaps 1.1 to 1.5 pu. Baeckeland *et al.* classify the remedies as direct limiters, which saturate the current reference and make the converter a current source while limited, and indirect limiters, which keep voltage-source behaviour by subtracting a virtual-impedance drop from the voltage reference [25], [26]:

$$e^* = E \angle \theta - Z_{vi} i, |Z_{vi} + jX| \geq \frac{E_0}{I_{max}} \quad (6.75)$$

The magnitude condition in Equation (6.75) sizes the virtual impedance for a bolted terminal fault with fixed E_0 . It does not by itself limit the transient current, because sampling delay, the initial inductor current and modulation saturation are outside it. Figure 6.53 compares the ideal steady currents for aligned E and V_g . Direct current saturation uses the full current capability but gives up voltage-source behaviour; a static virtual impedance keeps an impedance-shaped source but under-uses the current capability in a shallow sag; a threshold impedance lies between the two. The ratio X_{vi}/R_{vi} affects both damping and transient stability, and neither improves monotonically with it in every grid, so variable-impedance schemes are tested against the actual network and limiter dynamics [24]–[26].

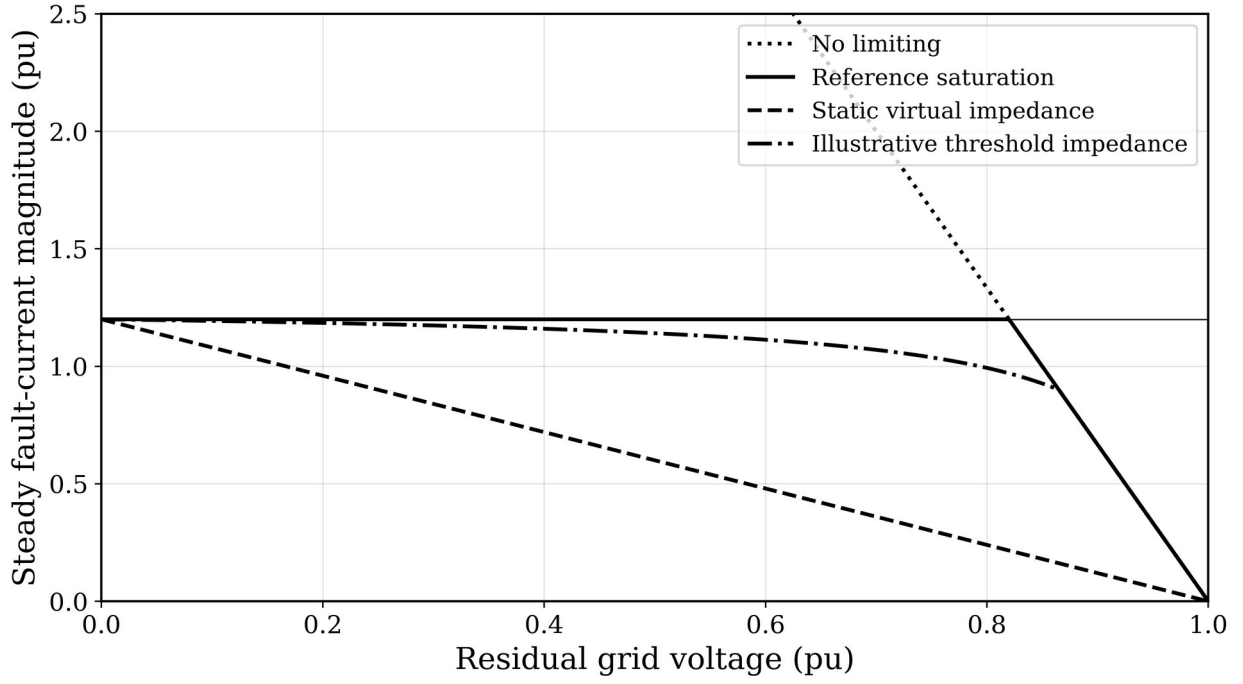


Figure 6.53. Ideal steady fault-current magnitudes for aligned $E = 1$ pu and V_g , $X = 0.15$ pu, $I_{max} = 1.2$ pu. The threshold example uses $|Z_{eff}| = X + K \max(1 - V_g - XI_{th}, 0)$, $I_{th} = 0.9$ pu and $K = 0.790$, an illustrative static law.

Open questions. Preserving voltage-source behavior during unbalanced faults; angle windup and loss of synchronism while limited; coordination among devices using different limiters; and demonstrating performance against stated interoperability and test requirements. UNIFI Version 3 provides an updated functional specification framework [33]; it is not a substitute for the applicable connection agreement or grid code.

6.7.5 Converter interaction and impedance-based stability

A STATCOM near a wind farm, an HVDC terminal or another FACTS device can interact with them at frequencies far above the electromechanical range. The standard tool is the impedance-based criterion [30]. Around an operating point, the STATCOM is represented by its small-signal output admittance $Y_c(s)$ and the network by its impedance $Z_g(s)$. The terminal voltage then obeys

$$V(s) = \frac{V_g(s)}{1 + Z_g(s)Y_c(s)} \quad (6.76)$$

Equation (6.76) uses an input-admittance convention, with positive current drawn into the converter. For stable scalar subsystems the Nyquist criterion is applied to $Z_g Y_c$ with its actual pole count; an intersection of the two magnitude curves alone does not indicate instability. Coupled dq or sequence models need the matrix return ratio with the generalized Nyquist criterion, or an equivalent state-space analysis. A strictly passive converter admittance connected to a passive grid is a sufficient condition for stability, but a scan showing $\text{Re } Y_c \geq 0$ over a limited band does not prove it [29], [30].

Digital control spoils passivity. For a current-controlled converter with filter inductance L , proportional gain K_p , voltage feedforward through a filter $H_{ff}(s)$ and a computation-plus-modulation delay T_d ,

$$Y_c(s) = \frac{1 - H_{ff}(s)e^{-sT_d}}{sL + K_p e^{-sT_d}} \quad (6.77)$$

For the scalar model of Figure 6.54 ($T_d = 1.5T_s$, $f_s = 5$ kHz, 400 Hz current-loop bandwidth and a 100 Hz first-order voltage-feedforward filter), $\text{Re } Y_c$ turns negative near 0.84 kHz, close to the familiar $f_s/6$

boundary of delay-dominated models; the exact crossing depends on the gains and the feedforward. Negative conductance marks a risk of adverse interaction rather than a proven instability: a grid resonance in this band, such as the 1.2 kHz example, has to be assessed with a return-ratio or coupled-model stability test.

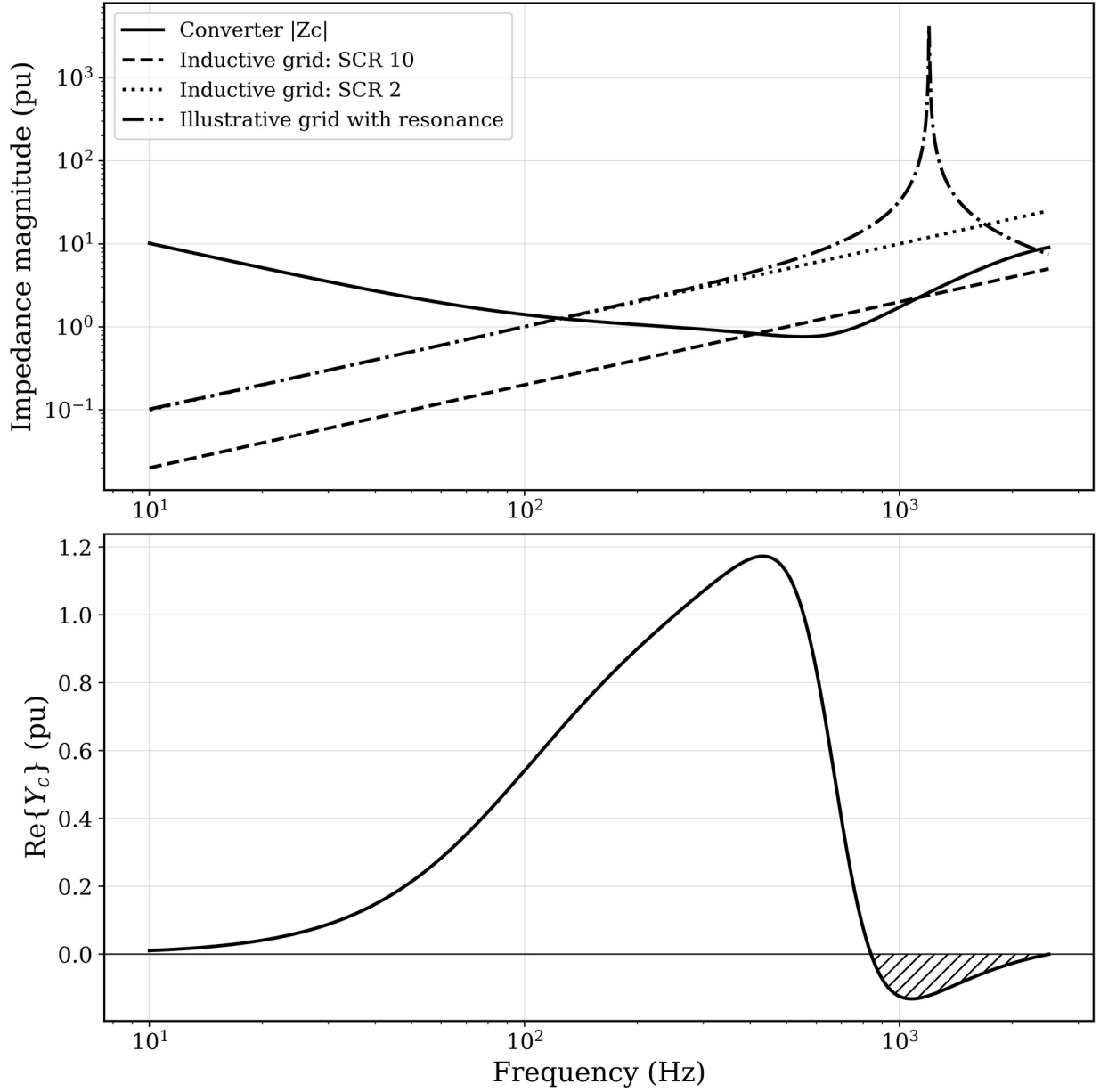


Figure 6.54. Scalar grid impedances and converter input conductance from Equation (6.77): $f_s = 5$ kHz, $T_d = 1.5 T_s$, coupling $X = 0.15$ pu at 50 Hz, 400 Hz proportional current-loop bandwidth and 100 Hz first-order feedforward. The hatched band marks negative conductance.

Figure 6.55 places typical phenomena on a frequency axis. Fundamental-frequency phasor models suit electromechanical and outer voltage-control studies but cannot resolve switching harmonics or fast current-loop interactions. Average-value EMT models can represent fast controls if their states, delays and bandwidth are included, and switching EMT models are needed for switching ripple and device stress. The frequency boundaries between these model classes are approximate.

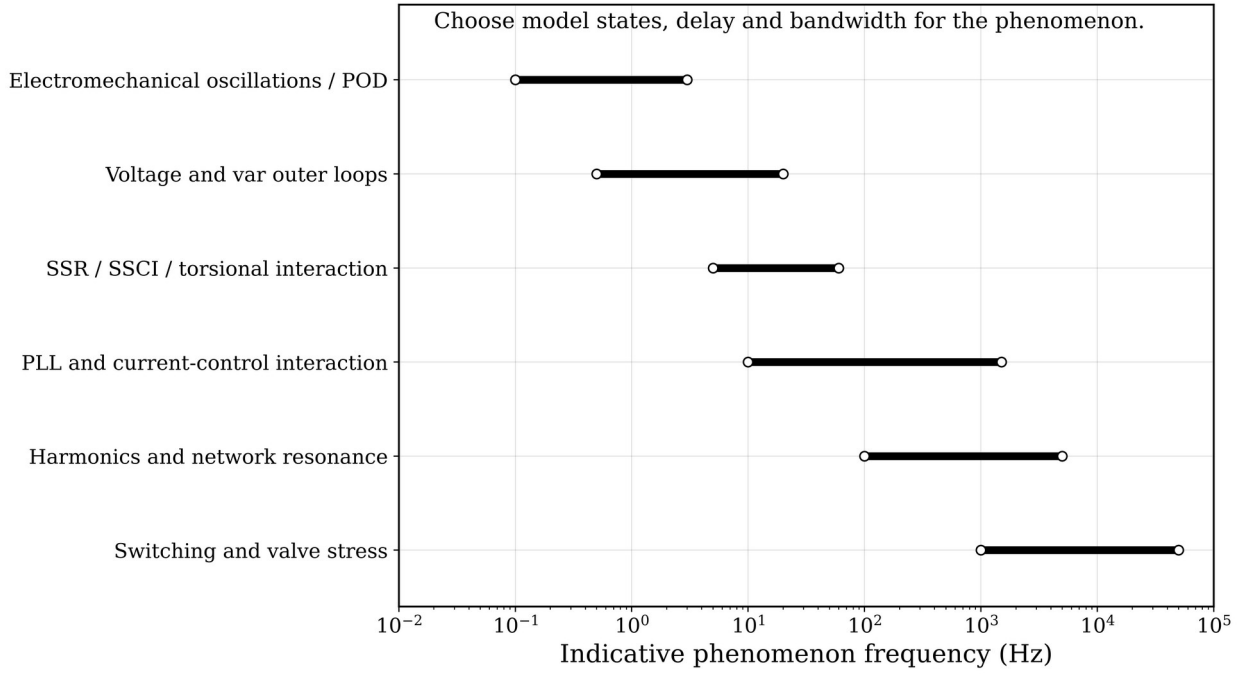


Figure 6.55. Indicative frequency ranges of shunt-compensator phenomena and the model classes suited to them; the boundaries are approximate.

Open questions. Impedance models that remain valid when limiters and mode switching are active; frequency-coupled (dq) impedance models of MMC STATCOMs including cluster-energy control; online impedance estimation for adaptive tuning; and design rules that make passivity hold over the widest band at acceptable cost.

6.7.6 Hybrid, coordinated and data-assisted control

Commercial hybrid compensators combine a converter with switched capacitor or reactor branches so that the converter supplies fast, continuous control while the passive part carries the steady base (Section 6.6.8). The dispatch problem can be written as minimizing losses subject to keeping dynamic reserve:

$$\min_{Q_{conv}, N_k} P_{loss, conv}(Q_{conv}) + \sum_k P_{loss, k}(N_k), Q_{conv} + Q_{passive}(V) = Q_{ref},$$

$$Q_{conv, min}(V, P) + Q_{res, ind} \leq Q_{conv} \leq Q_{conv, max}(V, P) - Q_{res, cap}. \quad (6.78)$$

Positive Q is delivered to the network. Here $Q_{passive}(V) = \sum_k N_k B_k V^2$ on common per-unit bases; capacitor stages have $B_k > 0$, reactors $B_k < 0$. N_k is integer within the installed stage count. The separate nonnegative margins $Q_{res, cap}$ and $Q_{res, ind}$ reserve upward capacitive and downward inductive capability. Both limits depend on voltage, active-power duty, modulation and current capability. Equivalently impose these two bounds on $Q_{ref} - Q_{passive}(V)$ for each proposed stage state. Switching limits, dwell times and capacitor transients are additional constraints in a time-horizon dispatch.

Model-predictive, adaptive and machine-learning controllers are being studied for these tasks. Their value has to be demonstrated against a well-tuned conventional controller, with stability margins, constraint handling, tolerance of network changes, communication needs and real-time implementability stated. A data-driven controller is not better merely because it is newer, and a controller that cannot be certified for protection and grid-code compliance will not be installed.

6.7.7 Validation and evidence

Claims about a new STATCOM controller should climb a ladder of evidence (Figure 6.56). An analytical model gives insight; RMS simulation tests the system-level effect; sensitivity studies check behaviour across

operating points and network strengths; EMT simulation, first with average-value and then switching converter models, tests fast dynamics; controller-hardware-in-the-loop (C-HIL) testing exposes sampling, delay, saturation and protection logic; and power-hardware-in-the-loop or field testing closes the loop. CIGRE Technical Brochure 663 gives guidance on STATCOM specification and testing [21]. Digital twins and online parameter estimation can support commissioning and condition monitoring, but they do not replace validated physical models and protection studies.

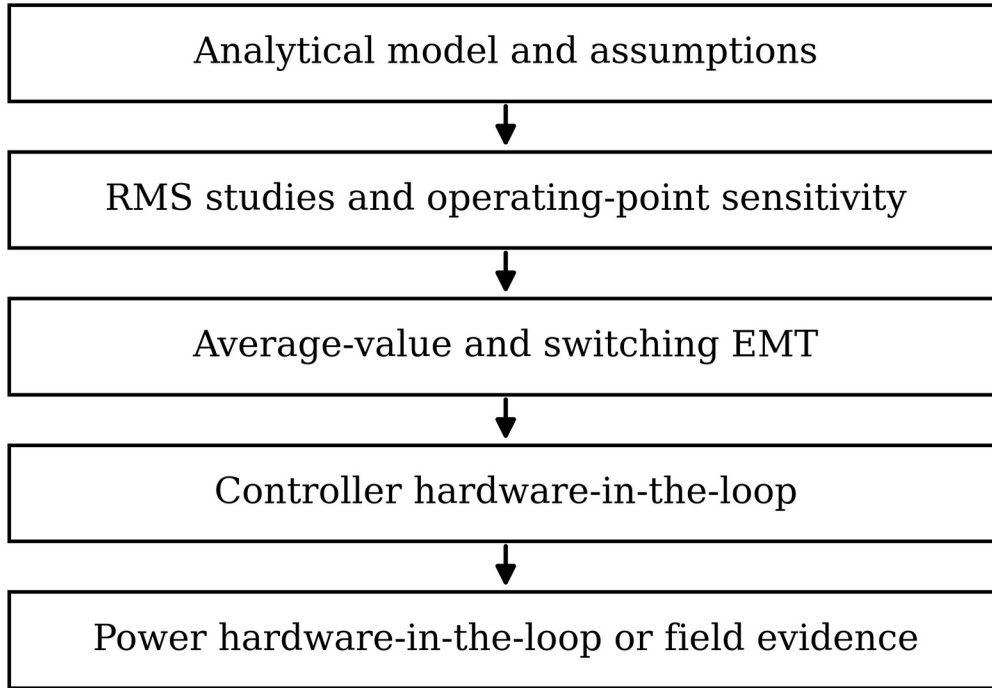


Figure 6.56. Evidence ladder for shunt-compensator controls. Each stage should cover the relevant operating points, limits, delays and protection logic.

6.7.8 Devices and packaging

Semiconductor technology affects switching frequency, conduction loss, valve redundancy, cooling and fault behaviour, but no device family is best at every rating. Silicon IGBTs, press-pack IGBTs and IGCTs dominate high-power STATCOMs, while silicon-carbide devices are most relevant at lower voltage and higher switching frequency. Comparisons should be made at the level of the complete converter and station, including losses, insulation, cooling, redundancy and maintainability, rather than device by device.

6.7.9 From classical concepts to research questions

Table 6.8 maps each classical concept of this chapter onto the state or constraint that recent work adds, and Table 6.9 lists gaps where a measurable contribution is possible.

Table 6.8. Extension of classical shunt-compensation concepts to current research questions.

Classical concept	Added state or constraint	Research question	Suitable validation
Voltage source behind X (Section 6.6.2)	Current saturation, weak-grid dynamics	How can voltage-forming behaviour be kept during faults?	EMT and C-HIL
DC capacitor (Section 6.6.7)	Cluster and cell energies, I_0 or V_0	How can energy ripple and imbalance be reduced?	Average model, then switching EMT
Reactive current control	P-Q sharing with storage, state of charge	How should priorities change during disturbances?	Dynamic simulation with energy constraints
Single compensator	Converter-to-converter impedance interaction	Which controller and grid combinations give poorly damped modes?	Impedance scan, eigenvalues, EMT
Fixed regulator gains (Section 6.5.7)	Operating point and network-strength variation	Can adaptation keep performance across network strengths without losing certifiability?	Sensitivity study, real-time tests

Table 6.9. Research gaps and possible contributions in shunt compensation.

Theme	Limitation of current practice	Potential contribution	Measurable impact
SVC gain tuning	Fixed gain chosen for the weakest ESCR; sluggish in strong grids	Online ESCR estimation with gain scheduling	Constant t_{95} across the ESCR range
Chain-link cell sizing	Capacitance set by worst-case ripple	Ripple-shaping injection; joint optimization with losses	kJ/MVA reduced at fixed ripple
Negative-sequence duty	Delta needs internal balancing current; star needs voltage headroom under stated sequence assumptions	Topologies or controls with lower balancing cost	Cluster current or voltage margin saved
Energy-storage STATCOM	Separate P and Q controllers	Unified, current-limited P-Q damping control with adaptive mode tracking	Damping ratio gained per MJ of storage
Grid-forming STATCOM	Current limiting removes voltage-source behaviour	Virtual-impedance or hybrid limiting, also for unbalanced faults	Fault ride-through with preserved voltage-source behaviour
Hybrid SVC/STATCOM	Heuristic coordination	Optimal dispatch of passive and converter parts	Loss reduction at equal dynamic reserve
Wide-band interaction	Studies at fundamental frequency only	Passivity-based design from 1 Hz to a few kHz	Positive damping across the band

6.7.10 Worked examples on recent topics

Example 6.13: Energy-storage STATCOM: P-Q capability at depressed voltage. A 100 MVA energy-storage STATCOM with $I_{max}=1.0$ pu operates at $V=0.8$ pu. During a fault it is commanded to deliver $P=0.30$ pu. Find the largest capacitive Q it can supply, and the result if the voltage falls to 0.5 pu.

Solution. The current limit gives $\sqrt{P^2+Q^2} \leq V I_{max}=0.8$ pu, so $Q_{max}=\sqrt{0.8^2-0.30^2}=0.742$ pu, or 74.2 Mvar. At 0.5 pu, $Q_{max}=\sqrt{0.5^2-0.30^2}=0.400$ pu, or 40 Mvar.

Note. The circle shrinks in proportion to voltage, so reserving real power for frequency support costs proportionally more reactive capability in a deep sag. A priority rule must be chosen, unless a short-time overcurrent rating is available.

Example 6.14: Negative-sequence duty of delta and star chain-link STATCOMs. A chain-link STATCOM supplies rated positive-sequence reactive current $|I^+|=1.0$ pu and is asked to supply negative-sequence current $|I^-|=0.3$ pu to balance a load. Find (a) the circulating current needed in a delta (SDBC) arrangement and the worst-case cluster current; (b) the range of zero-sequence voltage needed in a star (SSBC) arrangement; (c) the same for $|I^-|=0.8$ pu.

Solution. Use positive- and negative-sequence cluster currents normalized on consistent cluster bases; in delta the physical cluster-current base is the line-current base divided by $\sqrt{3}$. (a) Under the assumptions of (6.69), $|I_0|=0.3$ pu. The worst cluster current over negative-sequence angle is $1+2(0.3)=1.6$ pu, a 60% increase over its positive-sequence base. (b) For $r=0.3$, Equation (6.70) gives $0.231 \leq |V_0|/\hat{V}$. The sum-of-magnitudes bound on cluster voltage is therefore $1+0.429=1.429$ pu; it is a conservative headroom allowance, not the exact voltage in every phase. (c) At $r=0.8$, delta requires $|I_0|=0.8$ pu and up to 2.6 pu cluster current. Star requires $0.444 \leq |V_0|/\hat{V}$; whether its voltage limits are exceeded depends on sequence angle and available cells.

Note. Delta and star distribute the unbalance burden differently: delta adds internal current duty, while star adds voltage headroom. This often favors delta for strong load-balancing duty, but neither topology is an unconditional choice. Compare actual current, voltage, energy ripple, cell count and losses at the required phasor operating points.

Example 6.15: Sizing a static virtual impedance for current limiting. A grid-forming STATCOM has internal voltage $E_0=1.0$ pu, coupling reactance $X=0.15$ pu and current limit $I_{max}=1.2$ pu. Size a static virtual impedance with $X_{vi}/R_{vi}=5$ so that a bolted terminal fault is limited to I_{max} , and find the fault current when the grid voltage falls to 0.5 pu.

Solution. (6.75) requires $|R_{vi} + j(5R_{vi} + 0.15)| = 1/1.2 = 0.833$ pu. Solving, $R_{vi}=0.134$ pu and $X_{vi}=0.672$ pu. Without limiting the bolted-fault current would be $1/0.15=6.67$ pu. At $V_g=0.5$ pu, with E and V_g in phase, the current is $0.5/0.833=0.60$ pu.

Note. The static virtual impedance protects the converter but delivers only half of the available current in this shallow fault (Figure 6.53). Threshold and variable virtual impedances exist to recover that capability.

Example 6.16: Storage needed for inertia-like support. A 100 MVA energy-storage STATCOM on a 50 Hz system emulates $H=5$ s. Find (a) injected power at $df/dt = -1$ Hz/s; (b) delivered energy during a monotonic 0.8 Hz fall; (c) remaining reactive capability at 1.0 pu with $I_{max}=1.0$ pu; (d) minimum supercapacitor maximum-voltage energy if voltage may fall from 100% to 50% of rated voltage.

Solution. (a) $P_{inj} = -(2 \times 5 \times 100/50)(-1) = 20$ MW, positive injection. (b) $E_{del} = (2 \times 5 \times 100/50)(0.8) = 16$ MJ. (c) $Q_{max} = \sqrt{1^2 - 0.2^2} = 0.980$ pu, or about 98.0 Mvar, under the ideal current limit. (d) The 2:1 voltage window uses $1 - 0.5^2 = 0.75$ of maximum stored energy, so at least $16/0.75 = 21.3$ MJ is required before losses, initial-state reserve and ageing margins. The store and DC interface must also be capable of 20 MW.

Note. The emulated machine would store $HS=500$ MJ of kinetic energy, but only 16 MJ is delivered over the excursion. Sizing storage from HS overstates it about thirty-fold.

Key Equations Recap

Table 6.10. Key equations of Chapter 6.

Equation	Expression	Conditions
(6.4)	$P_c = (2V^2/X) \sin(\delta/2)$	Ideal midpoint regulation, lossless symmetric line
(6.6)	$Q_{sh} = (4V^2/X) [1 - \cos(\delta/2)]$	Ideal midpoint injection
(6.7)	$\beta = \alpha - 90^\circ$	Peak- versus zero-crossing-referenced firing
(6.9)	$i = (V_m/\omega L) (\cos \alpha - \cos \omega t)$	$\alpha \leq \omega t \leq 2\pi - \alpha$
(6.13)	$B_{TCR} = B_{max} [2(\pi - \alpha) + \sin 2\alpha] / \pi$	Fundamental, $90^\circ \leq \alpha \leq 180^\circ$
(6.17)	$I_{rms} = (V_m/\omega L) \sqrt{[(\pi - \alpha)(1 + 2\cos^2 \alpha) + 1.5 \sin 2\alpha] / \pi}$	Total RMS branch current
(6.20)	$I_n = \frac{4V_m}{\pi \omega L} \cdot \frac{\cos \alpha \sin n\alpha - n \sin \alpha \cos n\alpha}{n(n^2 - 1)}$	Odd $n \geq 3$, peak magnitude
(6.22)	$i = (V_m/\omega L) (\cos \theta_c - \cos \theta)$	TSR closed at θ_c
(6.25)	$k = n^2 / (n^2 - 1), n = \sqrt{X_C/X_L}$	TSC magnification
(6.29)	$\cos \theta_c = 0, V_{C0} = \pm k V_m$	Transient-free TSC closing
(6.30)	$I_{osc}/I_{AC} = \sqrt{1 - \rho^2(1 - 1/n^2)}$	Scheme 1, $-1 \leq \rho \leq 1$
(6.31)	$I_{osc}/I_{AC} = n 1 - (1 - 1/n^2)\rho $	Positive-crest firing; signed ρ ; same-polarity crest for trapped charge
(6.34)	$V = V_{ref} + X_{SL} I_{SVC}$	Within control range
(6.36)	$I_{SVC} = (E_{th} - V_{ref}) / (X_{th} + X_{SL})$	Operating point
(6.38)	$Q_{TCR} = Q_{cap} + Q_{ind}$	FC-TCR sizing
(6.41)	$B_{eq} = B / (1 - X_\sigma B)$	SVC behind transformer
(6.43)	$X_C = \frac{h^2}{h^2 - 1} \frac{V^2}{Q_F}, X_L = X_C / h^2$	Single-tuned filter
(6.47)	$T_\omega = 2T_b (K_{SL} + K_{N,max}) / (K_{SL} + K_N)$	SVC regulator response
(6.49)	$P_{c,max} / P_{u,max} = 2 \cos(\theta/2)$	Midpoint compensation of long line
(6.51)	$B_{bc}^c = -B_{bc}^L + (G_{ab} - G_{ca}) / \sqrt{3}$, etc.	Load balancing
(6.53)	$P = V V_c \sin \delta_c / X, Q = V (V_c \cos \delta_c - V) / X$	STATCOM exchange
(6.57)	$THD_V = \sqrt{\sum [(kp - 1)^{-2} + (kp + 1)^{-2}]}$	p -pulse converter
(6.60)	$P = \frac{3}{2} (v_d i_d + v_q i_q), Q = \frac{3}{2} (v_q i_d - v_d i_q)$	Amplitude-invariant dq

Equation	Expression	Conditions
(6.63)	$C = 2 \Delta P \Delta t / (V_{dc,max}^2 - V_{dc,min}^2)$	DC capacitor, transient deficit
(6.65)	$P = (V^2/X) \sin \delta + (V I_Q/2) \sin(\delta/2)$	Current-limited midpoint STATCOM

Table 6.11. Key equations of Section 6.7.

Eq.	Expression	Conditions
(6.68)	$E_{stored}/S \geq 1/(2\omega\epsilon)$	Chain-link cell energy, reactive output
(6.69)	$ I_0 = I_\Delta $	Delta cluster balancing
(6.70)	$r/(1+r) \leq V_0 /\dot{V}$	$r \neq 1, I^+ \neq 0$; balanced positive-sequence voltage
(6.71)	$P^2 + Q^2 \leq (V I_{max})^2$; $E(t) = E(0) - \int_0^t P_{ES} dt$	$E_{min} \leq E \leq E_{max}$; bidirectional power bounds
(6.72)	$P_{inj} = -(2HS/f_0) df/dt$	Positive power injected; small frequency excursion
(6.74)	$2H d\omega_{pu}/dt = P^*_{pu} - P_{pu} - D(\omega_{pu} - 1)$; $d\theta/dt = \omega_b \omega_{pu}$	Powers and speed per unit; θ radians; $\omega_b = 2\pi f_0$
(6.75)	$ Z_{vi} + jX \geq E_0/I_{max}$	Static virtual-impedance sizing
(6.76)	$V = V_g / (1 + Z_g Y_c)$	Impedance-based stability

Common Misconceptions

“Firing-angle formulas can be used whichever reference the problem gives.” They cannot. A TCR fired at “30°” conducts at 39% of full susceptance if the angle is peak-referenced ($\alpha = 120^\circ$), whereas a zero-crossing-referenced 30° lies outside the permitted range and would produce asymmetric current with a DC component. Always convert with Equation (6.7) first.

“TCR current THD has a maximum.” Individual harmonics have maxima; total branch THD increases monotonically towards blocking. The figure of about 36% often quoted is the THD at $\alpha = 120^\circ$, where the third harmonic peaks.

“Line current RMS of a delta TCR is $\sqrt{3}$ times the branch RMS.” This is true for the fundamental and for non-triplen harmonics only. Triplens circulate in the delta, so the line RMS current is $\sqrt{3}$ times the branch RMS excluding triplens.

“Firing a discharged TSC at the voltage zero is transient-free.” With a series reactor it minimizes, but does not eliminate, the oscillatory current, which then equals the steady-state current amplitude. Exactly transient-free closing needs crest firing with the capacitor at $\pm k V_m$.

“The series reactor of a TSC is chosen only for di/dt .” It also fixes the branch resonance, which must avoid characteristic TCR harmonics, and it magnifies capacitor voltage and output by k .

“A coupling transformer reduces the capacitive range of an SVC.” It does the opposite. In series with a net capacitive branch it enlarges the effective susceptance and raises the secondary voltage; only the inductive end shrinks.

“A midpoint compensator always doubles transfer capability.” Doubling requires an ideal, unlimited compensator on a short lossless symmetric line. A long line gives $2\cos(\theta/2)$, and any real rating limits the gain further.

“The STATCOM’s DC capacitor stores its reactive energy.” A common DC link need not buffer balanced fundamental three-phase reactive power, but it buffers real-power mismatch and ripple. Individual CHB/MMC cell capacitors still carry internal phase or arm energy ripple, including during purely reactive operation.

“A STATCOM supplies rated *Mvar* at any voltage.” It supplies rated *current*; its var output still falls linearly with voltage, merely more slowly than an SVC’s.

Industry Spotlight

Industrial shunt compensation has developed along two complementary technology paths. Thyristor-based SVCs remain appropriate where large steady-state and dynamic reactive-power ranges, proven high-voltage equipment and economical bulk compensation are required. Converter-based STATCOMs are particularly attractive where rapid current control, compact installation, strong low-voltage reactive-current support, weak-grid performance or modular multilevel implementation is important. These technologies should therefore be compared by duty rather than by age. A practical specification must state the required V-I characteristic, continuous and short-time current ratings, harmonic performance, response time, transformer and filter duties, protection and bypass philosophy, losses, availability and validation tests. The preferred solution is the one that satisfies the required network performance over the specified operating envelope.

Chapter Summary

Shunt compensation controls voltage by injecting reactive current, and the effectiveness of that current is set by the Thevenin reactance of the bus. Ideal midpoint regulation doubles the steady-state transfer limit of a short symmetric line and raises it by $2\cos(\theta/2)$ on a long line, at the cost of a reactive injection that grows rapidly with angle.

The TCR varies fundamental susceptance continuously through a nonlinear firing law and generates odd harmonics. A balanced delta traps triplens, and phase-shifted twelve-pulse groups cancel selected orders. TSRs and TSCs switch complete stages. TSC detuning determines fundamental magnification, harmonic impedance and switching transients; the firing rule must account for trapped charge. An SVC coordinates these branches behind a droop characteristic, with consistent current signs and controls validated across the expected network strengths.

The STATCOM synthesizes voltage behind a coupling impedance and uses rapid current control within its voltage, current and energy limits. Its maximum reactive output falls approximately linearly with depressed voltage, compared with the quadratic limit of an SVC. Common-link and individual-cell energy requirements are different. Hybrid arrangements assign steady compensation to passive branches and dynamic reserve to the converter. Storage and grid-forming control add active-power services, but also require bidirectional energy management, current limiting and interaction studies.

Solved Numerical Practice Problems

Worked examples have already been placed throughout the chapter immediately after the concepts they clarify. The examples collected here give broader design practice and cross-topic synthesis. Unless stated otherwise, firing angles are measured from the voltage zero crossing (Section 6.1.5), the supply frequency is 50 Hz ($\omega = 314.159$ rad/s), and per-unit quantities are on the stated three-phase base. Per-unit current and three-phase reactive power must use one consistent base: if V is the per-unit line-to-line voltage on a three-phase MVA base, a purely reactive shunt current has $|I| = |Q|/V$ in per unit. The factor $\sqrt{3}$ belongs only to conversions between physical line quantities, a distinction that matters in the midpoint-STATCOM examples.

Example 6.17: Single-phase TCR: currents, susceptance and valve stress.

A single-phase TCR with $L = 10$ mH is supplied at 240 V, 50 Hz and fired at $\beta = 30^\circ$ from the voltage peak. Find (a) the maximum var rating; (b) α and the total RMS current; (c) the RMS fundamental current, equivalent susceptance and reactive power; (d) the 3rd, 5th and 7th harmonic currents; (e) the THD; (f) the peak instantaneous current; (g) the valve voltage at the firing instant.

Solution. (a) $X_L = \omega L = 3.1416 \Omega$; $Q_{max} = V^2 / X_L = 240^2 / 3.1416 = 18.335$ kvar. Base $V_m / \omega L = 339.41 / 3.1416 = 108.04$ A.

(b) $\alpha = \beta + 90^\circ = 120^\circ$. Equation (6.17): $I_{rms} = 0.29411 \times 108.04 = 31.776$ A.

(c) $B / B_{max} = F(120^\circ) = 0.39100$, so $B_{TCR} = 0.391 / 3.1416 = 0.12446$ S, $I_1 = 240 \times 0.12446 = 29.870$ A and $Q = V^2 B_{TCR} = 7.169$ kvar.

(d) At 120° , $I_3 = 10.530$ A, $I_5 = 2.106$ A, $I_7 = 0.752$ A RMS.

(e) $THD = \sqrt{(31.776/29.870)^2 - 1} = 36.3\%$. This is the value at the third-harmonic peak, not a maximum THD.

(f) $i_{pk} = (V_m / \omega L)(1 + \cos \alpha) = 108.04 \times 0.5 = 54.02$ A.

(g) $v_T = V_m \sin \alpha = 339.41 \times 0.866 = 293.9$ V; the valve must nevertheless be rated for the full peak $V_m = 339.4$ V plus margin, since smaller α is used near full output.

Example 6.18: Six-pulse TCR at partial conduction: phase and line quantities.

A three-phase six-pulse TCR with 5.0 mH per delta branch operates at 415 V, 50 Hz with $\beta = 15^\circ$. Find (a) the maximum rating; (b) the branch RMS and fundamental currents; (c) the reactive power absorbed; (d) the branch harmonics up to the 13th; (e) the line-current harmonics; (f) the line RMS current and line THD.

Solution. (a) $X_L = 1.5708 \Omega$; $Q_{max} = 3 V^2 / X_L = 328.93$ kvar. Base $V_m / \omega L = 586.89 / 1.5708 = 373.63$ A.

(b) $\alpha = 105^\circ$. $I_{rms} = 180.65$ A; $F(105^\circ) = 0.6742$, so $I_1 = 0.6742 \times 373.63 / \sqrt{2} = 178.12$ A. Branch THD = 16.9%.

(c) $Q = 3 V^2 F / X_L = 221.75$ kvar.

(d) $I_3 = 26.15$, $I_5 = 12.89$, $I_7 = 6.61$, $I_9 = 2.96$, $I_{11} = 0.765$, $I_{13} = 0.462$ A.

(e) Triplens are absent. $I_{5\ell} = 22.33$ A (negative sequence), $I_{7\ell} = 11.44$ A (positive sequence), $I_{11\ell} = 1.32$ A, $I_{13\ell} = 0.80$ A; line fundamental 308.51 A.

(f) The triplen content of the branch current is $\sqrt{\sum I_{3,9,15,\dots}^2} = 26.35$ A, so the line RMS current is $\sqrt{3} \sqrt{180.65^2 - 26.35^2} = 309.54$ A, not $\sqrt{3} \times 180.65 = 312.9$ A. Line THD = $\sqrt{(309.54/308.51)^2 - 1} = 8.2\%$, about half the branch THD.

Example 6.19: Inverting the susceptance law.

A 100 Mvar TCR is controlled through a linearizer. Find (a) the firing angles for 20, 40, 60 and 80% output; (b) the peak-referenced equivalents; (c) the incremental gain $d(B/B_{max})/d\alpha$ at each point; (d) the ratio of the largest to smallest gain; (e) three Newton iterations for 40%, starting at 120° ; (f) the error if a peak-referenced $\beta = 30^\circ$ is mistakenly inserted as α .

Solution. (a) Solving $F(\alpha) = q$: $\alpha = 133.40^\circ$, 119.46° , 108.64° , 99.08° .

(b) $\beta = 43.40^\circ$, 29.46° , 18.64° , 9.08° .

- (c) From Equation (6.15), $-4 \sin^2 \alpha / \pi$ per radian: -0.01173 , -0.01685 , -0.01995 , -0.02167 per degree.
- (d) $0.02167/0.01173=1.85$; over the full 95° to 165° range the variation exceeds 14, which is why a linearizer is essential.
- (e) $\alpha_{k+1} = \alpha_k - [F(\alpha_k) - 0.4]/F'(\alpha_k)$: $F(120^\circ) = 0.3910$, $F' = -0.9549/\text{rad}$, giving $\alpha_1 = 119.460^\circ$; the next iterates are 119.463° and 119.463° . Convergence is quadratic because F is smooth.
- (f) $\alpha = 30^\circ$ is outside 90° to 180° ; the formula would return $F = 1.94$, an impossible 194% of rating. Using the correct conversion, $\beta = 30^\circ$ gives 39.1%.

Example 6.20: Segmented TCR.

A 100 kvar, 415 V delta TCR is built as $m=4$ sections of 25 kvar, only one of which is phase controlled. Find (a) the inductance per section; (b) the maximum third-harmonic branch current of one section; (c) the same for an unsegmented 100 kvar TCR; (d) the configuration and firing angle for 60 kvar; (e) the 3rd and 5th branch harmonics at 60 kvar for both designs; (f) the reduction factor.

Solution. (a) $X = 3(415)^2/25\,000 = 20.667\ \Omega$, $L = 65.79\ \text{mH}$.

- (b) $I_{3,max} = 0.13783 \times (586.89/20.667)/\sqrt{2} = 2.768\ \text{A}$ at $\alpha = 120^\circ$.
- (c) Four times larger: 11.07 A.
- (d) Two sections fully on (50 kvar) and the third supplying 10 kvar, i.e. 40% of its rating: $\alpha = 119.46^\circ$.
- (e) Segmented: $I_3 = 2.767\ \text{A}$, $I_5 = 0.589\ \text{A}$. Single TCR at 60% ($\alpha = 108.64^\circ$): $I_3 = 9.271\ \text{A}$, $I_5 = 4.046\ \text{A}$.
- (f) The worst-case reduction is exactly $m=4$. At this particular operating point it is 3.35 for the 3rd and 6.9 for the 5th.

Example 6.21: TCR valve and reactor design.

A 60 Mvar, 13.8 kV, delta-connected TCR uses thyristors with a repetitive peak voltage of 8 kV, applied at 60% of rating, and reactors with $Q=80$. Find (a) branch reactance and inductance; (b) branch RMS and peak currents at full conduction; (c) RMS and average current per thyristor; (d) the peak valve voltage with 10% overvoltage; (e) the number of series thyristors with one redundant device; (f) the reactor losses at full conduction.

Solution. (a) $X = 3(13.8\ \text{kV})^2/60\ \text{Mvar} = 9.522\ \Omega$; $L = 30.31\ \text{mH}$.

- (b) $I_b = 13\,800/9.522 = 1449.3\ \text{A}$; $I_{pk} = 2049.6\ \text{A}$.
- (c) $I_{T,rms} = I_{pk}/2 = 1024.8\ \text{A}$; $I_{T,av} = I_{pk}/\pi = 652.4\ \text{A}$.
- (d) $1.1 \times \sqrt{2} \times 13.8\ \text{kV} = 21.47\ \text{kV}$.
- (e) $21.47/(0.6 \times 8) = 4.47$, rounded to 5, plus one redundant: six thyristors per string in each direction.
- (f) $R = X/Q = 0.1190\ \Omega$; loss per branch $1449.3^2 \times 0.119 = 250\ \text{kW}$; 750 kW in total, 1.25% of rating.

Example 6.22: TCR quality factor and losses at partial conduction.

A single-phase TCR with $L = 18\ \text{mH}$ and $Q = 60$ is fed at 230 V and operated at $\alpha = 120^\circ$. Find (a) σ ; (b) the peak current; (c) the RMS current; (d) the fundamental current and vars; (e) the reactor loss; (f) the THD.

Solution. (a) $\sigma = 2(180^\circ - 120^\circ) = 120^\circ$; each thyristor conducts for 120° .

- (b) $X = 5.6549\ \Omega$; $i_{pk} = (325.27/5.6549)(1 + \cos 120^\circ) = 28.76\ \text{A}$ at $\omega t = 180^\circ$.
- (c) $I_{rms} = 0.29411 \times 57.52 = 16.92\ \text{A}$.

- (d) $I_1 = 0.391 \times 57.52 / \sqrt{2} = 15.90$ A; $Q = 230^2 \times 0.391 / 5.6549 = 3.658$ kvar.
- (e) $R = X / Q_f = 0.09425 \Omega$. Using the ideal lossless waveform as a first-order estimate, $P = I_{rms}^2 R \approx 26.97$ W. With finite resistance the extinction angle and waveform change slightly; an exact valve/conduction study must solve the R-L differential equation. The printed current and THD values in this example use the lossless waveform.
- (f) 36.3%.

Example 6.23: Binary-weighted TSR bank.

A three-phase delta TSR bank must provide 0 to 350 kvar in 50 kvar steps at 415 V. Find (a) the section ratings; (b) their inductances; (c) the branch currents; (d) the thyristor currents of the largest section; (e) the list of levels.

Solution. (a) Seven steps need three binary sections: 50, 100 and 200 kvar.

- (b) $X = 3 V^2 / Q$: 10.334, 5.167 and 2.583 Ω ; $L = 32.89, 16.45$ and 8.22 mH.
- (c) 40.16, 80.32 and 160.64 A per branch.
- (d) $I_{pk} = 227.18$ A; $I_{T,rms} = 113.59$ A; $I_{T,av} = 72.31$ A; blocking 586.9 V peak.
- (e) 0, 50, 100, 150, 200, 250, 300 and 350 kvar (eight states).

Example 6.24: Overcharged TSC: scheme 2.

A 12 kvar, 230 V TSC with $n=5$ is found charged to $V_{C0} = 380$ V. Find (a) the scheme and angle; (b) I_{osc} / I_{AC} ; (c) the steady peak current; (d) the peak current bound; (e) the ratio if the opposite-polarity crest were used; (f) the valve stress after the next turn-off.

Solution. (a) $\rho = 1.168 > 1$: scheme 2, crest firing at $\theta_c = 90^\circ$ (positive crest, same polarity as V_{C0}).

- (b) $5|0.96 \times 1.168 - 1| = 0.608$.
- (c) $\sqrt{2} \times 12\,000 / 230 = 73.79$ A.
- (d) $73.79 \times 1.608 = 118.6$ A.
- (e) At the negative crest ρ effectively changes sign: $5|-1.121 - 1| = 10.6$, a destructive inrush.
- (f) After settling to sinusoidal steady state, the next turn-off leaves $|V_C| = k V_m$ and peak valve stress $(1+k)V_m = 664.1$ V. An immediate turn-off during the oscillatory transient need not leave this trapped voltage; use the complete LC state to calculate its actual blocking stress.

Example 6.25: Medium-voltage TSC: detuning and di/dt .

A star-connected 20 Mvar TSC on a 33 kV bus uses $n=4.5$. Find (a) the net, capacitive and inductive reactances per phase; (b) C and L ; (c) the steady peak current; (d) the characteristic impedance; (e) the di/dt for a mistimed closure at the worst instant; (f) the minimum L for a 20 A/ μ s valve.

Solution. (a) $X_{net} = 33^2 / 20 = 54.45 \Omega$; $X_C = 57.28 \Omega$; $X_L = 2.829 \Omega$.

- (b) $C = 55.57 \mu$ F; $L = 9.00$ mH.
- (c) $\sqrt{2} \times 20 \text{ MVA} / (3 \times 19.05 \text{ kV}) = 494.8$ A.
- (d) $Z_n = \sqrt{L/C} = 12.73 \Omega$.
- (e) $V_m = 26.94$ kV, worst $(1+k)V_m = 55.29$ kV across L : $di/dt = 6.14$ A/ μ s.
- (f) $L_{min} = 55.29 \text{ kV} / 20 \text{ A} / \mu\text{s} = 2.76$ mH; the detuning reactor already exceeds it by a factor of 3.3.

Example 6.26: Binary TSC bank at low voltage.

A 415 V, delta-connected binary TSC bank has stages of 25, 50, 100 and 200 kvar with $n=4.5$. Find (a) the number of levels; (b) C and L for each stage; (c) branch and line currents of each stage; (d) the hysteresis band needed on a bus with $X_{th}=0.02$ pu on a 1 MVA base.

Solution. (a) $2^4=16$ states, 0 to 375 kvar in 25 kvar steps.

- (b) $X_{net}=3V^2/Q$; $C=1/[\omega k X_{net}]$; $L=k X_{net}/(n^2 \omega)$: 25 kvar: 146.4 μ F, 3.417 mH; 50 kvar: 292.8 μ F, 1.709 mH; 100 kvar: 585.7 μ F, 0.854 mH; 200 kvar: 1171.3 μ F, 0.427 mH.
- (c) Branch: 20.08, 40.16, 80.32, 160.64 A; line: 34.78, 69.56, 139.12, 278.24 A.
- (d) One step changes voltage by $X_{th} \Delta Q=0.02 \times 0.025=0.0005$ pu; the band must exceed this. A practical band of $\pm 0.5\%$ (1% wide) is twenty times the step.

Example 6.27: FC-TCR design at 11 kV.

An FC-TCR on an 11 kV bus must provide +40 Mvar to -20 Mvar. The TCR is delta connected and the capacitance is star connected. Find (a) Q_C and Q_{TCR} ; (b) TCR inductance; (c) the single equivalent capacitance; (d) the components if Q_C is split into a 25 Mvar 5th filter ($h=4.9$) and a 15 Mvar 7th filter ($h=6.9$); (e) firing angles for 0, +20 and -10 Mvar.

Solution. (a) Equation (6.38): $Q_C=40$ Mvar, $Q_{TCR}=60$ Mvar.

- (b) $X_L=3(11 \text{ kV})^2/60 \text{ Mvar}=6.05 \Omega$, $L=19.26$ mH per branch.
- (c) $X_C=V_{LL}^2/Q_C=3.025 \Omega$ per phase, $C=1052 \mu$ F.
- (d) Equation (6.43): 5th: $X_C=5.050 \Omega$, $C=630.3 \mu$ F, $X_L=0.2103 \Omega$, $L=0.670$ mH. 7th: $X_C=8.240 \Omega$, $C=386.3 \mu$ F, $X_L=0.1731 \Omega$, $L=0.551$ mH.
- (e) $Q_{TCR}=Q_C-Q$: 40, 20 and 50 Mvar, i.e. $F=0.667$, 0.333, 0.833, giving $\alpha=105.36^\circ$, 123.59° and 97.54° .

Example 6.28: TSC-TCR dispatch and dead zones.

A TSC-TCR SVC has three TSCs of 50 Mvar each and one TCR of 60 Mvar. Find (a) the range; (b) the dispatch for -45, 0, +80 and +130 Mvar; (c) the firing angles; (d) whether the overlap condition is met; (e) what happens if the TCR is reduced to 40 Mvar.

Solution. (a) -60 to +150 Mvar.

- (b) Equation (6.39): -45: no TSC, TCR absorbs 45; 0: no TSC, TCR blocked; +80: 2 TSCs, TCR 20; +130: 3 TSCs, TCR 20.
- (c) $F=0.75$: $\alpha=101.40^\circ$; $F=0.333$: $\alpha=123.59^\circ$; blocked: 180° .
- (d) $60>50$ Mvar, so each sub-range overlaps the next by 10 Mvar, enough for a hysteresis band of a few Mvar.
- (e) With 40 Mvar the TCR covers only $N \times 50 - 40$ to $N \times 50$; the intervals 0 to 10, 50 to 60 and 100 to 110 Mvar become dead zones that the SVC cannot produce continuously (Figure 6.32).

Example 6.29: Complete design of a ± 150 Mvar SVC.

Design a ± 150 Mvar SVC for a 400 kV bus with a 400/11 kV transformer, using two delta TCRs, two star TSCs ($n=4.3$), a 40 Mvar 5th filter ($h=4.9$) and a 10 Mvar 7th filter ($h=6.9$), all on the 11 kV side (transformer reactance neglected for sizing). Find (a) TSC and TCR ratings; (b) TCR inductance; (c) TSC capacitance and inductance; (d) filter components; (e) TSC valve peak voltage after turn-off; (f) verification of both limits.

Solution. (a) Capacitive: filters 50 Mvar, so TSCs supply 100 Mvar, 50 Mvar each. Inductive: TCRs must absorb $150+50=200$ Mvar, 100 Mvar each.

- (b) $X_L = 3(11)^2/100 = 3.63 \Omega$, $L = 11.55$ mH per branch.
- (c) $X_{net} = 11^2/50 = 2.42 \Omega$; $k = 18.49/17.49 = 1.0572$; $X_C = 2.558 \Omega$, $C = 1244 \mu\text{F}$; $X_L = 0.1384 \Omega$, $L = 0.440$ mH.
- (d) 5th: $X_C = 3.156 \Omega$, $C = 1008 \mu\text{F}$, $L = 0.418$ mH. 7th: $X_C = 12.36 \Omega$, $C = 257.5 \mu\text{F}$, $L = 0.826$ mH.
- (e) $V_m = 8.98$ kV phase peak; $(1+k)V_m = 18.48$ kV per valve.
- (f) Full capacitive: $50 + 2 \times 50 = 150$ Mvar with TCRs blocked. Full inductive: $2 \times 100 - 50 = 150$ Mvar with TSCs off. Both met at rated voltage.

Example 6.30: Midpoint SVC on a 500 km, 400 kV line.

A lossless 400 kV, 50 Hz line, 500 km long, has $Z_c = 300 \Omega$ and $\beta = 0.06^\circ/\text{km}$. End voltages are 1.0 pu. Find

- (a) θ and SIL; (b) the uncompensated maximum power; (c) the midpoint voltage without compensation at $\delta = 60^\circ$ and 90° ; (d) the maximum power with the midpoint held at 1.0 pu and the ratio; (e) the midpoint reactive injection at $\delta = 60^\circ$, 90° and 120° ; (f) the midpoint Thevenin reactance.

Solution. (a) $\theta = 30^\circ$; $P_0 = 400^2/300 = 533.3$ MW.

- (b) $P_0/\sin 30^\circ = 1066.7$ MW.
- (c) Equation (6.48): $\cos 30^\circ/\cos 15^\circ = 0.8966$ pu and $\cos 45^\circ/\cos 15^\circ = 0.7321$ pu.
- (d) $P_0/\sin 15^\circ = 2060.6$ MW; ratio $2 \cos 15^\circ = 1.932$, slightly less than the lumped-line value of 2.
- (e) Equation (6.50): 411.7 Mvar at 60° (1030 MW), 1066.7 Mvar at 90° (1457 MW) and 1920 Mvar at 120° (1785 MW). The injection grows much faster than the power, which is why practical ratings stop well short of the ideal.
- (f) $(Z_c/2) \tan 15^\circ = 40.19 \Omega$, or 0.0251 pu on 100 MVA, 400 kV.

Example 6.31: Load balancing (Steinmetz compensation).

A 500 kVA, 0.8 lagging single-phase load is connected between lines a and b of an 11 kV, 50 Hz system. Find

- (a) G and B^L of the load; (b) the compensator susceptances; (c) element values; (d) branch var ratings; (e) the balanced line current; (f) a phasor check.

Solution. (a) $P = 400$ kW, $Q = 300$ kvar: $G = P/V^2 = 3.306$ mS, $B^L = -Q/V^2 = -2.479$ mS.

- (b) Equation (6.51): $B_{ab}^c = +2.479$ mS (capacitive), $B_{bc}^c = +G/\sqrt{3} = +1.909$ mS (capacitive), $B_{ca}^c = -1.909$ mS (inductive).
- (c) $C_{ab} = 7.89 \mu\text{F}$, $C_{bc} = 6.08 \mu\text{F}$, $L_{ca} = 1.668$ H.
- (d) 300 kvar, 230.9 kvar and 230.9 kvar (reactive).
- (e) $I = P/(\sqrt{3}V) = 20.99$ A in each line.
- (f) With all compensator branches, KCL gives $|I_a| = |I_b| = |I_c| = 20.99$ A, with each current in phase with its own phase voltage. The a-b compensator cancels the load's 300 kvar. The equal and opposite b-c and c-a reactive currents cancel the negative-sequence current caused by the single-phase conductance. Neither reactive branch vanishes individually; their reactive powers sum to zero. This verifies both balance and unity input power factor.

Example 6.32: STATCOM, SVC and hybrid at depressed voltage.

An SVC and a STATCOM are each rated 100 Mvar capacitive at 1.0 pu. A hybrid consists of a ± 50 Mvar STATCOM and a 30 Mvar fixed capacitor. Find (a) both outputs at 0.7 pu; (b) at 0.5 pu; (c) the voltage at which the SVC gives half rating, and the STATCOM output there; (d) the hybrid range at 1.0 pu; (e) at 0.6 pu.

Solution. (a) 49 and 70 Mvar; ratio 1.43.

(b) 25 and 50 Mvar; ratio 2.0.

(c) $v^2 = 0.5$; $v = 0.707$; STATCOM 70.7 Mvar.

(d) +80 to -20 Mvar.

(e) Capacitive $50(0.6) + 30(0.36) = 40.8$ Mvar; inductive $50(0.6) - 10.8 = 19.2$ Mvar. The capacitor contributes only 10.8 Mvar here, so hybridization costs low-voltage support.

Example 6.33: Indirect control: energy needed for a reactive step.

A STATCOM with fixed modulation ($V_c = k V_{dc}$) has $X = 0.12$ pu on 100 MVA and stores 4 MJ at nominal V_{dc} (where $V_c = V = 1$ pu). Find (a) V_c for $Q = 0.5$ pu; (b) the required rise in V_{dc} ; (c) the energy to be added; (d) the average active power to do this in 20 ms; (e) the angle required; (f) its sign.

Solution. (a) $V_c = 1 + 0.5 \times 0.12 = 1.06$ pu.

(b) 6%.

(c) $\Delta W = 4(1.06^2 - 1) = 0.494$ MJ.

(d) 24.7 MW = 0.247 pu.

(e) For a first estimate use mean converter magnitude 1.03 pu: $\sin \delta_c \approx -0.247(0.12)/(1 \times 1.03)$, giving $\delta_c \approx -1.65^\circ$. This assumes the required AC absorption is approximately the average capacitor-charging power and neglects losses and the small phase-shift correction to Q . A time-domain model is needed for the exact 20 ms trajectory.

(f) The negative angle makes the converter absorb active power from the bus. Indirect control is constrained by that charging power, current headroom and the changing DC voltage; it cannot be assessed from the requested reactive step alone.

Example 6.34: Midpoint STATCOM with a current limit.

A lossless line with $V_s = V_r = 1.0$ pu and total $X = 0.5$ pu has a midpoint STATCOM. Find (a) P , Q and I at $\delta = 60^\circ$ with ideal regulation; (b) the same at 90° ; (c) δ_{lim} and P for $I_{max} = 1.0$ pu; (d) P at 90° , 120° and 150° in current limit and the midpoint voltage; (e) the comparison with an SVC of $B_C = 1$ pu; (f) the maximum power in each case.

Solution. (a) Equations (6.4) and (6.6): $P = 4 \sin 30^\circ = 2.0$ pu (uncompensated 1.732); $Q = I = 8(1 - \cos 30^\circ) = 1.072$ pu.

(b) $P = 2.828$ pu; $Q = I = 2.343$ pu.

(c) $\cos(\delta_{lim}/2) = 1 - I_{max} X/4 = 0.875$; $\delta_{lim} = 57.91^\circ$, $P = 1.936$ pu.

(d) Equation (6.65): 2.354, 2.165 and 1.483 pu, with $V_m = 0.832$, 0.625 and 0.384 pu.

(e) Equation (6.66): 2.286, 1.979 and 1.143 pu: the STATCOM is 3%, 9% and 30% higher.

(f) STATCOM 2.361 pu at 94.9° ; SVC 2.286 pu at 90° ; uncompensated 2.0 pu. In per unit $I = Q/V$; writing $Q/(\sqrt{3}V)$ mixes line and phase quantities.

Example 6.35: Voltage restoration at a distribution bus.

A 415 V, 50 Hz feeder with $Z_s = 0.8 + j4 \Omega$ per phase supplies a star load $Z_L = 35 + j27 \Omega$ per phase (constant impedance). Find (a) the current and load voltage without compensation; (b) the drop across the feeder; (c) the shunt capacitive susceptance that restores 415 V at the load; (d) the compensator var rating; (e) its current; (f) the load power after restoration.

Solution. (a) $I = 239.6/|35.8 + j31| = 5.060$ A; $V_L = 223.65$ V (387.4 V line).

- (b) $|I Z_s| = 20.64 \text{ V}$.
- (c) Solving $|V_L(B)| = 239.6 \text{ V}$ numerically: $B = 18.11 \text{ mS}$ per phase, capacitive.
- (d) $B V^2 = 1039 \text{ var}$ per phase, 3.12 kvar three-phase.
- (e) 4.34 A .
- (f) 1028 W per phase, 3.08 kW total. (If the load were modelled as constant power at its depressed-voltage value, a slightly different rating results; the load model must always be stated.)

Unsolved Practice Problems

- 6.1** A 35 kvar , 400 V , 50 Hz delta TCR is to be designed. Find the branch reactance and inductance, the branch and line currents at full conduction, and the firing angle for 40% of rated output.
- 6.2** For a TCR at $\alpha = 130^\circ$, find the 3rd, 5th and 7th branch harmonics as percentages of the full-conduction fundamental, the branch-current THD and the delta line-current THD.
- 6.3** A single-phase TCR of 20 mH on 230 V is fired at $\alpha = 140^\circ$. Find the total RMS current, the fundamental current, the reactive power and the peak instantaneous current.
- 6.4** For a twelve-pulse TCR at $\alpha = 135^\circ$, which harmonics appear in the primary line current? Give the 11th, 13th, 23rd and 25th as percentages of the full-conduction fundamental.
- 6.5** Design a three-section binary TSR bank giving 0 to 175 kvar in 25 kvar steps at 415 V , delta connected. Give the reactance and inductance of each section.
- 6.6** A 60 kvar , 415 V , delta TSC with $n = 4.8$ must deliver its rated output. Find X_{net} , X_C , C , L , k , the peak capacitor voltage and the peak valve voltage after turn-off.
- 6.7** A TSC with $n = 4$ has a residual voltage of $+0.3 V_m$. Find the two scheme-1 closing angles in the positive half cycle, I_{osc}/I_{AC} for scheme 1, and I_{osc}/I_{AC} for crest firing.
- 6.8** A discharged TSC with $n = 4$ is closed by mistake at a voltage crest. Find I_{osc}/I_{AC} and an upper bound on the peak current in multiples of I_{AC} .
- 6.9** An SVC with $V_{ref} = 1.0 \text{ pu}$ and $X_{SL} = 0.04 \text{ pu}$ sits on a bus with $X_{th} = 0.08 \text{ pu}$. Find the SVC current, bus voltage and reactive power for $E_{th} = 1.04 \text{ pu}$ and for $E_{th} = 0.95 \text{ pu}$.
- 6.10** An FC-TCR on a 13.8 kV bus must cover $+60$ to -30 Mvar . Find Q_{TCR} , the delta TCR inductance, the star capacitance and the firing angle for zero output.
- 6.11** A TSC-TCR has four 25 Mvar TSCs and a 30 Mvar TCR. Give the dispatch (TSCs in service and firing angle) for $+10$, $+55$, $+90$ and -20 Mvar .
- 6.12** An FC-TCR with $B_C = 1.0 \text{ pu}$ and $B_{L,max} = 1.4 \text{ pu}$ on its secondary is connected through a transformer with $X_\sigma = 1/6 \text{ pu}$. Find the exact capacitive and inductive limits seen from the primary.
- 6.13** A STATCOM has $X = 0.1 \text{ pu}$. Find V_c for 0.6 pu capacitive output at $V = 1.0 \text{ pu}$, the output at $V = 0.85 \text{ pu}$ at $I_{max} = 1.0 \text{ pu}$, and V_c for 0.4 pu inductive output at 1.0 pu .
- 6.14** A midpoint STATCOM with $I_{max} = 1.5 \text{ pu}$ serves a line with $X = 0.6 \text{ pu}$ and 1.0 pu end voltages. Find δ_{lim} , the power at δ_{lim} , the power at 120° and the ideal maximum.
- 6.15** A STATCOM DC link must cover 1.5 MW for 15 ms while falling from 18 to 16 kV . Find C , and the ripple for $\hat{P}_2 = 0.6 \text{ MW}$ at a mean of 17 kV .
- 6.16** For a 24-pulse STATCOM with coupling reactance 0.15 pu , find the 23rd and 25th harmonic currents as a percentage of rated current.

6.17 A 10 kW resistive load is connected between lines b and c of a 415 V system. Design the balancing compensator and find the line current.

6.18 A utility must choose shunt compensation for a weak 220 kV bus feeding an arc furnace and a wind farm. Fault-induced sags to 0.5 pu occur, flicker limits are tight, and the site is space-constrained. Compare an FC-TCR, a TSC-TCR, a STATCOM and a hybrid STATCOM, and justify a recommendation.

Answer Key (Unsolved Problems)

6.1 13.714 Ω ; 43.65 mH; 29.17 A; 50.52 A; 119.46°.

6.2 12.26%, 0.75%, 2.04%; branch THD 51.6%; line THD 9.7%.

6.3 5.86 A; 4.79 A; 1.103 kvar; 12.11 A.

6.4 Only 12 k $\pm$ 1: 0.58%, 0.35%, 0.13%, 0.10%.

6.5 25 kvar: 20.67 Ω , 65.8 mH; 50 kvar: 10.33 Ω , 32.9 mH; 100 kvar: 5.17 Ω , 16.4 mH.

6.6 8.611 Ω ; 9.002 Ω ; 353.6 μ F; 1.244 mH; 1.0454; 613.5 V; 1200 V.

6.7 17.46° and 162.54° (choose the instant at which the valve with the correct polarity is forward biased); 0.957; 2.875.

6.8 4.0; 5 I_{AC} .

6.9 +0.333 pu, 1.0133 pu, 0.338 pu absorbed; -0.417 pu, 0.9833 pu, 0.410 pu injected.

6.10 90 Mvar; 20.21 mH; 1003 μ F; 105.36°.

6.11 +10: 1 TSC, $\alpha = 113.83^\circ$; +55: 3 TSCs, 105.36°; +90: 4 TSCs, 123.59°; -20: none, 105.36°.

6.12 +1.200 pu and -0.375 pu (using $B_{eq} = B / (1 - X_\sigma B)$).

6.13 1.06 pu; 0.85 pu; 0.96 pu.

6.14 78.39°; 2.107 pu; 2.093 pu; 3.333 pu (ideal); in current limit the maximum is 2.217 pu at 98.4°.

6.15 0.662 mF; 170 V peak-to-peak.

6.16 1.26% and 1.07%.

6.17 $G = 58.06$ mS; capacitive $B_{ca} = 33.52$ mS (106.7 μ F) across c - a ; inductive $B_{ab} = 33.52$ mS (94.95 mH) across a - b ; each 5.77 kvar; line current 13.91 A.

6.18 Open-ended. A STATCOM or suitable hybrid is a plausible choice because its current-limited reactive output falls more slowly during a sag and its current control can suit flicker duty. A complete answer must assess transformer and modulation limits, required negative-sequence duty, filter/resonance study, current-loop and voltage-loop response, available space and whole-station cost. An FC-TCR remains an option where its voltage-sag performance and losses satisfy the specified duty.

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